

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwOTAwNDBfNTM0Nw%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

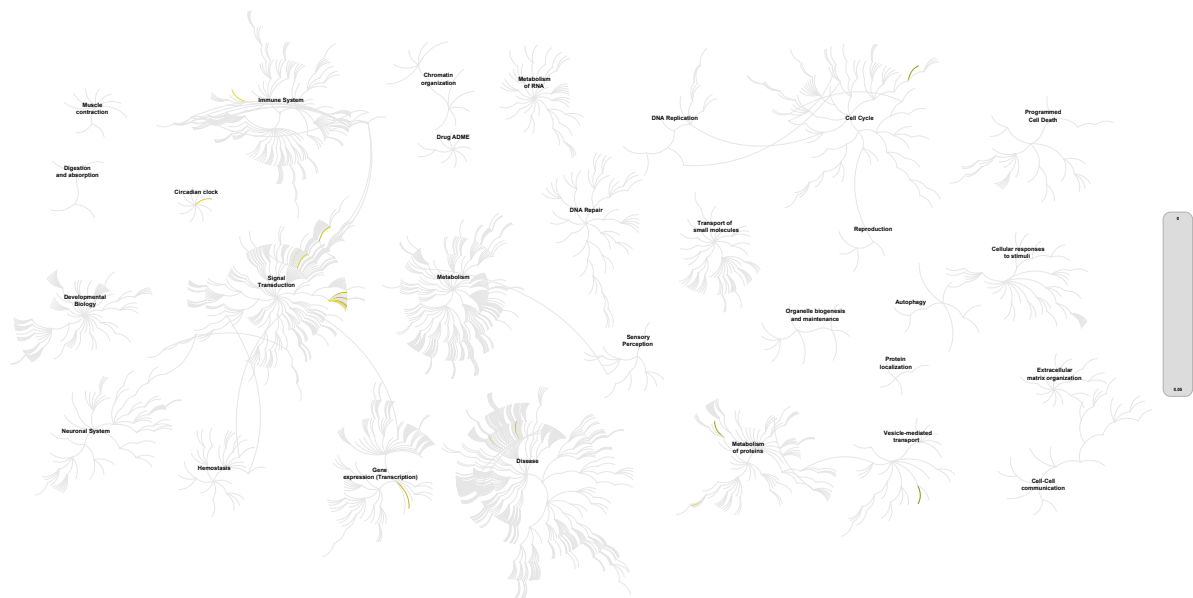
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 2600 out of 3457 identifiers in the sample were found in Reactome, where 2505 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwOTAwNDBfNTM0Nw%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
GRB2 events in ERBB2 signaling	13 / 21	8.46e-04	0.004	1	4 / 4	2.52e-04
Antigen processing: Ubiquitination & Proteasome degradation	104 / 307	0.012	0.005	1	9 / 9	5.67e-04
RHOV GTPase cycle	19 / 39	0.002	0.007	1	1 / 2	1.26e-04
Formation of the ternary complex, and subsequently, the 43S complex	24 / 54	0.002	0.008	1	3 / 3	1.89e-04
The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex	12 / 21	8.46e-04	0.009	1	4 / 4	2.52e-04
Insulin receptor recycling	21 / 44	0.002	0.012	1	4 / 6	3.78e-04
SOS-mediated signalling	7 / 10	4.03e-04	0.016	1	2 / 2	1.26e-04
RHOJ GTPase cycle	24 / 59	0.002	0.021	1	3 / 3	1.89e-04
RHOB GTPase cycle	29 / 75	0.003	0.022	1	4 / 6	3.78e-04
RNA Polymerase II Transcription Termination	27 / 69	0.003	0.023	1	4 / 4	2.52e-04
CDC42 GTPase cycle	54 / 159	0.006	0.027	1	4 / 6	3.78e-04
Signaling by PDGFRA transmembrane, juxtamembrane and kinase domain mutants	10 / 19	7.65e-04	0.027	1	6 / 7	4.41e-04
RAS signaling downstream of NF1 loss-of-function variants	6 / 9	3.62e-04	0.03	1	1 / 1	6.30e-05
Association of Tric/CCT with target proteins during biosynthesis	18 / 44	0.002	0.039	1	2 / 2	1.26e-04
RHOQ GTPase cycle	24 / 63	0.003	0.04	1	4 / 5	3.15e-04
Separation of Sister Chromatids	72 / 205	0.008	0.04	1	7 / 8	5.04e-04
RAB GEFs exchange GTP for GDP on RABs	50 / 121	0.005	0.049	1	21 / 21	0.001
ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	38 / 109	0.004	0.057	1	4 / 4	2.52e-04
RHOA GTPase cycle	50 / 154	0.006	0.059	1	4 / 6	3.78e-04
Signaling by PDGFRA extracellular domain mutants	9 / 19	7.65e-04	0.06	1	5 / 7	4.41e-04
Formation of a pool of free 40S subunits	42 / 113	0.005	0.061	1	2 / 2	1.26e-04
Signalling to RAS	13 / 28	0.001	0.062	1	10 / 10	6.30e-04

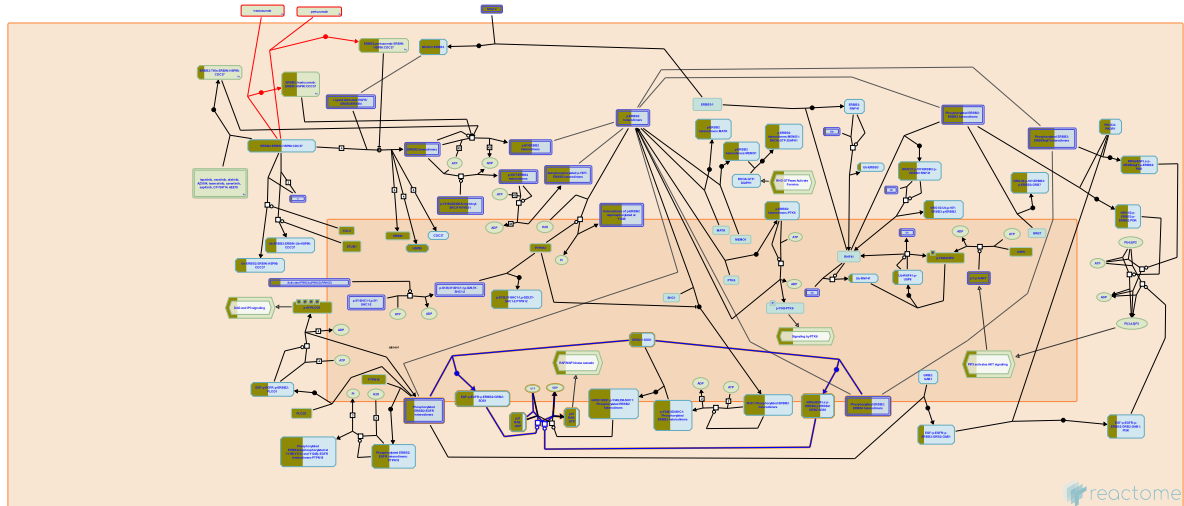
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Ribosomal scanning and start codon recognition	29 / 74	0.003	0.072	1	2 / 2	1.26e-04
Prefoldin mediated transfer of substrate to CCT/TriC	12 / 29	0.001	0.075	1	2 / 2	1.26e-04
Regulation of RAS by GAPs	24 / 65	0.003	0.081	1	5 / 5	3.15e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. GRB2 events in ERBB2 signaling (R-HSA-1963640)



ERBB2:EGFR and ERBB2:ERBB4 can directly recruit GRB2:SOS1 complex through phosphorylated C-tail tyrosines of EGFR (Y1068 and Y1086) and ERBB2 (Y1139) that serve as docking sites for GRB2 (Xie et al. 1995, Sepp-Lorenzino et al. 1996), which, again, results in SOS1-mediated guanyl-nucleotide exchange on RAS and activation of RAF and MAP kinases (Janes et al. 1994, Sepp-Lorenzino et al. 1996).

References

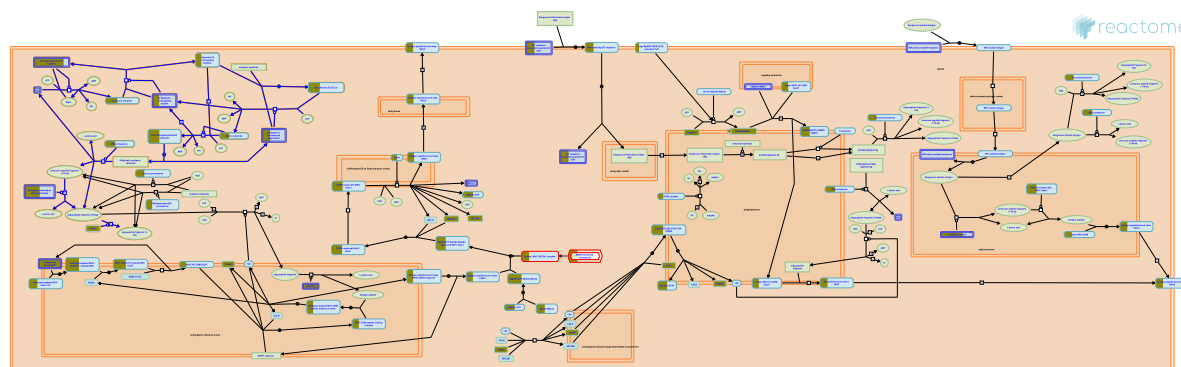
Edit history

Date	Action	Author
2011-11-04	Authored	Orlic-Milacic M
2011-11-04	Created	Orlic-Milacic M
2011-11-07	Edited	Matthews L, D'Eustachio P
2011-11-11	Reviewed	Xu W, Neckers LM

10 submitted entities found in this pathway, mapping to 13 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BTC	P35070	EGF	P01133	ERBB4	Q15303-1, Q15303-2, Q15303-3
HBEGF	Q99075	HRAS	P01112	KRAS	P01116-1, P01116-2
NRAS	P01111	NRG1	Q02297	NRG2	O14511
SOS1	Q07889				

2. Antigen processing: Ubiquitination & Proteasome degradation (R-HSA-983168)



Cellular compartments: cytosol.

Intracellular foreign or aberrant host proteins are cleaved into peptide fragments of a precise size, such that they can be loaded on to class I MHC molecules and presented externally to cytotoxic T cells. The ubiquitin-26S proteasome system plays a central role in the generation of these class I MHC antigens.

Ubiquitination is the mechanism of adding ubiquitin to lysine residues on substrate protein leading to the formation of a polyubiquitinated substrate. This process involves three classes of enzyme, an E1 ubiquitin-activating enzyme, an E2 ubiquitin-conjugating enzyme, and an E3 ubiquitin ligase. Polyubiquitination through lysine-48 (K48) generally targets the substrate protein for proteasomal destruction. The protease responsible for the degradation of K48-polyubiquitinated proteins is the 26S proteasome. This proteasome is a two subunit protein complex composed of the 20S (catalytic core particle) and 19S (regulatory particle) proteasome complexes. The proteasome eliminates most of the foreign and non-functional proteins from the cell by degrading them into short peptides; only a small fraction of the peptides generated are of the correct length to be presented by the MHC class I system. It has been calculated that between 994 and 3122 protein molecules have to be degraded for the formation of a single, stable MHC class I complex at the cell surface, with an average efficiency of 1 in 2000 (Kloetzel et al. 2004, Princiotta et al. 2003).

References

- Loureiro J & Ploegh HL (2006). Antigen presentation and the ubiquitin-proteasome system in host-pathogen interactions. *Adv Immunol*, 92, 225-305. [🔗](#)
- Hearn A, York IA, Bishop C & Rock KL (2010). Characterizing the specificity and cooperation of aminopeptidases in the cytosol and endoplasmic reticulum during MHC class I antigen presentation. *J Immunol*, 184, 4725-32. [🔗](#)
- Kloetzel PM & Ossendorp F (2004). Proteasome and peptidase function in MHC-class-I-mediated antigen presentation. *Curr Opin Immunol*, 16, 76-81. [🔗](#)
- Princiotta MF, Finzi D, Qian SB, Gibbs J, Schuchmann S, Buttgerit F, ... Yewdell JW (2003). Quantitating protein synthesis, degradation, and endogenous antigen processing. *Immunity*, 18, 343-54. [🔗](#)
- Glickman MH & Ciechanover A (2002). The ubiquitin-proteasome proteolytic pathway: destruction for the sake of construction. *Physiol Rev*, 82, 373-428. [🔗](#)

Edit history

Date	Action	Author
2010-10-29	Edited	Garapati P V
2010-10-29	Authored	Garapati P V
2010-10-29	Created	Garapati P V
2011-02-11	Reviewed	Elliott T

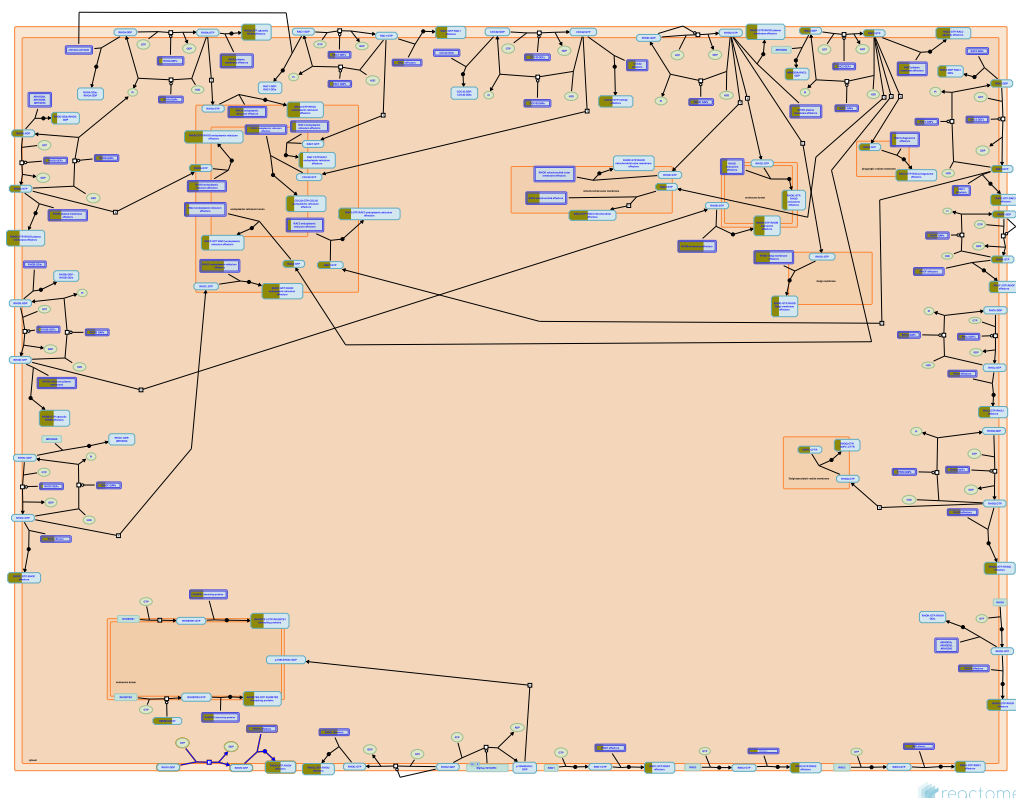
100 submitted entities found in this pathway, mapping to 107 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ANAPC2	Q9UJX6	ANK2	Q8WXI3	ANKRD44	Q9H672
ASB13	Q8W XK3	ASB3	A6NK59, Q9Y575	BTBD6	Q96KE9
CBLB	Q13191	CDC16	Q13042	CDC26	Q8NHZ8
CDC27	P30260	CDC34	P49427	CUL2	Q13617
CUL5	Q93034	DTX3L	Q8TDB6	DZIP3	Q86Y13
ELOB	Q15370	FBXL14	Q8N1E6	FBXL15	Q9H469
FBXL18	Q96ME1	FBXL19	Q6PCT2	FBXL3	Q9UKT7
FBXL5	Q9UKA1	FBXO10	Q9UK96	FBXO17	Q96EF6
FBXO27	Q8NI29, Q96EF6	FBXO30	Q8TB52	FBXO7	Q9Y3I1
FBXW5	Q969U6	HECTD1	Q9ULT8	HECTD2	Q5U5R9
HECTD3	Q5T447	HERC1	Q15751	HERC3	Q15034
HERC6	Q8IVU3, Q9UII4	ITCH	Q96J02	KLHL20	Q9Y2M5
KLHL21	Q9UJP4	KLHL3	Q9UH77	LMO7	Q8WWI1
LNPEP	Q9UIQ6	LTN1	Q94822	MIB2	Q96AX9
MTPN	Q96Q27	NEDD4	P46934, Q96PU5	NEDD4L	Q96PU5
PJA2	Q43164	PRKN	Q60260	PSMA1	P25786
PSMA3	P25788	PSMB4	P28070	PSMC1	P62191
PSMC2	P35998	PSMC6	P62333	PSMD1	Q99460
PSMD12	Q00232	PSMD14	Q00487	PSMD6	Q15008
RLIM	Q9NVW2	RNF14	Q9UBS8	RNF144B	Q7Z419
RNF182	Q8N6D2	RNF19A	Q9NV58	RNF220	Q5VTB9
RNF7	Q9UBF6	RPN1	Q13200	SMURF2	Q9HAU4
SPSB1	Q96BD6	SPSB2	Q99619	STUB1	Q9UNE7
THOP1	P52888	TPP2	P29144	TRAF7	Q6Q0C0
TRIM32	Q13049	TRIM36	Q9NQ86	TRIM41	Q8WV44
TRIM47	Q8WV44	TRIP12	Q14669	UBA3	Q8TBC4
UBA5	Q9GZZ9	UBA52	P62979, P62987	UBA6	A0AVT1
UBA7	P22314, P41226	UBAC1	Q9BSL1	UBE2D3	P61077
UBE2D4	Q9Y2X8	UBE2J2	Q8N2K1	UBE2K	P61086
UBE2Q1	Q7Z7E8	UBE2Q2	Q8WVN8	UBE2W	Q96B02
UBE3A	Q05086	UBE4A	Q14139	UBOX5	Q94941
UBR1	Q8I WV7	UBR2	Q8I WV7, Q8I WV8	UBR4	Q5T4S7
UFL1	Q94874	UNK	Q9H9P5	WSB1	Q9Y6I7
WWP1	Q9H0M0				

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CALM2	P62161	P52888			

3. RHOV GTPase cycle (R-HSA-9013424)



RHOV (also known as Chp) is an atypical RHO GTPase that is thought to be constitutively active due to its high intrinsic guanine nucleotide exchange activity. No guanine nucleotide exchange factors (GEFs) nor GTPase activator proteins (GAPs) that act on RHOV have been identified. RHOV is expressed at very low levels. The expression of RHOV is detected during embryonic development in fish (Tay et al. 2010), frog (Guémar et al. 2007) and chicken (Notarnicola et al. 2008). RHOV is involved in neural crest formation, where its expression is induced downstream of WNT signaling. RHOV is thought to regulate cell adhesion, as its zebrafish orthologue is required for proper localization of E-cadherin and beta-catenin at adherens junctions. RHOV activates JNK and induces apoptosis in rat pheochromocytoma cell line PC12 (Shepelev et al 2011) and in macrophages (Song et al. 2015).

RHOV gene overexpression is a molecular marker of human lung adenocarcinoma (Shepelev and Korobko 2013, Shukla et al. 2017, Ma et al. 2020, Zhang et al. 2020), where RHOV is likely to act as an oncogene (Chen et al. 2021).

For review, please refer to Faure and Fort 2015, and Hodge and Ridley 2020.

References

- Hodge RG & Ridley AJ (2020). Regulation and functions of RhoU and RhoV. *Small GTPases*, 11, 8-15. [🔗](#)
- Faure S & Fort P (2015). Atypical RhoV and RhoU GTPases control development of the neural crest. *Small GTPases*, 6, 174-7. [🔗](#)
- Tay HG, Ng YW & Manser E (2010). A vertebrate-specific Chp-PAK-PIX pathway maintains E-cadherin at adherens junctions during zebrafish epiboly. *PLoS One*, 5, e10125. [🔗](#)

Notarnicola C, Le Guen L, Fort P, Faure S & de Santa Barbara P (2008). Dynamic expression patterns of RhoV/Chp and RhoU/Wrch during chicken embryonic development. *Dev Dyn*, 237, 1165-71.



Guémar L, de Santa Barbara P, Vignal E, Maurel B, Fort P & Faure S (2007). The small GTPase RhoV is an essential regulator of neural crest induction in *Xenopus*. *Dev Biol*, 310, 113-28.



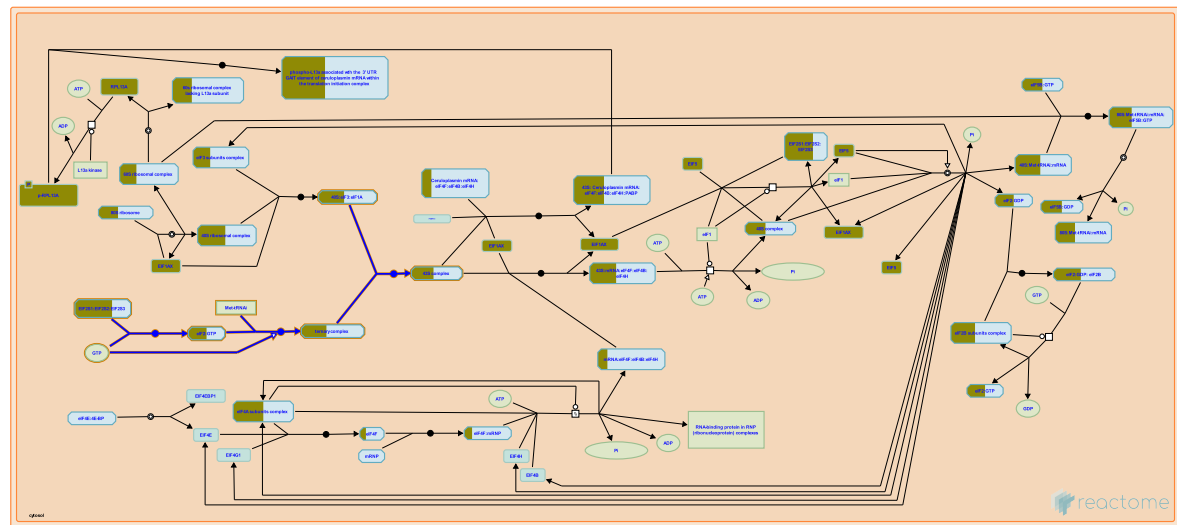
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-03-30	Reviewed	Shepelev MV
2021-04-15	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

19 submitted entities found in this pathway, mapping to 20 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP12	Q8IWW6	CCP110	O43303	CLTC	Q00610
DST	Q03001	GIT1	Q9Y2X7	IQGAP1	P46940
MAP3K11	Q16584	MYO9A	B2RTY4	NCK1	O43639, P16333
NCK2	O43639	PAK3	Q13153	PARD6A	Q9NPB6
PARD6B	Q9BYG5	PEAK1	Q9H792	PIK3R1	P27986
SPTAN1	Q13813	USP9X	Q93008	VANGL1	Q8TAA9
WDR6	Q9NNW5				

4. Formation of the ternary complex, and subsequently, the 43S complex (R-HSA-72695)



Cellular compartments: cytosol.

Binding of the methionyl-tRNA initiator to the active eIF2:GTP complex results in the formation of the ternary complex. Subsequently, this Met-tRNAi:eIF2:GTP (ternary) complex binds to the complex formed by the 40S subunit, eIF3 and eIF1A, to form the 43S complex.

References

Peterson DT, Merrick WC & Safer B (1979). Binding and release of radiolabeled eukaryotic initiation factors 2 and 3 during 80 S initiation complex formation. J Biol Chem, 254, 2509-16. [↗](#)

Safer B, Adams SL, Anderson WF & Merrick WC (1976). Binding of MET-TRNAf and GTP to homogeneous initiation factor MP. J Biol Chem, 250, 9076-82. [↗](#)

Trachsel H, Erni B, Schreier MH & Staehelin T (1978). Initiation of mammalian protein synthesis. II. The assembly of the initiation complex with purified initiation factors. J Mol Biol, 116, 755-67. [↗](#)

McCarthy JE & Tuite M (1990). *New insights into an old problem: ternary complex (Met-tRNAf.eIF.GTP) formation in animal cells., Post-Transcriptional Control of Gene Expression*, 521-526.

Benne R & Hershey JW (1978). The mechanism of action of protein synthesis initiation factors from rabbit reticulocytes. J Biol Chem, 253, 3078-87. [↗](#)

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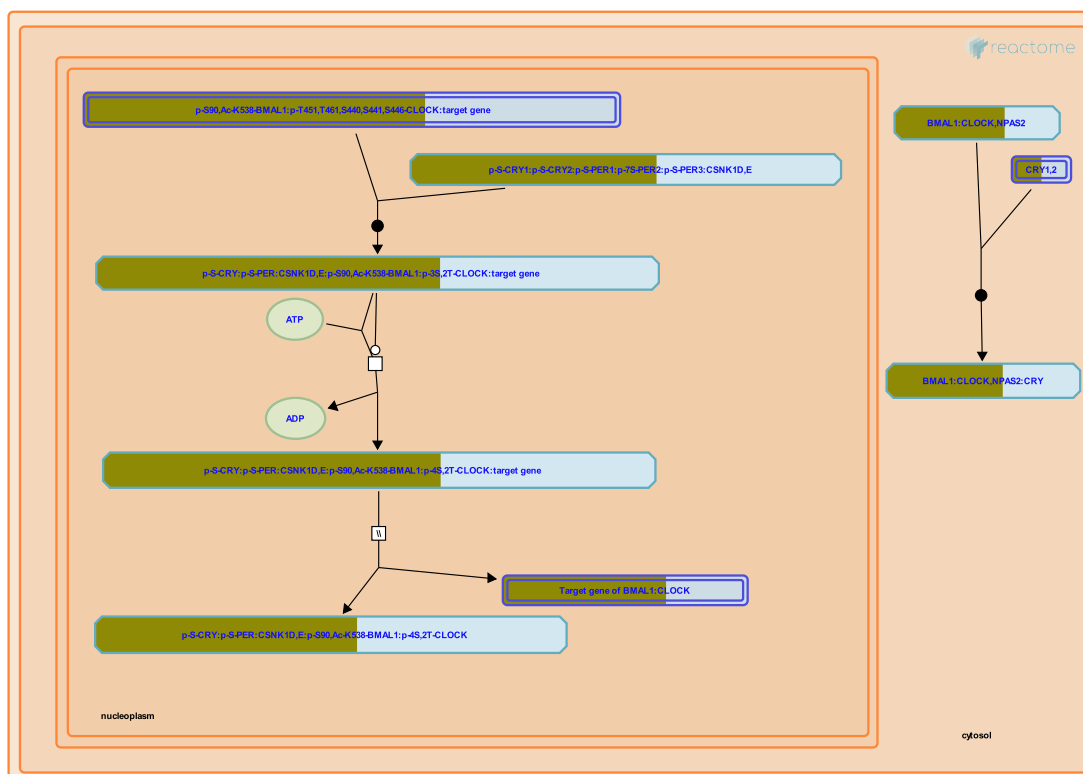
Date	Action	Author
2025-11-15	Modified	Weiser JD

24 submitted entities found in this pathway, mapping to 26 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EIF1AY	P47813	EIF2A	P05198
EIF2S3	P41091	EIF3A	Q14152	EIF3E	P60228
EIF3J	O75822	EIF3M	Q7L2H7	NAB1	P08865

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPS10	P46783	RPS11	P62280	RPS14	P62263
RPS15	P62841	RPS16	P62249	RPS17	P08708
RPS19	P39019	RPS20	P60866	RPS21	P62841, P63220
RPS27	P42677, Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS5	P46782	UBA52	P62979

5. The CRY:PER:kinase complex represses transactivation by the BMAL:CLOCK (ARNTL:CLOCK) complex (R-HSA-9931521)



A complex containing Cryptochrome proteins CRY1, CRY2, Period proteins PER1, PER2, PER3, and kinases CSNK1D and/or CSNK1E binds a heterodimer containing phosphorylated CLOCK and phosphorylated, acetylated BMAL1 bound to E-boxes in the promoters of target genes (inferred from mouse homologs in Griffin et al. 1999, Langmesser et al. 2008, Ye et al. 2011, Ye et al. 2014, Aryal et al. 2017, Cao et al. 2021). This effectively delivers the kinases CSNK1D and/or CSNK1E to the proximity of CLOCK, resulting in further phosphorylation of CLOCK (inferred from mouse homologs in Cao et al. 2021). NPAS2, which acts redundantly with CLOCK, may be similarly phosphorylated. The phosphorylation of CLOCK causes the dissociation of the BMAL1:CLOCK heterodimer from DNA (inferred from mouse homologs in Ye et al. 2014, Chiou et al. 2016, Cao et al. 2021), terminating the diurnal activation of gene expression by BMAL1:CLOCK.

References

- Ye R, Selby CP, Ozturk N, Annayev Y & Sancar A (2011). Biochemical analysis of the canonical model for the mammalian circadian clock. *J Biol Chem*, 286, 25891-902. [🔗](#)
- Aryal RP, Kwak PB, Tamayo AG, Gebert M, Chiu PL, Walz T & Weitz CJ (2017). Macromolecular Assemblies of the Mammalian Circadian Clock. *Mol Cell*, 67, 770-782.e6. [🔗](#)
- Ye R, Selby CP, Chiou YY, Ozkan-Dagliyan I, Gaddameedhi S & Sancar A (2014). Dual modes of CLOCK:BMAL1 inhibition mediated by Cryptochrome and Period proteins in the mammalian circadian clock. *Genes Dev*, 28, 1989-98. [🔗](#)
- Cao X, Yang Y, Selby CP, Liu Z & Sancar A (2021). Molecular mechanism of the repressive phase of the mammalian circadian clock. *Proc Natl Acad Sci U S A*, 118. [🔗](#)

Langmesser S, Tallone T, Bordon A, Rusconi S & Albrecht U (2008). Interaction of circadian clock proteins PER2 and CRY with BMAL1 and CLOCK. BMC Mol Biol, 9, 41. [🔗](#)

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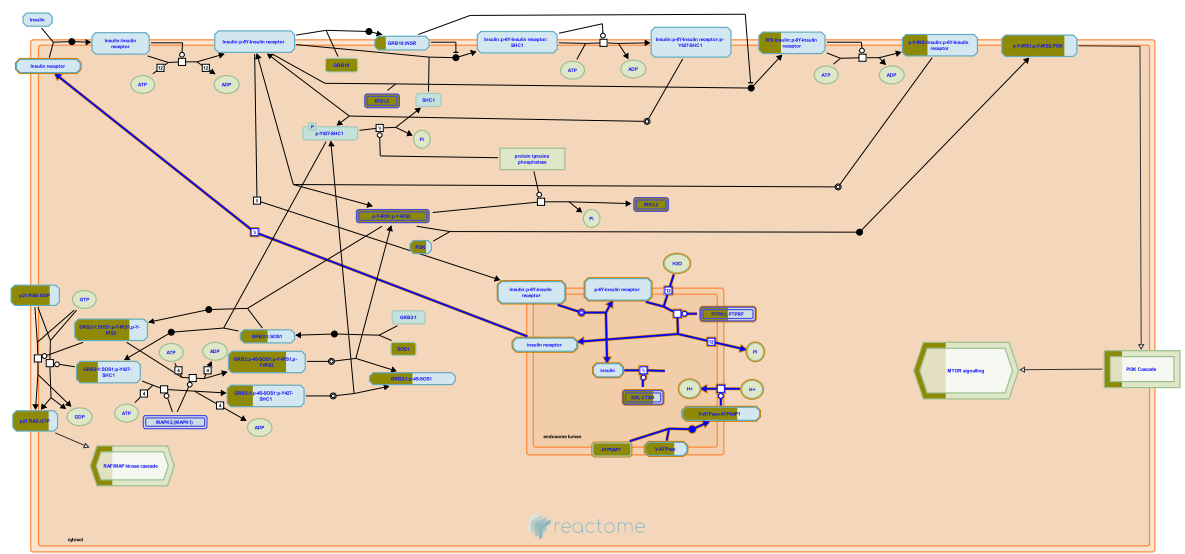
Date	Action	Author
2024-12-09	Edited	May B
2024-12-09	Authored	May B
2024-12-12	Created	May B
2025-03-04	Reviewed	Sancar A, Albrecht U

9 submitted entities found in this pathway, mapping to 12 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CLOCK	O15516	CRY2	Q49AN0	DBT	P49674
KMT2A	Q03164	NPAS2	Q99743	PER1	O15534
PER2	O15055				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
CRY2	ENSG00000121671	NR1D1	ENSG00000126368	PER1	ENSG00000179094
PER2	ENSG00000132326	RORA	ENSG00000069667		

6. Insulin receptor recycling (R-HSA-77387)



Triggered by acidification of the endosome, insulin dissociates from the receptor and is degraded. The receptor is dephosphorylated and re-integrated into the plasma membrane, ready to be activated again by the binding of insulin molecules.

References

Edit history

Date	Action	Author
2003-07-31	Authored	Bevan AP
2003-07-31	Created	Bevan AP

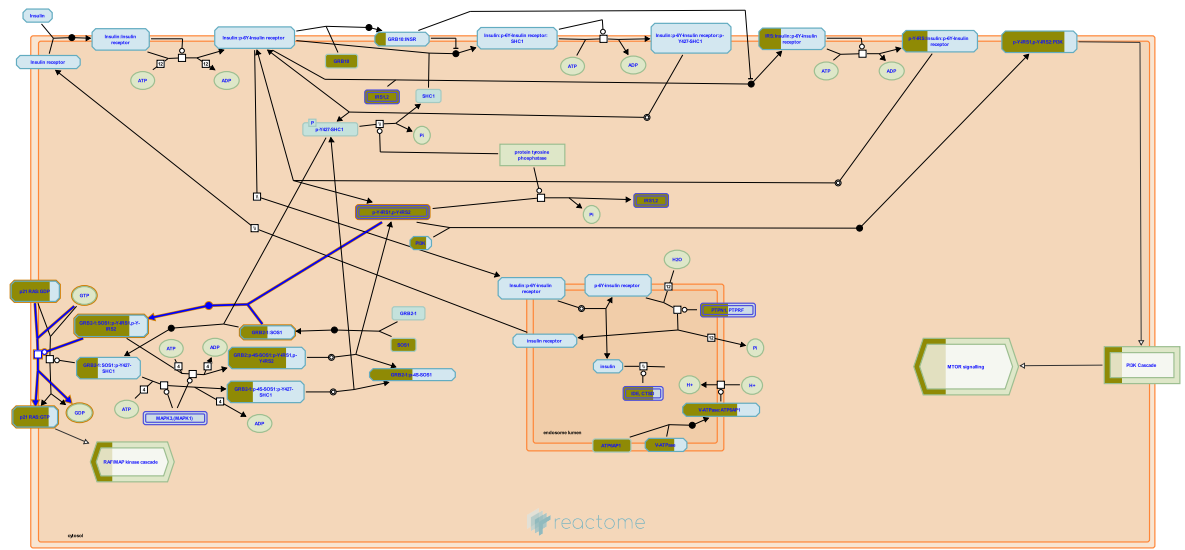
19 submitted entities found in this pathway, mapping to 20 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ATP6AP1	Q15904	ATP6V0A1	Q93050	ATP6V0A2	Q9Y487
ATP6V0B	Q99437	ATP6V0C	P27449	ATP6V0D1	P61421
ATP6V0E1	O15342	ATP6V1A	P38606	ATP6V1B2	P21281
ATP6V1C1	P21283	ATP6V1C2	Q8NEY4	ATP6V1E1	P36543
ATP6V1F	Q16864	ATP6V1G1	O75348, Q96LB4	ATP6V1H	Q9UI12
CTSD	P07339	IDE	P14735	PTPN1	P18031
TCIRG1	Q13488				

Interactors found in this pathway (6)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATP6V0A1	Q93050	Q15904	ATP6V0A2	Q9Y487	Q15904
ATP6V0C	P27449	Q15904	ATP6V0D1	P61421	Q15904
CANX	P27824	Q15904	TCIRG1	Q13488	Q15904

7. SOS-mediated signalling (R-HSA-112412)



Cellular compartments: plasma membrane, cytosol.

SOS is recruited to the plasma membrane and mediates activation of Ras.

References

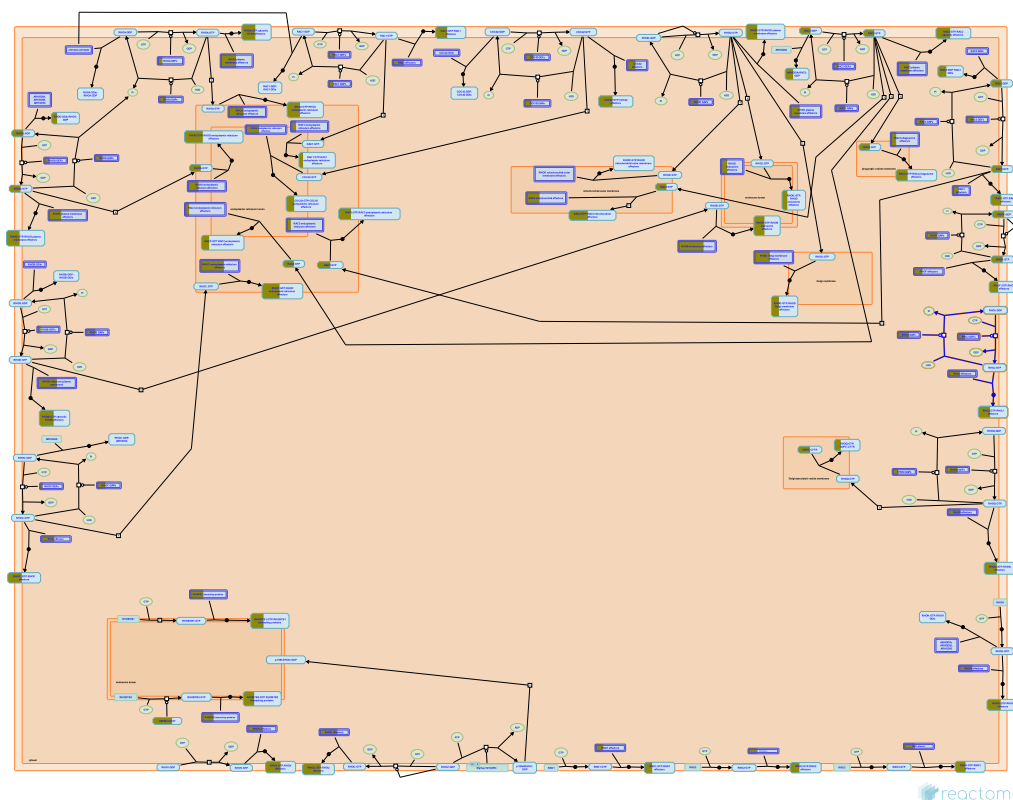
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Date	Action	Author
2004-04-29	Authored	Charalambous M
2004-04-29	Created	Charalambous M

6 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HRAS	P01112	IRS1	P35568	IRS2	Q9Y4H2
KRAS	P01116-1, P01116-2	NRAS	P01111	SOS1	Q07889

8. RHOJ GTPase cycle (R-HSA-9013409)



This pathway catalogues RHOJ (also known as TCL or RHOI) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOJ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOJ. RHOJ, together with RHOQ (TC10), belongs to the CDC42 subfamily of RHO GTPases and shares 85% and 78% amino acid similarity with RHOQ and CDC42, respectively (Vignal et al. 2000). RHOJ regulates the cytoskeleton, including formation of lamellipodia and actin filaments (Vignal et al. 2000, Abe et al. 2003, Aspenstrom et al. 2004, Shi et al. 2016). RHOJ is highly expressed in endothelial cells, regulating their motility and, consequently, vascular morphogenesis (Leszczynska et al. 2011; Shi et al. 2016). As a part of the VEGF signaling cascade, RHOJ promotes retinal angiogenesis (Fukushima et al. 2013). RHOJ plays a role in cancer cell migration and cancer-related angiogenesis (Shi et al. 2016). RHOJ is involved in adipocyte differentiation (Nishizuka 2003; Shi et al. 2016).

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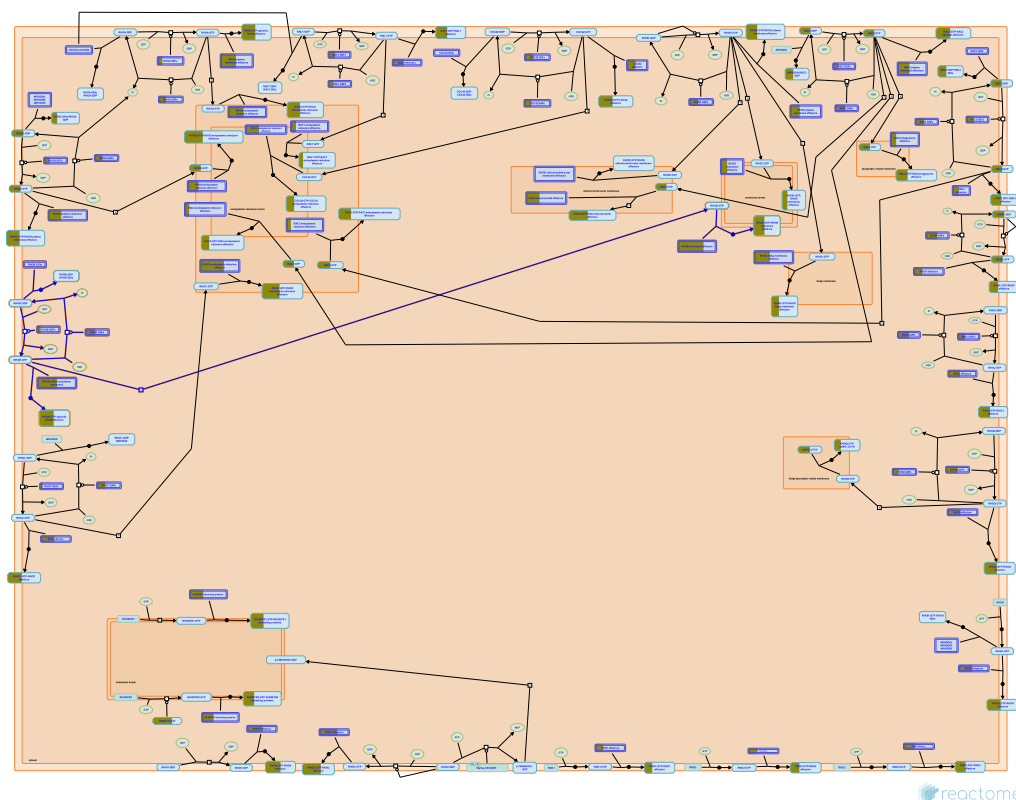
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

23 submitted entities found in this pathway, mapping to 24 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP21	Q5T5U3	ARHGAP33	A7KAX9	CDC42BPA	Q5VT25
DIAPH3	Q9NSV4	FNBP1L	Q5T0N5	GIT1	Q9Y2X7
GJA1	P17302	IQGAP3	Q86VI3	JUP	P14923
NIPSNAP2	O75323	OCRL	Q01968	PAK3	O75914, Q13153
PIK3R1	P27986	PIK3R2	O00459	RAB7A	P51149
SCRIB	Q14160	SNAP23	O00161	STEAP3	Q658P3
STOM	P27105	SYDE1	Q6ZW31	TRIO	O75962
VANGL1	Q8TAA9	WWP2	O00308		

9. RHOB GTPase cycle (R-HSA-9013026)



This pathway catalogues RHOB guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOB effectors. RHOB belongs to the RHOA subfamily of RHO GTPases and shares 85% sequence identity with RHOA and RHOC. Posttranslational modifications of the unique C-terminus of RHOB regulate its subcellular localization. Similar to RHOA and RHOC, RHOB regulates the cytoskeleton and plays a role in cell migration. RHOB plays a role in regulation of membrane trafficking (Vega and Ridley 2018), cell proliferation (Vega and Ridley 2018), cell adhesion (Vega and Ridley 2018), DNA repair (Vega and Ridley 2018), and apoptosis (Prendergast 2001; Vega and Ridley 2018). RHOB-regulated cellular processes are important for the immune system function (Vega and Ridley 2018), pulmonary gas exchange (Wojciak-Stothard et al. 2012; Vega and Ridley 2018), and angiogenesis and vascular function (Vega and Ridley 2018). RHOB has been implicated both as a tumor suppressor and as an oncogene in cancer (Prendergast 2001; Huang and Prendergast 2006; Ju and Gilkes 2018; Calvayrac et al. 2018).

References

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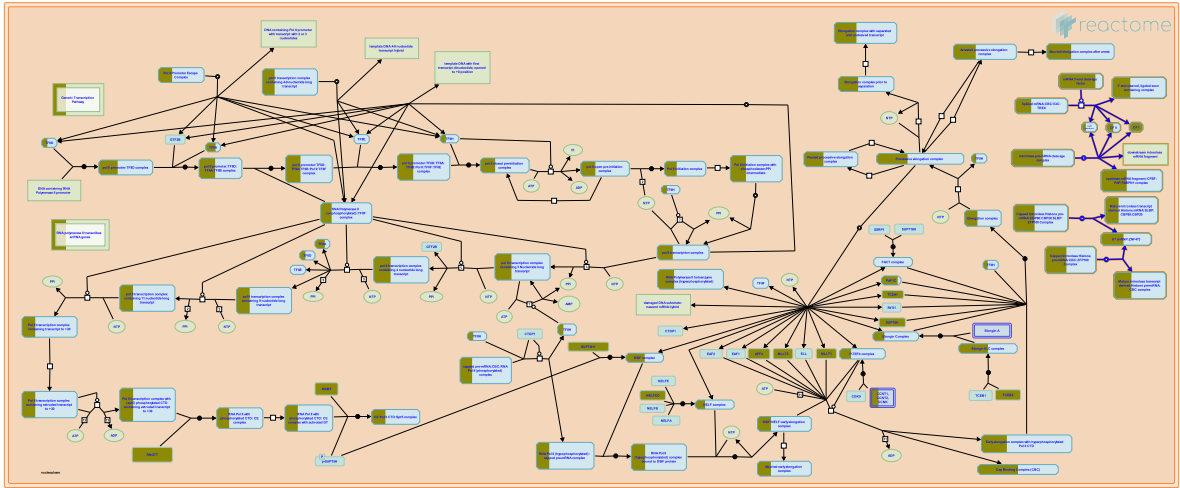
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2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

28 submitted entities found in this pathway, mapping to 29 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABR	Q12979	ANLN	Q9NQW6	ARHGAP21	Q5T5U3
ARHGAP33	A7KAX9	ARHGEF11	O15085	ARHGEF12	Q9NZN5
ARHGEF3	Q9NR81	CAVIN1	Q6NZI2	CIT	O14578, O14578-3
DIAPH3	Q9NSV4	ERBIN	Q96RT1	IQGAP3	Q86VI3
JUP	P14923	MYO9A	B2RTY4	PCDH7	O60245
PIK3R1	P27986	PIK3R2	O00459	PKN1	Q16512
PKN2	Q16513	RHPN2	Q8IUC4	ROCK1	Q13464
ROCK2	O75116	SLK	Q9H2G2	SNAP23	O00161
STARD13	Q9Y3M8	STARD8	Q92502	STOM	P27105
VANGL1	Q8TAA9				

10. RNA Polymerase II Transcription Termination (R-HSA-73856)



Cellular compartments: nucleoplasm.

This section includes the cleavage of both polyadenylated and non-polyadenylated transcripts.

In the former case polyadenylation has to precede transcript cleavage, while in the latter case there is no polyadenylation.

References

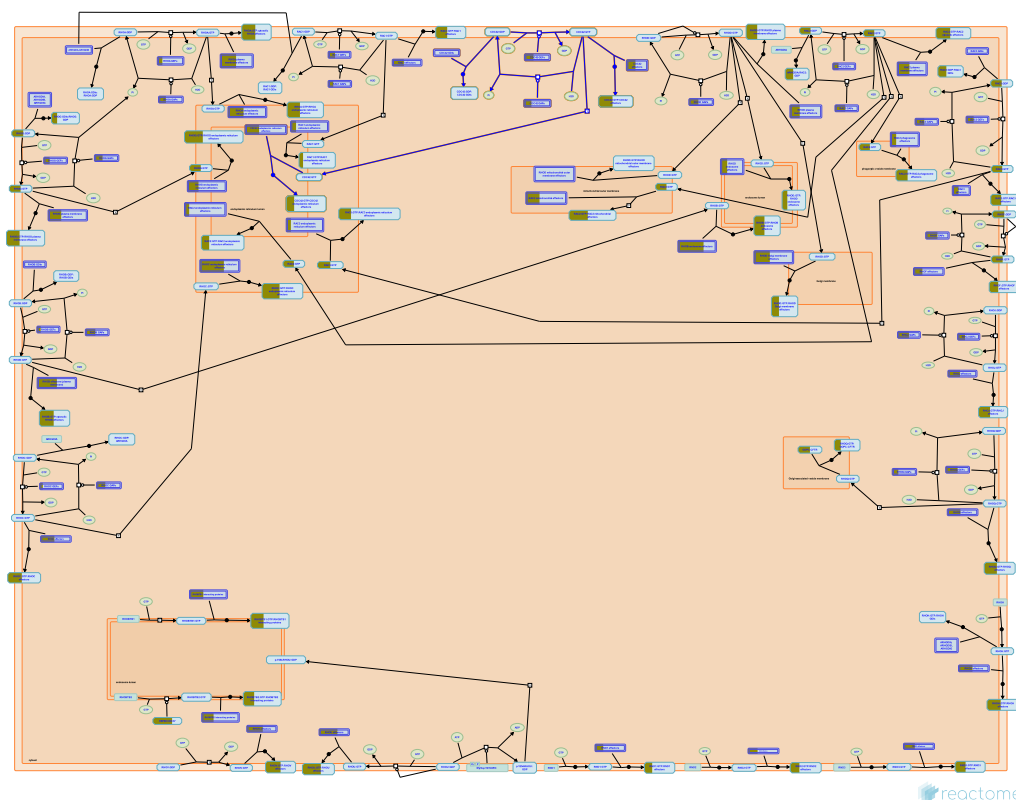
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Date	Action	Author
2003-09-11	Authored	Proudfoot NJ
2003-09-11	Created	Proudfoot NJ
2025-11-12	Edited	Joshi-Tope G
2025-11-15	Modified	Weiser JD

27 submitted entities found in this pathway, mapping to 28 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDC40	O60508	CHTOP	Q9Y3Y2	CLP1	Q92989
CPSF1	Q10570	CPSF2	Q9P2I0	CPSF6	Q16630-1, Q16630-2
CPSF7	Q8N684	CSTF3	Q12996	DDX39A	O00148
DDX39B	Q13838	DHX38	Q92620	FIP1L1	Q6UN15
FYTDD1	Q96QD9	LSM10	Q969L4	LSM11	P83369
NCBP1	Q09161	NUDT21	O43809	PAPOLA	P51003
POLDIP3	Q9BY77	RSF1	Q16629	SLU7	O95391
SMG1	P62308	SNRNPB	P14678	SRP54	Q05519
SRSF11	Q05519	THOC3	Q96J01	THOC6	Q86W42

11. CDC42 GTPase cycle (R-HSA-9013148)



This pathway catalogues CDC42 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and CDC42 effectors. CDC42 is one of the three best characterized RHO GTPases, the other two being RHOA and RAC1. By regulating the cytoskeleton, CDC42 regulates cell polarity across different species, from yeast to humans (Pichaud et al. 2019, Woods and Lew 2019). CDC42 is an essential regulator of polarized morphogenesis in epithelial cells, where it coordinates formation of the apical membrane and lumen formation, as well as junction maturation (Pichaud et al. 2019). CDC42 plays a role in cell-to-cell adhesion and cell cycle regulation (Xiao et al. 2018). CDC42 takes part in the regulation of membrane trafficking. Dysfunction of several CDC42-specific GEFs has been shown to impair intracellular trafficking (Egorov and Polishchuk 2017). CDC42 participates in insulin synthesis and secretion and contributes to the pathogenesis of insulin resistance and diabetic nephropathy (Huang et al. 2019). CDC42 is often dysregulated in cancer because a number of GEFs and GEF activators that act upstream of RAC1 and CDC42 are known oncogenes (Aguilar et al. 2017; Maldonado et al. 2018; Zhang et al. 2019; Maldonado et al. 2020). CDC4 promotes cancer cell proliferation, survival, invasion, migration and metastasis (Xiao et al. 2018), especially under hyperglycemia (Huang et al. 2019).

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Maldonado MDM & Dharmawardhane S (2018). Targeting Rac and Cdc42 GTPases in Cancer. *Cancer Res.*, 78, 3101-3111. [↗](#)

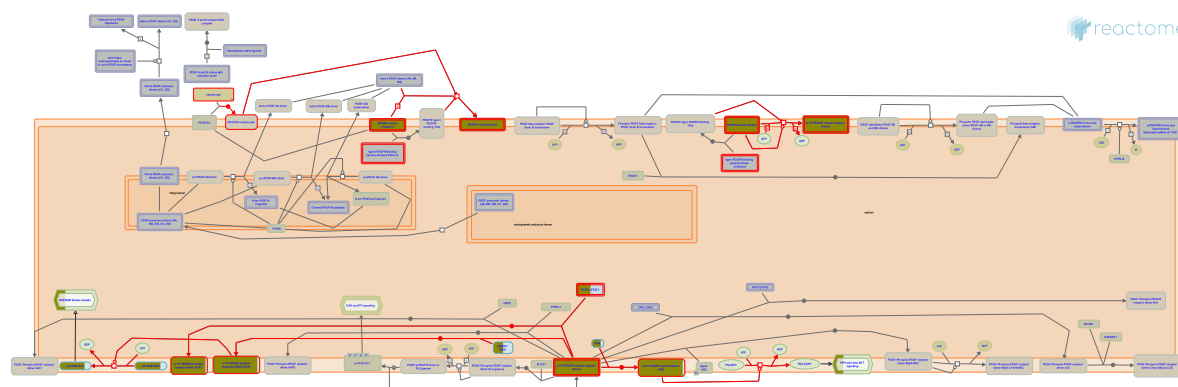
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2017-07-24	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

52 submitted entities found in this pathway, mapping to 55 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABR	Q12979	ARAP2	Q8WZ64	ARFGAP1	Q9NP61
ARFGAP2	Q8N6H7	ARFGAP3	Q9NP61	ARHGAP21	Q5T5U3
ARHGAP31	Q2M1Z3	ARHGAP33	A7KAX9, O14559	ARHGAP42	A6NI28
ARHGAP45	Q92619	ARHGEF11	O15085	ARHGEF12	Q9NZN5
ARHGEF26	Q96DR7	ARHGEF4	Q9NR80	ARHGEF6	Q15052
BAIAP2	Q9UQB8	CDC42BPA	Q5VT25	CDC42EP2	O14613
DIAPH3	Q9NSV4	DOCK10	Q96BY6	DOCK7	Q96N67
DOCK9	Q9BZ29	FGD1	P98174	FGD4	Q96M96
FNBPI1L	Q5T0N5	GIT1	Q9Y2X7	IQGAP1	P46940
IQGAP3	Q86VI3	ITSN1	Q15811	JUP	P14923
KCTD3	Q9Y597	KTN1	Q86UP2	MAP3K11	Q16584
PAK3	O75914, Q13153	PARD6A	Q9NPB6	PIK3R1	P27986
PIK3R2	O00459	PLD1	Q13393	PLEKHG3	A1L390
RAB7A	P51149	SCRIB	Q14160	SNAP23	O00161
SPATA13	Q96N96	STARD13	Q9Y3M8	STARD8	Q92502
STEAP3	Q658P3	STOM	P27105	SYDE1	Q6ZW31
TRIO	O75962, Q96PX9	VANGL1	Q8TAA9	WDR81	Q562E7
WDR91	A4D1P6				

12. Signaling by PDGFRA transmembrane, juxtamembrane and kinase domain mutants (R-HSA-9673767)



Diseases: cancer.

PDGFRA is a type III transmembrane receptor tyrosine kinase. The extracellular domain consists of 5 immunoglobulin (IG) domains that contribute to dimerization and ligand binding. The intracellular region has a juxtamembrane domain that plays a role in autoinhibiting the receptor in the absence of ligand, and a bi-lobed kinase region with an activation loop and the catalytic cleft (reviewed in Klug et al, 2018). Upon ligand binding, PDGFRA undergoes dimerization and transautophosphorylation at at least 11 tyrosine residues in the intracellular domain. These phosphorylated residues are binding sites for downstream effectors of PDGFRA-responsive signaling pathways (reviewed in Klug et al, 2018; Roskoski, 2018).

PDGFRA is subject to activating mutations in a number of cancers, including gastrointestinal stromal tumors (GIST), melanoma and haematological cancers (reviewed in Corless et al, 2011; Wang et al, 2016; Roskoski, 2018). The most prevalent mutations in PDGFRA are at residue V561 in the juxtamembrane domain, N659 in the small lobe of the kinase domain and D842 in the activation loop of the kinase domain. PDGFRA is also subject to short deletions in the activation loop segment (reviewed in Roskoski, 2018). Activated forms of the protein may signal from the plasma membrane, similar to the wild type receptor, however there is also evidence that some mutants, notably D842V and V561D localize primarily to the Golgi membrane (Bahlawane et al, 2014). Activated PDGFRA mutants signal constitutively in the absence of ligand (reviewed in Roskoski, 2018; Wang et al, 2016; Klug et al, 2018).

References

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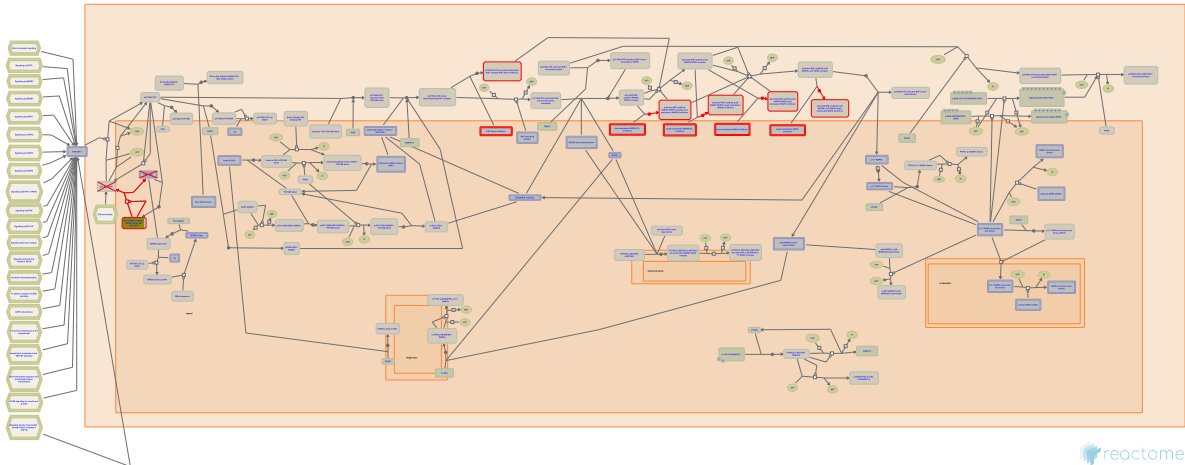
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Date	Action	Author
2020-01-08	Created	Rothfels K
2020-02-06	Reviewed	Ip CKM
2020-02-25	Edited	Rothfels K
2020-02-25	Authored	Rothfels K
2023-03-08	Modified	Matthews L

9 submitted entities found in this pathway, mapping to 10 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HRAS	P01112	KRAS	P01116-1, P01116-2	NRAS	P01111
PIK3CB	P42338	PIK3R1	P27986	PIK3R2	O00459
PVR	P16234	SOS1	Q07889	STAT1	P42224

13. RAS signaling downstream of NF1 loss-of-function variants (R-HSA-6802953)



Diseases: cancer, neurofibromatosis.

NF1 is a RAS GAP that stimulates the intrinsic RAS GTPase activity, thereby shifting the RAS pathway towards the inactive state (reviewed in King et al, 2013). Loss-of-function mutations in NF1 have been identified both in germline diseases like neurofibromatosis 1 and in a range of sporadically occurring cancers. These mutations, which range from complete gene deletions to missense or frameshift mutations, generally decrease NF1 protein levels and abrogate RAS GAP activity in the cells, resulting in constitutive RAS pathway activation (reviewed in Maertens and Cichowski, 2014; Tidyman and Rauen, 2009; Ratner and Miller, 2015).

References

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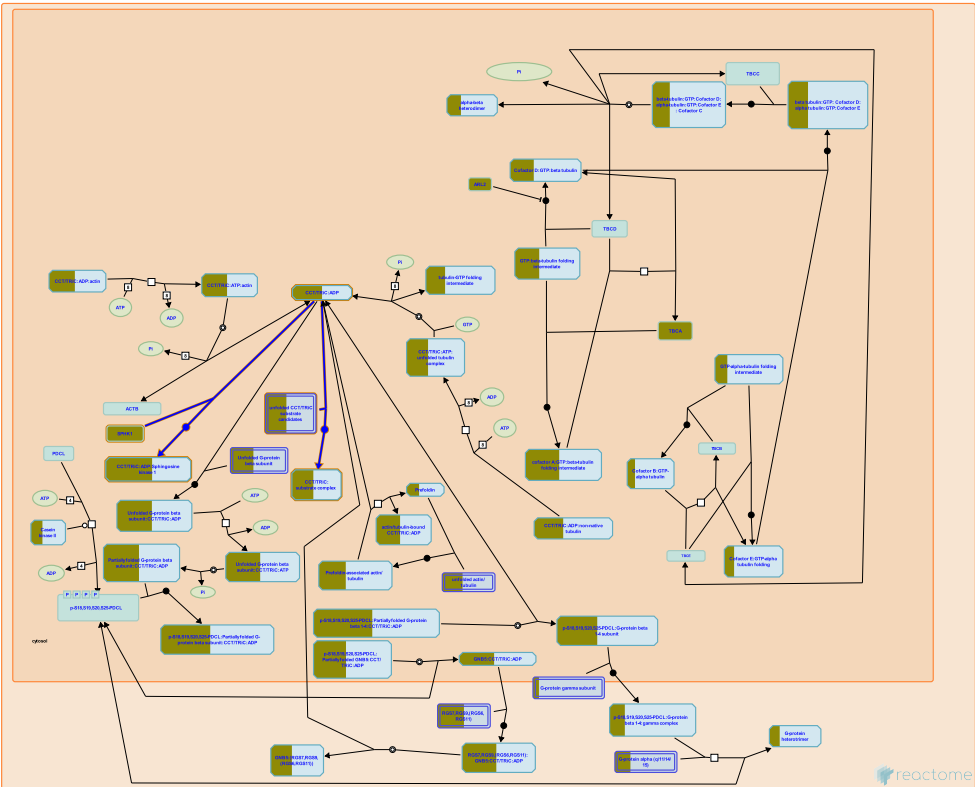
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Date	Action	Author
2015-08-10	Edited	Rothfels K
2015-08-10	Authored	Rothfels K
2015-10-02	Created	Rothfels K
2016-08-05	Reviewed	Stephens RM

5 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HRAS	P01112	KRAS	P01116-1, P01116-2	NF1	P21359
NRAS	P01111	SPRED2	Q7Z698		

14. Association of TriC/CCT with target proteins during biosynthesis (R-HSA-390471)



Cellular compartments: cytosol.

TRiC has broad recognition specificities, but in the cell it interacts with only a defined set of substrates (Yam et al. 2008). Many of its substrates that are targeted during biosynthesis are conserved between mammals and yeast (Yam et al. 2008).

References

Yam AY, Xia Y, Lin HT, Burlingame A, Gerstein M & Frydman J (2008). Defining the TRiC/CCT interactome links chaperonin function to stabilization of newly made proteins with complex topologies. Nat Struct Mol Biol, 15, 1255-62. [🔗](#)

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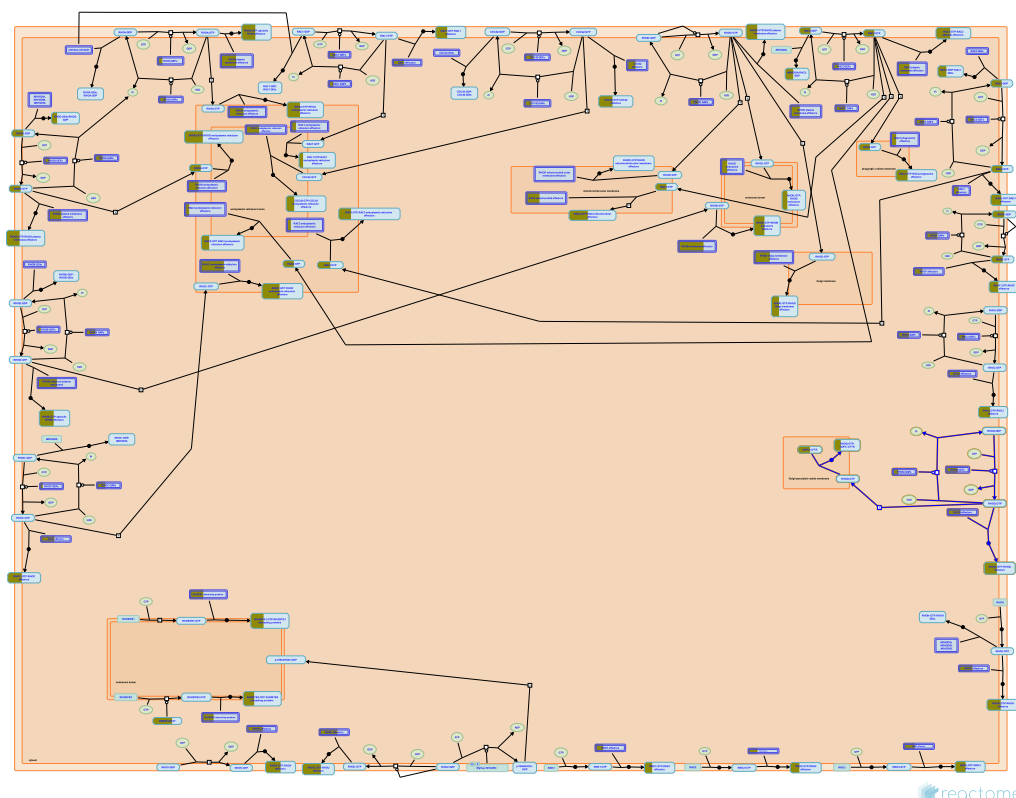
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2009-01-21	Reviewed	Cowan NJ
2009-02-09	Created	Matthews L
2009-02-21	Edited	Matthews L
2025-11-15	Modified	Weiser JD

17 submitted entities found in this pathway, mapping to 18 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AP3M1	Q9Y2T2	CCNE2	O96020	CCT4	P50991

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCT6A	P40227, Q92526	CCT8	P50990	FBXL3	Q9UKT7
FBXL5	Q9UKA1	FBXW5	Q969U6	GBA1	P04062
KIFC3	Q9BVG8-2	SIK1	O00567	SPHK1	Q9NYA1
TCP1	P17987	TP53	P04637	TST	Q15477
USP11	P51784	WRAP53	Q9BUR4		

15. RHOQ GTPase cycle (R-HSA-9013406)



This pathway catalogues RHOQ (also known as TC10) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOQ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOQ. Two GDIs, ARHGDIA (also known as Rho-GDI alpha) (Michaelson et al. 2001; Murphy et al. 2001) and ARHGDIB (also known as D4-GDI) (Zhang et al. 2009), were specifically shown not to interact with RHOQ. RHOQ, together with RHOJ (TCL), belongs to the CDC42 subfamily of RHO GTPases, with the three family members sharing 65-85% of homology (Murphy et al. 1999, Vignal et al. 2000, Donnelly et al. 2014; Shi et al. 2016). RHOQ is highly activated on exocytosing vesicles and recycling endosomes. RHOQ GTPase activity is necessary for fusion of vesicles with the plasma membrane (Kawase et al. 2006, Donnelly et al. 2014). RHOQ is required for insulin-stimulated glucose uptake (Kanzaki and Pessin 2003; Saltiel and Pessin 2003) and is involved in insulin signaling in adipocytes (Satoh 2014). While some studies reported RHOQ involvement in neurite outgrowth (Abe et al. 2003, Gonzalez-Billault et al. 2012) and regulation of membrane addition during axon formation (Dupraz et al. 2009, Gonzalez-Billault et al. 2012), other studies failed to identify RHOQ as a regulator of neurite outgrowth (Murphy et al. 1999, Murphy et al. 2001).

References

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Shi TT, Li G & Xiao HT (2016). The Role of RhoJ in Endothelial Cell Biology and Tumor Pathology. Biomed Res Int, 2016, 6386412. [🔗](#)

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Satoh T (2014). Rho GTPases in insulin-stimulated glucose uptake. Small GTPases, 5, e28102. [🔗](#)

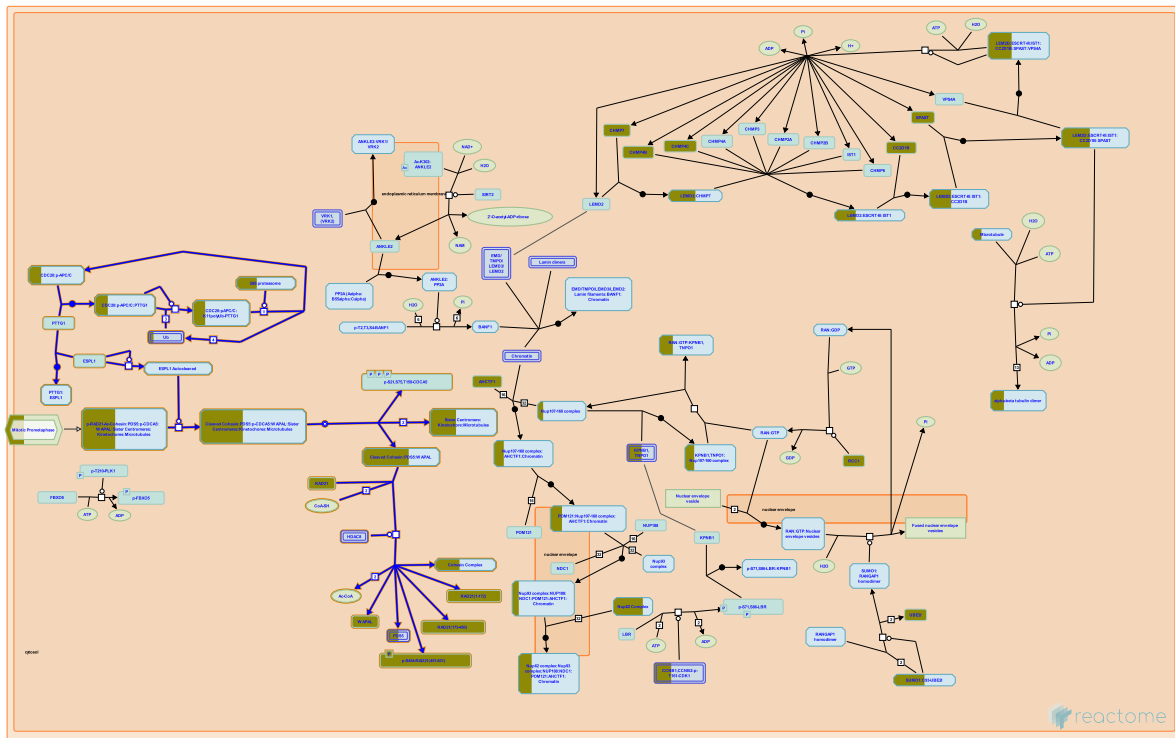
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Date	Action	Author
2017-07-25	Created	Orlic-Milacic M
2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

23 submitted entities found in this pathway, mapping to 24 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP21	Q5T5U3	ARHGAP33	A7KAX9, O14559	CDC42BPA	Q5VT25
CDC42EP2	O14613	DIAPH3	Q9NSV4	GIT1	Q9Y2X7
GJA1	P17302	GOPC	Q9HD26	IQGAP1	P46940
IQGAP3	Q86VI3	ITSN1	Q15811	JUP	P14923
OBSL1	Q5VST9	PAK3	Q13153	PLEKHG3	A1L390
RAB7A	P51149	SCRIB	Q14160	SNAP23	O00161
STEAP3	Q658P3	STOM	P27105	SYDE1	Q6ZW31
VANGL1	Q8TAA9	WWP2	O00308		

16. Separation of Sister Chromatids (R-HSA-2467813)



Cellular compartments: cytosol.

While sister chromatids resolve in prometaphase, separating along chromosomal arms, the cohesion of sister centromeres persists until anaphase. At the anaphase onset, the anaphase promoting complex/cyclosome (APC/C) ubiquitinates PTTG1 (securin), targeting it for degradation (Hagting et al. 2002). PTTG1 acts as an inhibitor of ESPL1 (known as separin i.e. separase). Hence, PTTG1 removal initiated by APC/C, enables ESPL1 to become catalytically active (Zou et al. 1999, Waizenegger et al. 2002). ESPL1 undergoes autolysis (Waizenegger et al. 2002) and also cleaves RAD21 subunit of centromeric cohesin (Hauf et al. 2001). RAD21 cleavage promotes dissociation of cohesin complexes from sister centromeres, leading to separation of sister chromatids. Subsequent movement of sister chromatids to opposite poles of the mitotic spindle segregates replicated chromosomes to two daughter cells (Waizenegger et al. 2000, Hauf et al. 2001, Waizenegger et al. 2002).

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Zou H, McGarry TJ, Bernal T & Kirschner MW (1999). Identification of a vertebrate sister-chromatid separation inhibitor involved in transformation and tumorigenesis. *Science*, 285, 418-22. [🔗](#)

Edit history

Date	Action	Author
2012-09-13	Created	Orlic-Milacic M
2012-10-02	Authored	Orlic-Milacic M
2012-10-05	Edited	Matthews L, Gillespie ME
2012-10-22	Reviewed	Zhang N
2012-11-20	Reviewed	Tanno Y, Watanabe Y

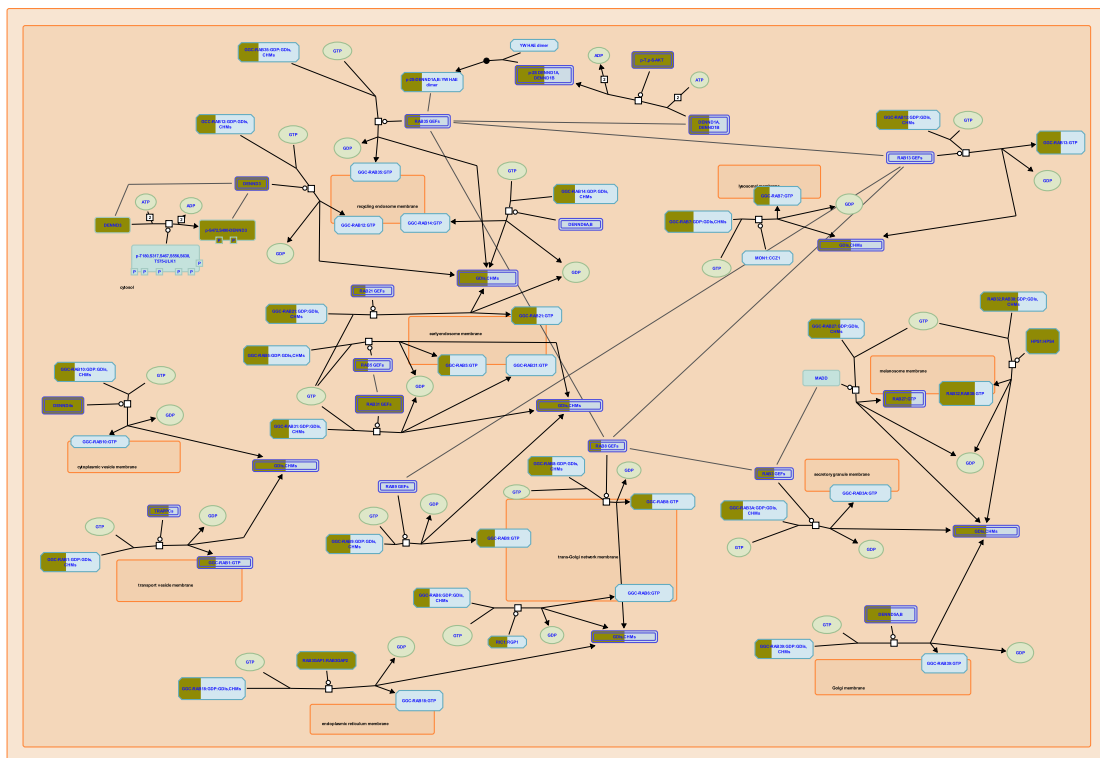
65 submitted entities found in this pathway, mapping to 70 Reactome entities

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BUB1B	O60566	CDC16	Q13042	CDC26	Q8NHZ8
CDC27	P30260	CENPC	Q03188	CENPE	Q02224
CENPU	Q71F23	CKAP5	Q14008	CLASP1	Q7Z460
CLASP2	O75122	CLIP1	P30622	DYNC1H1	Q14204
DYNC1I1	O14576	DYNC1I2	Q13409	DYNC1LI2	O43237
DYNLL1	P63167	EMB	O14980	ITGB3BP	Q13352
KIF18A	Q8NI77	KIF2A	O00139, Q8N4N8	KNL1	Q8NG31
KNTC1	P50748	MAD1L1	Q9Y6D9	MAPRE1	Q15691
NDC80	O14777	NUF2	Q9BZD4	NUP133	Q8WUM0
PAFAH1B1	P43034	PDS5A	Q29RF7	PPP2R1B	P30154
PPP2R5C	Q13362	PPP2R5E	Q16537	PSMA1	P25786
PSMA3	P25788	PSMB4	P28070	PSMC1	P62191
PSMC2	P35998	PSMC6	P62333	PSMD1	Q99460
PSMD12	O00232	PSMD14	O00487	PSMD6	Q15008
RAD21	O60216	RANBP2	P49792	RCC2	Q9P258
RPN1	Q13200	RPS27	P42677	SGO2	Q562F6
SKA2	Q8WVK7	SPC25	Q9HBM1	SPDL1	Q96EA4
STAG2	Q8N3U4	TAOK1	Q7L7X3	TUBA1A	Q71U36
TUBA4A	P68366	TUBB2A	Q13885, Q9BVA1	TUBB2B	Q13885, Q9BVA1
TUBB4A	P04350	UBA52	P62979, P62987	WAPL	Q7Z5K2
XPO1	O14980	ZW10	O43264		

Interactors found in this pathway (10)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BMP1	P13497	O60216	DEC1	Q16698	O95997
PDS5A	Q29RF7	O60216, Q7Z5K2, Q96FF9	PSMC2	P35998	O60216
RAD21	O60216	Q7Z5K2, Q96FF9	SSU72	Q9NP77	O60216
STAG2	Q8N3U4, Q8N3U4-2	O60216, Q7Z5K2, Q96FF9	USP13	Q92995	O60216
USP47	Q96K76	O60216	WAPL	Q7Z5K2	O60216, Q7Z5K2

17. RAB GEFs exchange GTP for GDP on RABs ([R-HSA-8876198](#))

 reactome

Human cells have more than 60 RAB proteins that are key regulators of intracellular membrane trafficking. These small GTPases contribute to trafficking specificity by localizing to the membranes of different organelles and interacting with effectors such as sorting adaptors, tethering factors, kinases, phosphatases and tubular-vesicular cargo (reviewed in Stenmark et al, 2009; Wandinger-Ness and Zerial, 2014; Zhen and Stenmark, 2015).

RAB localization depends on a number of factors including C-terminal prenylation, the sequence of upstream hypervariable regions and what nucleotide is bound, as well as interaction with RAB-interacting proteins (Chavrier et al, 1991; Ullrich et al, 1993; Soldati et al, 1994; Farnsworth et al, 1994; Seabra, 1996; Wu et al, 2010; reviewed in Stenmark, 2009; Wandinger-Ness and Zerial, 2014). More recently, the activity of RAB GEFs has also been implicated in regulating the localization of RAB proteins (Blumer et al, 2003; Schoebel et al, 2009; Cabrera and Ungermann, 2013; reviewed in Barr, 2013; Zhen and Stenmark, 2015)

In the active, GTP-bound form, RAB proteins are membrane-associated, while in the inactive GDP-bound form, RABs are extracted from the target membrane and exist in a soluble form in complex with GDP dissociation inhibitors (GDIs) (Ullrich et al, 1993; Soldati et al, 1994; Gavriljuk et al, 2013). Conversion between the inactive and active form relies on the activities of RAB guanine nucleotide exchange factors (GEFs) and GTPase activating proteins (GAPs) (Yoshimura et al, 2010; Wu et al, 2011; Pan et al, 2006; Frasa et al, 2012; reviewed in Stenmark, 2009; Wandinger-Ness and Zerial, 2014; Ishida et al, 2016).

Newly synthesized RABs are bound to a RAB escort protein, CHM (also known as REP1) or CHML (REP2) (Alexandrov et al, 1994; Shen and Seabra, 1996). CHM/REP proteins are the substrate-binding component of the trimeric RAB geranylgeranyltransferase enzyme (GGTaseII) along with the two catalytic subunits RABGGTA and RABGGTB (reviewed in Gutkowska and Swiezewska, 2012; Pal-suledesai and Distefano, 2015). REP proteins recruit the unmodified RAB in its GDP-bound state to the GGTase for sequential geranylgeranylation at one or two C-terminal cysteine residues (Alexandrov et al, 1994; Seabra et al 1996; Shen and Seabra, 1996; Baron and Seabra, 2008). After geranylation, CHM/REP proteins remain in complex with the geranylated RAB and escort it to its target membrane, where RAB activity is regulated by GAPs, GEFs, GDIs and membrane-bound GDI displacement factors (GDFs) (Sivars et al, 2003; reviewed in Stenmark, 2009; Wandinger-Ness and Zerial, 2014).

Unlike the RAB GAPs, which (to date) all contain a shared TBC domain, RAB GEFs are structurally diverse and range from monomeric to multisubunit complexes (reviewed in Fukuda et al, 2011; Frasa et al, 2012; Cherfils and Zeghouf, 2013; Ishida et al, 2016). While many GEFs contain one of three conserved GEF domains identified to date - the DENN (differentially expressed in normal and neoplastic cell) domain, the VPS9 domain and the SEC2 domain- other GEFs lack a conserved domain (reviewed in Ishida et al, 2016). Based on sequence conservation and subunit organization, GEFs can be grouped into 6 general classes: the DENND-containing GEFs, the VPS9-containing GEFs (both monomeric), the SEC2-containing GEFs (homodimeric), heterodimeric GEF complexes such as RIC1:RGPI, the multisubunit TRAPPC GEF, and others (reviewed in Barr and Lambricht, 2010; Marat et al, 2011; Ishida et al, 2016). GEFs for many RABs have still not been identified, however.

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Edit history

Date	Action	Author
2016-06-10	Created	Rothfels K
2016-06-29	Edited	Rothfels K
2016-06-29	Authored	Rothfels K
2016-08-03	Reviewed	Marat AL
2025-11-15	Modified	Weiser JD

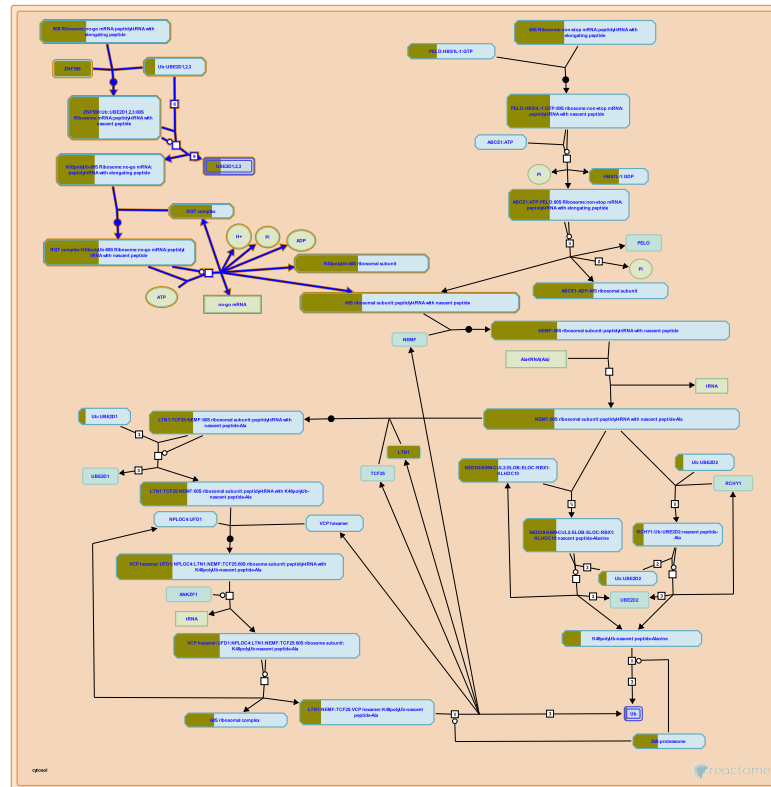
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AKT1	P31749	AKT2	P31751	AKT3	Q9Y243
CHM	P24386, P26374	DENND1B	Q6P3S1	DENND3	A2RUS2
DENND4A	Q7Z401	DENND4B	O75064	DENND4C	Q5VZ89
DENND5A	Q6IQ26	GAPVD1	Q14C86	HPS1	Q92902
HPS4	Q9NQG7	RAB13	P51153	RAB1B	Q9H0U4
RAB21	Q9UL25	RAB27A	P51159	RAB27B	O00194
RAB32	Q13637	RAB38	P57729	RAB3GAP1	Q15042
RAB3GAP2	Q9H2M9	RAB3IL1	Q8TBN0	RAB5C	P51148
RAB7A	P51149	RAB8B	Q92930	RAB9A	P51151
RABGEF1	Q9UJ41	RIC1	Q4ADV7	RIN3	Q8TB24
SBF2	Q86WG5	TRAPPC11	Q7Z392	TRAPPC13	A5PLN9
TRAPPC2L	Q9UL33	TRAPPC3	O43617	TRAPPC5	Q8IUR0
TRAPPC6B	O75865, Q86SZ2	TRAPPC8	Q9Y2L5	TRAPPC9	Q96Q05

Interactors found in this pathway (8)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
AMBRA1	Q9C0C7, Q9C0C7-3	O75385	ATG101	Q9BSB4	O75385
GABARAP	O95166	O75385	PRKAA1	Q13131	O75385
PRKAA2	P54646	O75385	PRKAB1	Q9Y478	O75385
RPTOR	Q8N122	O75385	WASHC3	Q9Y3C0	O75385

18. ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA (R-HSA-9954716)



Due to features such as damaged nucleotides, strong secondary structure, presence of stretches of more than four uninterrupted AAA codons in a mRNA coding region (designated no-go mRNAs) a ribosome translating an mRNA can stall and cause collisions with the ribosomes trailing it on the mRNA (reviewed in Joazeiro 2017, Eisenack and Trentini 2022, Filbeck et al. 2022, McGirr et al. 2025). In cases in which the ribosome stalls internally on the mRNA and a 3' region of the mRNA protrudes from the ribosome, ZNF598, a ubiquitin E3 ligase that is the homolog of yeast HEL2, binds the ribosome (Garzia et al. 2017, Juszkievicz and Hegde 2017, Sundaramoorthy et al. 2017) and ubiquitinates the 40S subunit ribosomal proteins eS10 (RPS10) at residues K138 and K139 and uS10 (RPS20) at residues K4 and K8 to initiate splitting of the 40S and 60S ribosomal subunits (Juszkievicz and Hegde 2017, Sundaramoorthy et al. 2017, Garzia et al. 2017, Juszkievicz et al. 2018, Narita et al. 2022, Miścicka et al. 2024, also inferred from yeast homologs in Matsuo et al. 2017, Sitron et al. 2017, Ikeuchi et al. 2019, reviewed in Ford et al. 2024). ZAK also interacts with the collided ribosome at the 18S helix through its c-terminal domain, which triggers ZAK autophosphorylation and phosphorylation of p38/JNK in a MAPK cascade known as the ribotoxic stress response (Wu et al. 2020, Vind et al. 2020, Stoneley et al. 2022).

The ASCC2 subunit (Narita et al. 2022) of the ribosome quality control trigger complex (RQT complex, ASCC2:ASCC3:TRIP4) binds K63-linked polyubiquitin conjugated to the 40S protein uS10 (Hashimoto et al. 2020, Juskiewicz et al. 2020, Narita et al. 2022, and inferred from yeast homologs in Matsuo et al. 2023). The ASCC3 subunit of the RQT complex splits stalled 80S ribosomes with K63-polyubiquitinated uS10 into 60S and 40S subunits (Hashimoto et al. 2020, Juskiewicz et al. 2020, Narita et al. 2022, Miścicka et al. 2024) apparently by exerting a pulling force on the mRNA (inferred from the yeast homolog Slh1 in Best et al. 2023). The peptidyl-tRNA remains bound in the 60S subunit, with the tRNA positioned in the P site. The problematic mRNA dissociates after splitting and is thought to be degraded at this time. In the case of collided yeast ribosomes, the mRNA is first endonucleolytically cleaved and the cleavage products are exonucleolytically degraded by XRN1 and the exosome (Ikeuchi et al. 2019).

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Edit history

Date	Action	Author
2025-06-06	Edited	May B
2025-06-06	Authored	May B
2025-06-08	Created	May B
2025-08-09	Reviewed	Bennett EJ, Ford PW
2025-11-07	Reviewed	McGirr T, Jafarnejad SM

35 submitted entities found in this pathway, mapping to 42 Reactome entities

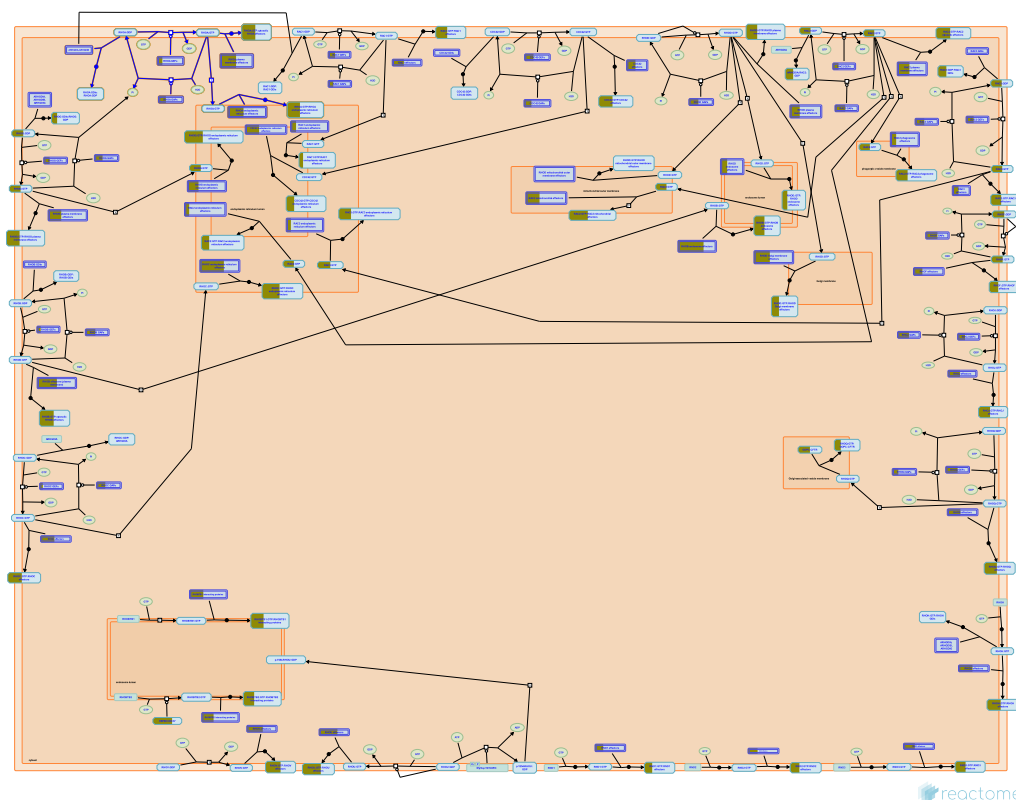
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ASCC3	Q8N3C0	CRY2	P62263	NAB1	P08865
RPL13	P26373, P40429	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL27	P61353	RPL32	P62888, P62910
RPL35A	P18077, P42766	RPL37A	P18077, P61513	RPL38	P63173
RPL39L	Q96EH5	RPLP1	P05386	RPLP2	P05387
RPS10	P46783	RPS11	P62280	RPS14	P62263
RPS15	P62841	RPS16	P62249	RPS17	P08708
RPS19	P39019	RPS20	P60866	RPS21	P62841, P63220
RPS27	P42677, Q71UM5	RPS28	P62857	RPS29	P62273

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPS3	P23396	RPS5	P46782	RRP1	P05386
SOS1	P42766	TRIP4	Q15650	UBA52	P62979, P62987
UBE2D3	P61077	ZNF598	Q86UK7		

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GIGYF1	O75420	Q86UK7			

19. RHOA GTPase cycle (R-HSA-8980692)



This pathway catalogues RHOA guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and RHOA effectors. RHOA is one of the three best characterized RHO GTPases, the other two being RAC1 and CDC42, and is the founding member of the RHO GTPase family (Zhou and Zheng 2013). RHOA regulates the cytoskeleton and cell contractility (Lessey et al. 2012), thus playing a role in a number of cellular functions, such as adhesion, migration, survival, division, vesicle trafficking and gene expression (Zhou and Zheng 2013). RHOA-regulated processes are involved in mechanotransduction (Lessey et al. 2012; Marjoram et al. 2014), neuronal development (Zhou and Zheng 2013; Fujita and Yamashita 2014; Hu and Selzer 2017), immune system development (Zhou and Zheng 2013; Ricker et al. 2016; Bros et al. 2019), and cardiovascular regulation (Zhou and Zheng 2013; Cai et al. 2016; Shimokawa et al. 2016). RHOA mutations are frequently found in cancer (Kataoka and Ogawa 2016). Toxins of numerous pathogens target RHOA to hijack the host cytoskeleton (Jamilloux et al. 2018).

References

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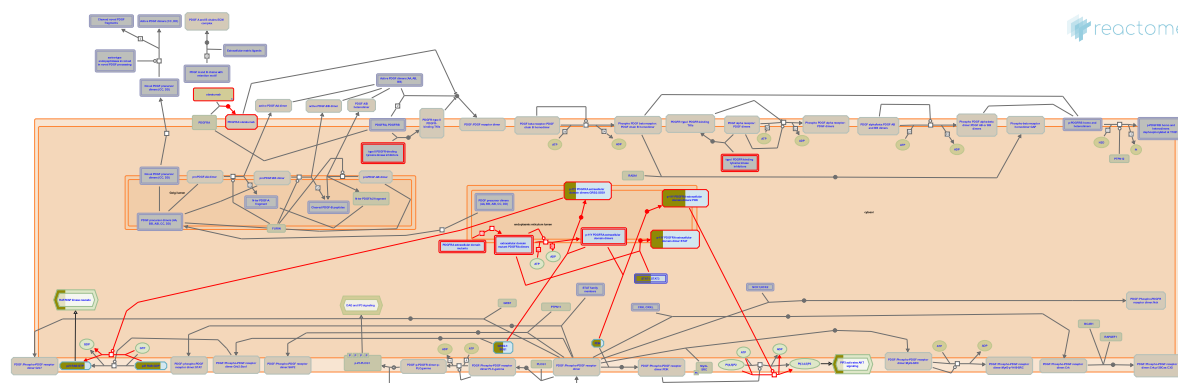
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2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M

49 submitted entities found in this pathway, mapping to 50 Reactome entities

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ABCD3	P28288	ABR	Q12979	ACBD5	Q5T8D3
ANLN	Q9NQW6	ARAP2	Q8WZ64	ARHGAP11A	Q6P4F7
ARHGAP18	Q8N392	ARHGAP21	Q5T5U3	ARHGAP31	Q2M1Z3
ARHGAP33	A7KAX9	ARHGAP42	A6NI28	ARHGAP45	Q92619
ARHGEF11	O15085	ARHGEF12	Q9NZN5	ARHGEF3	Q9NR81
ARHGEF4	Q9NR80	ATP6AP1	Q15904	CAVIN1	Q6NZI2
CIT	O14578, O14578-3	DIAPH3	Q9NSV4	ERBIN	Q96RT1
IQGAP1	P46940	IQGAP3	Q86VI3	JUP	P14923
KTN1	Q86UP2	LMAN1	P49257	MACO1	Q8N5G2
MYO9A	B2RTY4	OBSL1	Q5VST9	PCDH7	O60245
PIK3R1	P27986	PIK3R2	O00459	PKN1	Q16512
PKN2	Q16513	PLD1	Q13393	PLEKHG3	A1L390
RHPN2	Q8IUC4	ROCK1	Q13464	ROCK2	O75116
SCFD1	Q8WVM8	SLK	Q9H2G2	SNAP23	O00161
STARD13	Q9Y3M8	STARD8	Q92502	STOM	P27105
STX5	Q13190	TMEM87A	Q8NBN3	TRIO	O75962
VANGL1	Q8TAA9				

20. Signaling by PDGFRA extracellular domain mutants ([R-HSA-9673770](#))



Diseases: cancer.

Mutations in the extracellular domain of receptor tyrosine kinases like PDGFRA have the potential to interfere with glycosylation, which is required for proper trafficking to the cell surface. Versions of PDGFRA bearing mutations in the extracellular domain have been identified in some cancers, and while some of these are inactive, some have also been shown to signal constitutively from the endoplasmic reticulum (Ip et al, 2018; Velghe et al, 2014; Ozawa et al, 2010; Paugh et al, 2013; Clarke et al, 2003). PDGFRA Y288C carries a mutation in the extracellular domain of the receptor that prevents the normal trafficking of the protein to the plasma membrane. PDGFRA Y288C is trapped in the endoplasmic reticulum membrane, from where it is able to signal constitutively in the absence of ligand (Ip et al, 2018). Similarly, a version of PDGFRA bearing an in-frame deletion of exons 8 and 9 has been identified in glioblastoma. The resulting protein is constitutively active in the absence of ligand despite the low fraction of protein expressed at the cell surface (Clarke et al, 2003; Ozawa et al, 2010).

References

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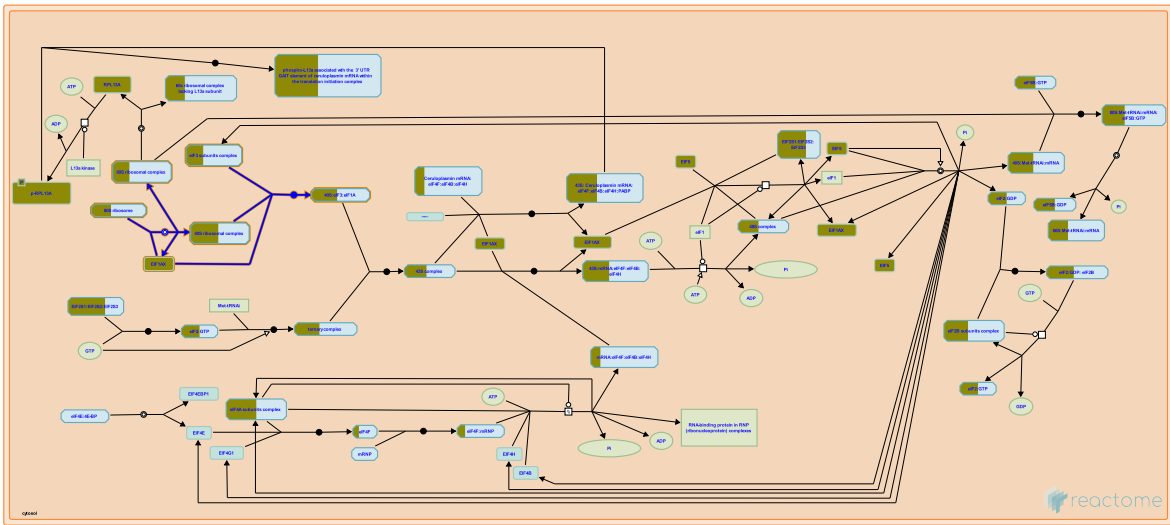
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Date	Action	Author
2020-02-06	Reviewed	Ip CKM
2020-02-25	Edited	Rothfels K
2020-02-25	Authored	Rothfels K
2023-03-08	Modified	Matthews L

8 submitted entities found in this pathway, mapping to 9 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HRAS	P01112	KRAS	P01116-1, P01116-2	NRAS	P01111
PIK3CB	P42338	PIK3R1	P27986	PIK3R2	O00459
SOS1	Q07889	STAT1	P42224		

21. Formation of a pool of free 40S subunits (R-HSA-72689)



Cellular compartments: cytosol.

The 80S ribosome dissociates into free 40S (small) and 60S (large) ribosomal subunits. Each ribosomal subunit is constituted by several individual ribosomal proteins and rRNA.

References

Edit history

Date	Action	Author
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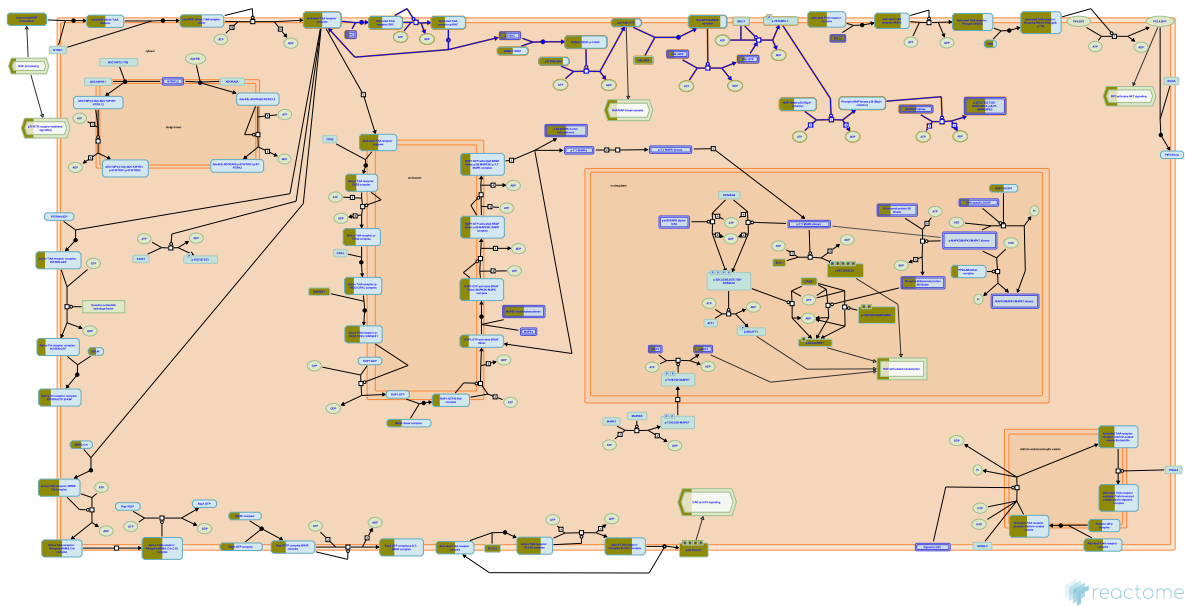
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EIF3E	P60228	EIF3J	O75822	EIF3M	Q7L2H7
NAB1	P08865	RPL13	P26373, P40429	RPL18	Q07020
RPL18A	Q02543	RPL19	P84098	RPL27	P61353
RPL32	P62888, P62910	RPL35A	P18077, P42766	RPL37A	P18077, P61513
RPL38	P63173	RPL39L	Q96EH5	RPLP1	P05386
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS14	P62263	RPS15	P62841	RPS16	P62249
RPS17	P08708	RPS19	P39019	RPS20	P60866
RPS21	P62841, P63220	RPS27	P42677, Q71UM5	RPS28	P62857
RPS29	P62273	RPS3	P23396	RPS5	P46782
RRP1	P05386	SOS1	P42766	UBA52	P62979, P62987

Interactors found in this pathway (5)

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EIF1AY	O14602	P47813	EIF3A	Q14152	P47813
EIF5	P55010	P47813	IPO13	O94829	P47813
NCR3LG1	Q68D85	P47813			

22. Signalling to RAS (R-HSA-167044)



Signalling through Shc adaptor proteins appears to be identical for both NGF and EGF. It leads to a fast, but transient, MAPK/ERK activation, which is insufficient to explain the prolonged activation of MAPK found in NGF-treated cells.

References

Sofroniew MV, Howe CL & Mobley WC (2001). Nerve growth factor signaling, neuroprotection, and neural repair. Annu Rev Neurosci, 24, 1217-81. [🔗](#)

Edit history

Date	Action	Author
2006-02-07	Created	
2006-10-10	Edited	Jassal B
2006-10-10	Authored	Annibali D, Nasi S
2007-11-08	Reviewed	Greene LA

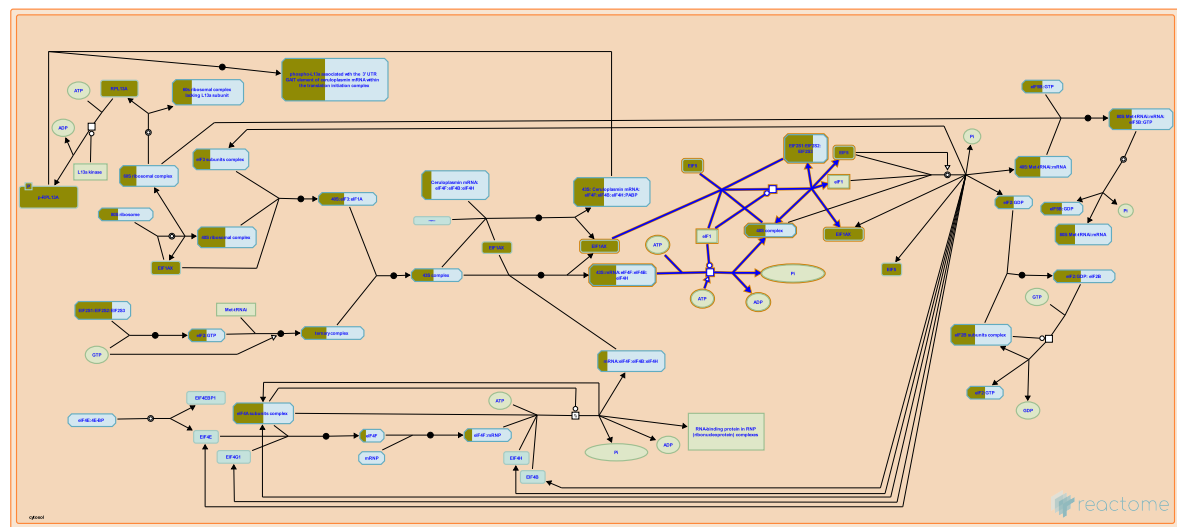
11 submitted entities found in this pathway, mapping to 12 Reactome entities

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MAPK13	O15264	MAPKAPK2	P49137	NGF	P01138
NRAS	P01111	RALA	P11233	RGL4	Q12967
SHC3	Q92529	SOS1	Q07889		

Interactors found in this pathway (2)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BRAF	P15056	Q12967	HRAS	P01112	Q12967

23. Ribosomal scanning and start codon recognition (R-HSA-72702)



The 80S ribosome bound to the mRNA moves along the mRNA molecule from its initial site to the initiation codon and forms a 48S complex, in which the initiation codon is base paired to the anticodon of the Met-tRNA_i. Proper recognition of the AUG initiation codon depends on base pairing with the anticodon of the Met-tRNA_i and requires eIF1, eIF1A, eIF2 and eIF5.

References

Pestova TV, Borukhov SI & Hellen CU (1999). Eukaryotic ribosomes require initiation factors 1 and 1A to locate initiation codons. *Nature*, 394, 854-9. [🔗](#)

Kozak M (1981). Evaluation of the "scanning model" for initiation of protein synthesis in eucaryotes. *Cell*, 22, 7-8. [🔗](#)

Sonenberg N, Hershey JW & Mathews MB (2001). *Genetic approaches to translation initiation in Saccharomyces cerevisiae.*, *Translational Control of Gene Expression*, 487-502.

Trachsel H, Erni B, Schreier MH & Staehelin T (1978). Initiation of mammalian protein synthesis. II. The assembly of the initiation complex with purified initiation factors. *J Mol Biol*, 116, 755-67. [🔗](#)

Benne R & Hershey JW (1978). The mechanism of action of protein synthesis initiation factors from rabbit reticulocytes. *J Biol Chem*, 253, 3078-87. [🔗](#)

Edit history

Date	Action	Author
2002-12-16	Created	Merrick WC
2025-11-15	Modified	Weiser JD

26 submitted entities found in this pathway, mapping to 28 Reactome entities

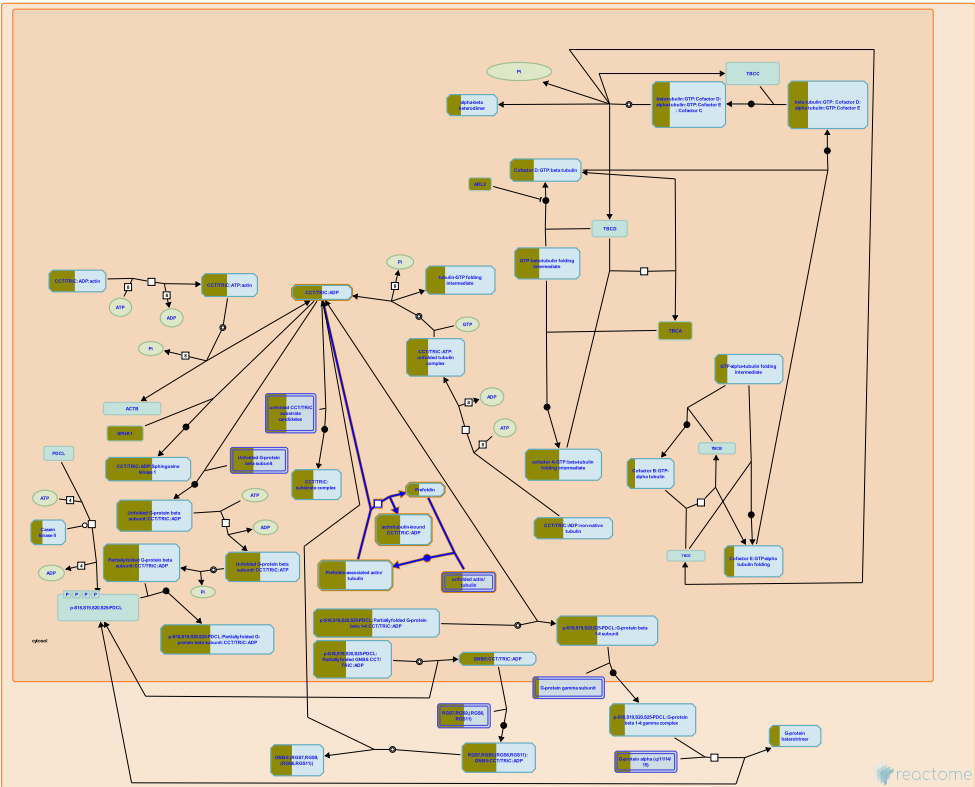
Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CRY2	P62263	EIF1AY	P47813	EIF2A	P05198
EIF2S3	P41091	EIF3A	Q14152	EIF3E	P60228
EIF3J	O75822	EIF3M	Q7L2H7	EIF4A2	Q14240
EIF5	P55010	NAB1	P08865	RPS10	P46783
RPS11	P62280	RPS14	P62263	RPS15	P62841

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPS16	P62249	RPS17	P08708	RPS19	P39019
RPS20	P60866	RPS21	P62841, P63220	RPS27	P42677, Q71UM5
RPS28	P62857	RPS29	P62273	RPS3	P23396
RPS5	P46782	UBA52	P62979		

Interactors found in this pathway (5)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EIF1AY	O14602	P47813	EIF3A	Q14152	P47813
EIF5	P55010	P47813	IPO13	O94829	P47813
NCR3LG1	Q68D85	P47813			

24. Prefoldin mediated transfer of substrate to CCT/TriC (R-HSA-389957)



Cellular compartments: cytosol.

Unfolded actins and tubulins bound to prefoldin are transferred to CCT via a docking mechanism (McCormack and Willison, 2001).

References

McCormack EA, Llorca O, Carrascosa JL, Valpuesta JM & Willison KR (2001). Point mutations in a hinge linking the small and large domains of beta-actin result in trapped folding intermediates bound to cytosolic chaperonin CCT. J Struct Biol, 135, 198-204.

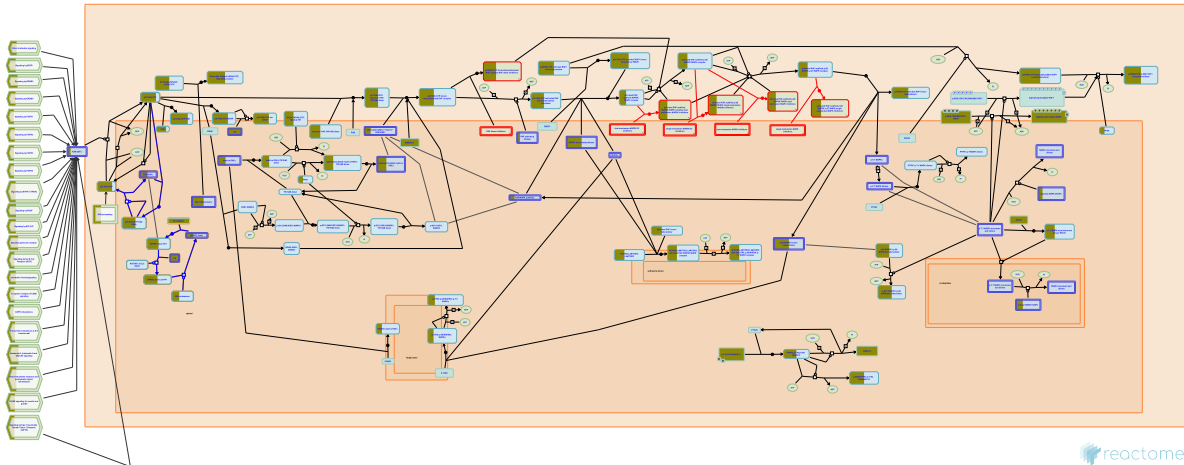
Edit history

Date	Action	Author
2008-12-01	Authored	Matthews L
2009-01-21	Reviewed	Cowan NJ
2009-01-22	Created	Matthews L
2009-02-21	Edited	Matthews L
2023-10-12	Modified	Weiser JD

11 submitted entities found in this pathway, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCT4	P50991	CCT6A	P40227, Q92526	CCT8	P50990
PFDN2	Q9UHV9	TCP1	P17987	TUBA1A	Q71U36
TUBA4A	P68366	TUBB2A	Q13885, Q9BVA1	TUBB2B	Q13885, Q9BVA1
TUBB4A	P04350	VBP1	P61758		

25. Regulation of RAS by GAPs (R-HSA-5658442)



The intrinsic GTPase activity of RAS proteins is stimulated by the GAP proteins, of which there are at least 10 in the human genome (reviewed in King et al, 2013).

References

King PD, Lubeck BA & Lapinski PE (2013). Nonredundant functions for Ras GTPase-activating proteins in tissue homeostasis. Sci Signal, 6, re1. [🔗](#)

Edit history

Date	Action	Author
2014-12-18	Authored	Rothfels K
2014-12-18	Created	Rothfels K
2015-02-12	Edited	Rothfels K
2015-04-29	Reviewed	Roskoski R Jr
2025-11-15	Modified	Weiser JD

20 submitted entities found in this pathway, mapping to 23 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HRAS	P01112	KRAS	P01116-1, P01116-2	NF1	P21359
NRAS	P01111	PSMA1	P25786	PSMA3	P25788
PSMB4	P28070	PSMC1	P62191	PSMC2	P35998
PSMC6	P62333	PSMD1	Q99460	PSMD12	O00232
PSMD14	O00487	PSMD6	Q15008	RASA1	P20936
RASA2	Q15283	RPN1	Q13200	SPRED2	Q7Z698
SYNGAP1	Q96PV0, Q9UJF2	UBA52	P62979, P62987		

Interactors found in this pathway (2)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HRAS	P01112	P21359	SCN3B	Q9NY72	P21359

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

2600 of the submitted entities were found, mapping to 3097 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ABCA1	O95477	ABCA3	Q99758	ABCA5	Q8IUA7, Q8WWZ7
ABCB5	O75027	ABCB9	Q9NP78	ABCC3	O15438
ABCC9	O60706	ABCD3	P28288	ABCF1	Q8NE71
ABCF2	Q9UG63	ABHD12	Q8N2K0	ABHD3	Q8WU67
ABHD6	Q9BV23	ABI1	Q8IZP0	ABI2	Q9NYB9
ABLM1	O14639	ABLM3	O94929	ABR	Q12979
ABRAXAS1	Q6UWZ7	ACAA2	P42765	ACAD11	Q709F0
ACADM	P11310	ACADSB	P45954	ACBD5	Q5T8D3
ACBD6	Q9BR61	ACCS	Q93063	ACER3	Q9NUN7
ACOT2	P49753	ACOX2	Q99424	ACP5	P13686
ACSL3	O95573	ACSL4	O60488	ACTN2	P35609
ACTR10	Q9NZ32	ACTR1B	P42025	ACTR2	P61160
ACTR3	P61158	ACTR5	Q9H9F9	ACY1	Q03154
ADAM10	O14672	ADAM9	Q99965	ADAMTS1	Q9UHI8
ADAMTS14	Q8WXS8	ADAMTSL1	Q8N6G6	ADAMTSL3	P82987
ADARB1	P78563	ADAT3	Q96EY9	ADCY9	O60503
ADD3	Q9UEY8	ADGRG3	Q86Y34	ADGRG6	Q86SQ4
ADI1	Q9BV57	ADIPOR1	Q96A54	ADIPOR2	Q86V24
ADK	P55263	ADM2	Q7Z4H4	ADO	Q96SZ5
ADORA2B	P29275	ADSS2	P30520	AFDN	P55196-2
AFF4	Q9UHB7	AFG3L2	Q9Y4W6	AGFG1	P52594
AGT	P01019	AGTPBP1	Q9UPW5	AHCTF1	Q8WYP5
AHCYL1	Q96HN2	AIMP2	Q13155	AK3	P27144
AK4	P27144	AKAP1	Q92667	AKAP9	Q99996
AKR1A1	P14550	AKR1B10	Q96JD6	AKT1	P31749
AKT1S1	Q96B36	AKT2	P31751	AKT3	Q9Y243
ALDH1L1	O75891	ALDH3B1	P43353	ALDH7A1	P49419
ALG1	Q9BT22	ALG11	Q2TAA5	ALG12	Q9BV10
ALG13	Q9NP73	ALG2	Q9H553	ALG9	Q9H6U8
ALKBH8	Q96BT7	ALMS1	Q8TCU4	ALOX12	P18054
ALOXE3	Q9BYJ1	AMBRA1	Q9C0C7	AMDHD2	Q9Y303
AMPD2	Q01433	AMT	P48728	ANAPC2	Q9UJX6
ANK2	Q8WXI3	ANK3	Q12955	ANKRD1	Q15327
ANKRD28	O15084	ANKRD44	Q9H672	ANKRD9	Q96BM1
ANLN	Q9NQW6	ANOS1	P23352	ANXA3	P12429
ANXA5	P08758	ANXA6	P08133	AOC2	O75106
AOX1	Q06278	AP1G1	O43747	AP1S2	P56377, Q96PC3
AP2B1	P63010	AP3B1	O00203	AP3M1	Q9Y2T2
AP4B1	Q9Y6B7	AP4E1	Q9UPM8	APAF1	O14727
APC	P25054	APEH	P13798	APEX1	P27695

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
APOOL	Q6UXV4	APPL1	Q9UKG1	AQP11	Q8NBQ7
AQR	O60306	ARAP2	Q8WZ64	ARC	Q7LC44
ARF5	P84077, P84085	ARFGAP1	Q9NP61	ARFGAP2	Q8N6H7
ARFGAP3	Q9NP61	ARFIP2	P53365	ARG2	P78540
ARHGAP11A	Q6P4F7	ARHGAP12	Q8IWW6	ARHGAP18	Q8N392
ARHGAP21	Q5T5U3	ARHGAP31	Q2M1Z3	ARHGAP33	A7KAX9
ARHGAP42	A6NI28	ARHGAP45	Q92619	ARHGEF11	O15085
ARHGEF12	Q9NZN5	ARHGEF26	Q96DR7	ARHGEF3	Q9NR81
ARHGEF39	Q8N4T4	ARHGEF4	Q9NR80	ARHGEF6	Q15052
ARID2	Q68CP9	ARID4A	P29374	ARID4B	Q4LE39
ARID5B	Q14865	ARIH1	Q9Y4X5	ARL2	P36404
ARL3	P36405	ARL4C	P56559	ARL6IP5	O75915
ARMC8	Q8IUR7	ARPP19	P56211	ARRB1	P49407
ARSB	P15848	ARSD	P51689	ARSI	Q5FYB1
AS3MT	Q9HBK9	ASAH1	Q13510	ASAP1	Q9ULH1
ASB13	Q8W XK3	ASB3	A6NK59, Q9Y575	ASCC3	Q8N3C0
ASF1A	Q9Y294	ASH1L	Q9NR48	ASIC2	Q16515
ASIC3	Q9UHC3	ASL	P04424	ASNS	P08243
ASPH	Q12797	ASPCR1	Q9BZE9	ATF6	P18850
ATF7IP	Q6VMQ6	ATG101	Q9BSB4	ATG16L2	Q676U5, Q8NAA4
ATG4A	Q8WYN0, Q9Y4P1	ATG9A	Q7Z3C6	ATM	Q13315
ATN1	P54259	ATP11B	Q9Y2G3	ATP11C	Q8NB49
ATP13A1	Q9HD20	ATP13A4	Q4VNC0, Q4VNC1	ATP1A1	P05023
ATP1B1	P05026	ATP1B3	P54709	ATP2B4	P23634
ATP2C1	P98194	ATP5F1D	P30049	ATP6AP1	Q15904
ATP6V0A1	Q93050	ATP6V0A2	Q9Y487	ATP6V0B	Q99437
ATP6V0C	P27449	ATP6V0D1	P61421	ATP6V0E1	O15342
ATP6V1A	P38606	ATP6V1B2	P21281	ATP6V1C1	P21283
ATP6V1C2	Q8NEY4	ATP6V1E1	P36543	ATP6V1F	Q16864
ATP6V1G1	O75348, Q96LB4	ATP6V1H	Q9UI12	ATP7A	Q04656
ATP9A	O75110	ATRNL1	Q9Y2I2	ATRX	P46100
ATXN7	O15265	ATXN7L3	Q14CW9	AUP1	Q9Y679
AXIN1	O15169	AZIN1	O14977	B3GALT1	Q9Y5Z6
B3GALT6	Q96L58	B3GLCT	Q6Y288	B3GNT9	Q6UX72
B4GALNT1	Q00973	B4GALT1	P15291	B4GALT2	O60909
B4GALT3	O60512	B4GALT5	O43286	B4GAT1	O43505
BABAM1	Q9NWW8	BACH1	Q9BX63	BAD	Q92934
BAG6	P46379	BAIAP2	Q9UQB8	BAMBI	Q13145
BARD1	Q99728	BASP1	P80723	BAX	Q07812
BBS1	Q8NFJ9	BBS10	Q8TAM1	BBS4	Q96RK4
BCAN	Q96GW7	BCAS2	O75934	BCDIN3D	Q7Z5W3
BCKDHA	P12694	BCKDHB	P21953	BCL10	O95999
BCL2L1	Q07817	BCL2L11	O43521	BCL7A	Q4VC05
BCL7B	Q9BQE9	BCL9L	Q86UU0	BDH1	Q02338
BDH2	Q9BUT1	BDP1	A6H8Y1	BEST1	O76090
BHLHE41	Q9C0J9	BICD1	Q96G01	BIRC2	Q13489, Q13490
BIRC3	Q13489	BIRC6	Q9NR09	BLOC1S3	Q6QNY0
BLOC1S4	Q9NUP1	BLOC1S6	Q9UL45	BLTP3B	A0JNW5
BLZF1	Q9H2G9	BMP1	P13497	BMP4	P12644

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BMPRI1A	P36894	BMPR2	Q13873	BMS1	Q14692
BNIP2	Q12982	BNIP3L	O60238	BOD1	Q8NFC6, Q96IK1
BOD1L1	Q8NFC6	BOLA2B	Q9H3K6	BRAF	P15056
BRCC3	P46736	BRD2	P25440	BRI3	O95415
BRIP1	Q9P0J6	BRPF1	P55201	BRPF3	Q9ULD4
BRWD1	Q9NSI6	BSG	P35613	BTBD6	Q96KE9
BTC	P35070	BUB1	O43683, O60566	BUB1B	O60566
BYSL	Q13895	C1S	P09871	C2CD5	Q86YS7
C2orf49	Q9BVC5	CA9	Q16790	CAB39	Q9Y376
CABLES1	Q8TDN4	CACNA2D1	Q9UIU0	CACTIN	Q8WUQ7
CAD	O76075	CALD1	Q05682	CALM2	P0DP24
CAMKK1	Q8N5S9	CAND1	Q86VP6	CANT1	Q8WVQ1
CANX	P27824	CAP2	P40123	CAPN10	Q9HC96
CAPN15	O75808	CAPN7	Q9Y6W3	CAPZA1	P52907
CARD11	Q9BXL7	CARNS1	A5YM72	CARS2	Q9HA77
CASK	P68400	CASP3	P42574	CASP4	P49662
CAST	O15446	CAT	P04040	CAV2	P51636
CAVIN1	Q6NZI2	CBLB	Q13191	CBR1	P16152
CBR3	P16152	CBX5	P45973	CBX6	O95503
CBX8	Q9HC52	CC2D1B	Q13415	CCAR2	Q8N163
CCDC88A	Q3V6T2	CCN1	P20248	CCN2	P29279
CCNA2	P20248	CCNB2	O95067	CCNC	P24863
CCNE2	O96020	CCNH	P51946	CCNT2	O60583
CCP110	O43303	CCRL2	O00421	CCT4	P50991
CCT6A	P40227, Q92526	CCT8	P50990	CD109	Q6YHK3
CD200	P41217	CD2AP	Q9Y5K6	CD2BP2	O95400
CD46	P15529	CD63	P08962	CD9	P21926
CDC14B	O60729	CDC16	Q13042	CDC25B	P30304
CDC26	Q8NHZ8	CDC27	P30260	CDC34	P49427
CDC37L1	Q7L3B6	CDC40	O60508	CDC42BPA	Q5VT25
CDC42EP2	O14613	CDC73	Q6P1J9	CDH2	P19022
CDH23	Q9H251	CDH4	P55283	CDIPT	O14735
CDK11A	Q9UQ88	CDK11B	P21127	CDK19	Q9BWU1
CDK4	P11802	CDK5	Q00535	CDK5RAP2	Q96SN8
CDK6	Q00534	CDK7	P50613	CEBPB	P17676
CENPC	Q03188	CENPE	Q02224	CENPU	Q71F23
CEP192	Q96EE3	CEP20	O75665	CEP250	Q9BV73
CEP290	O15078	CEP57	Q86XR8	CEP70	Q8NHQ1
CEP83	Q9Y592	CERK	Q8TCT0	CERS1	P27544
CERS4	Q9HA82	CERS6	Q6ZMG9	CERT1	Q9Y5P4-2
CFI	P05156	CFLAR	O15519-1, O15519-2	CHCHD1	Q96BP2
CHCHD3	Q9NX63	CHCHD4	Q8N4Q1	CHD1	O14646
CHD1L	Q86WJ1	CHD6	Q8TD26	CHD8	Q9HCK8
CHD9	Q3L8U1	CHDH	Q8NE62	CHERP	Q8IWX8
CHI3L1	P36222	CHKA	P35790	CHM	P24386, P26374
CHMP1A	Q9HD42	CHMP4B	Q96CF2, Q9H444	CHMP5	Q9NZZ3
CHMP7	Q8WUX9	CHPF2	Q9P2E5	CHPT1	Q8WUD6
CHRD1	Q9BU40	CHST10	O43529	CHST12	Q9NRB3
CHST15	Q7LFX5	CHST2	Q9Y4C5	CHTOP	Q9Y3Y2

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CITED2	Q99967	CKAP5	Q14008	CLASP1	Q7Z460
CLASP2	O75122	CLCF1	Q9UBD9	CLCN5	P51795
CLCN7	P51798	CLDN15	P56746	CLINT1	Q14677
CLIP1	P30622	CLIP3	Q96DZ5	CLN3	Q13286
CLOCK	O15516	CLP1	Q92989	CLPP	Q16740
CLTA	P09496	CLTC	Q00610	CLUAP1	Q96AJ1
CLVS1	Q8IUQ0	CMAS	Q8NFW8	CMC2	Q9NRP2
CMPK1	P30085	CNN2	Q99439	CNOT1	A5YKK6
CNOT2	Q9NZN8	CNOT3	O75175	CNOT6L	Q96LI5
CNOT8	Q9UFF9	CNOT9	Q92600	CNPY3	Q9BT09
CNTN1	Q12860	CNTRL	Q7Z7A1	COA1	Q9GZY4
COA6	Q5JTJ3	COASY	Q13057	COG3	Q96JB2
COG5	Q9UP83	COG6	Q9Y2V7	COL16A1	Q07092
COL4A5	P29400	COL5A1	P20908	COL5A2	P05997
COL8A1	P27658	COMMD4	Q9H0A8	COMMD5	Q9GZQ3
COMMD7	Q86VX2	COMMD9	Q9P000	COMT	P21964
COP1	P53621	COPA	P53621	COPB1	P53618
COPB2	P35606	COPG2	Q9UBF2	COPRS	Q9NQ92
COPS2	P61201	COPS4	Q9BT78	COPZ2	Q9P299
COQ10A	Q96MF6	COQ9	O75208	COX10	Q12887
COX14	Q96I36	COX17	Q14061	COX4I1	P13073
COX5A	P20674	COX6B1	P14854	COX7C	P15954
CPM	P14384	CPNE3	O75131	CPNE7	Q9UBL6
CPOX	P36551	CPS1	P31327	CPSF1	Q10570
CPSF2	Q9P2I0	CPSF6	Q16630-1, Q16630-2	CPSF7	Q8N684
CPT1A	P50416	CPT1B	Q92523	CPTP	Q5TA50
CRABP2	P29373	CRBN	Q96SW2	CREB1	P16220-1
CRISPLD2	Q9H0B8	CRMP1	Q14194	CROT	Q9UKG9
CRTAP	O75718	CRTC1	Q6UUV9	CRTC2	Q53ET0
CRY2	P62263	CRYL1	Q9Y2S2	CSF1	P09603
CSGALNACT2	Q8N6G5	CSK	P41240	CSKMT	A8MUP2
CSNK1A1	P48729	CSNK1G2	P78368	CSTB	P04080
CSTF3	Q12996	CTBP1	Q13363	CTCF	P49711
CTDNEP1	O95476	CTNNA1	P35221	CTNND1	O60716
CTNS	O60931	CTR9	Q6PD62	CTSA	P10619
CTSB	P07858	CTSD	P07339	CTSH	P09668
CTSK	P43235	CTSL	O60911, P07711	CTSO	P43235
CTU1	Q7Z7A3	CTU2	Q2VPK5	CUL2	Q13617
CUL5	Q93034	CWC27	Q6UX04	CXADR	P78310
CXXC1	Q9P0U4	CYB5RL	Q6IPT4	CYC1	P08574
CYP1A1	P04798	CYP1B1	Q16678	CYP2U1	Q7Z449
CYSTM1	Q9H1C7	DAP3	P51398	DAPK3	O43293
DARS1	P14868	DBI	P07108	DBNL	Q9UJU6
DBP	Q10586	DBT	P49674	DCP1A	Q9NPI6
DCP2	Q8IU60	DCPS	Q96C86	DCTN4	Q9UJW0
DCTN6	O00399	DCUN1D4	Q92564	DDB1	Q16531
DDB2	Q92466	DDHD1	Q8NEL9	DDIT4	Q9NX09

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DDO	Q99489	DDR1	Q08345	DDR2	Q16832
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DDX39B	Q13838	DDX42	Q86XP3	DDX5	P17844
DDX6	P26196	DECR1	Q16698	DEK	P51580
DENND1B	Q6P3S1	DENND3	A2RUS2	DENND4A	Q7Z401
DENND4B	O75064	DENND4C	Q5VZ89	DENND5A	Q61Q26
DERA	Q9Y315	DFFA	O00273	DFFB	O76075
DGAT1	O75907	DGKA	P23743, P52824	DGKH	Q86XP1
DGKI	O75912	DGKQ	P52824	DHCR24	Q15392
DHX36	Q9H2U1	DHX37	Q8IY37	DHX38	Q92620
DHX58	Q96C10	DIAPH2	O60879, O60879-3	DIAPH3	Q9NSV4
DICER1	Q9UPY3	DIS3	Q9Y2L1	DKK1	O94907
DLD	P09622	DLGAP5	Q15398	DMD	P11532
DMGDH	Q9UI17	DMPK	Q09013	DMTN	Q08495
DNAJA1	P31689, Q8WW22	DNAJB9	Q9UBS3	DNAJC10	Q8IXB1
DNAJC24	Q6P3W2	DNAJC3	Q13217	DNAJC6	O75061
DNAJC7	Q99615	DNAJC8	O75937	DNER	Q8NFT8
DNM1L	O00429	DOCK1	Q14185	DOCK10	Q96BY6
DOCK5	Q9H7D0	DOCK7	Q96N67	DOCK9	Q9BZ29
DOHH	Q9BU89	DOLK	Q9UPQ8	DOT1L	Q8TEK3
DPAGT1	Q9H3H5	DPH1	Q9BZG8	DPH3	Q96FX2
DPH5	Q9H2P9	DPH7	Q9BTV6	DPM3	Q9P2X0
DPP7	Q9UHL4	DPYD	Q12882	DRAP1	Q14919
DSC3	Q08554	DSEL	Q8IZU8	DSG2	Q14126
DST	Q03001	DTNBP1	Q96EV8	DTX3L	Q8TDB6
DUSP10	Q9Y6W6	DUSP3	P51452	DUSP5	Q16690
DVL1	O14640	DVL2	O14641	DXO	O77932
DYNC1H1	Q14204	DYNC1I1	O14576	DYNC1I2	Q13409
DYNC1LI2	O43237	DYNC2H1	Q8NCM8	DYNC2I1	Q8WVS4
DYNC2LI1	Q8TCX1	DYNLL1	P63167	DYNLRB1	Q9NP97
DYRK3	Q92630	DZIP3	Q86Y13	E2F4	Q16254
ECI2	O75521-2	ECSIT	Q9BQ95	EDC3	Q96F86
EDC4	Q6P2E9	EDEM1	Q92611	EDEM3	Q9BZQ6
EEF1AKMT2	Q5JPI9	EEF1AKMT4-ECE2	P0DPD6	EEF1D	P29692
EEPD1	Q7L9B9	EFEMP1	Q12805	EFNA1	P20827
EFNB2	P52799	EGF	P01133	EGLN2	Q96KS0
EGR2	P11161	EHD3	Q6NVY1	EHHADH	Q08426
EHMT2	Q96KQ7	EIF1AY	P47813	EIF2A	P05198
EIF2AK2	P19525	EIF2B4	Q9UI10	EIF2B5	Q13144
EIF2S3	P41091	EIF3A	Q14152	EIF3E	P60228
EIF3J	O75822	EIF3M	Q7L2H7	EIF4A2	Q14240
EIF5	P55010	EIF5B	O60841	ELF2	Q15723
ELK1	P19419	ELL2	O00472	ELOB	Q15370
ELOVL2	Q9NXB9, Q9NYP7	ELOVL5	Q9NYP7	ELOVL7	A1L3X0
ELP1	O95163	ELP2	Q6IA86	ELP5	Q8TE02
EMB	O14980	ENAH	Q8N8S7	ENDOD1	O94919
ENGASE	Q8NFI3	ENO4	A6NNW6	ENPP2	Q13822
ENTPD4	Q9Y227	ENTPD7	Q9NQZ7	EPAS1	Q99814
EPB41L1	Q9H4G0	EPB41L2	O43491	EPB41L3	Q9Y2J2

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EPHA4	P54764	EPHB1	P54762	EPHB2	P29323
EPHX1	P07099	EPHX2	P34913	EPN1	Q9Y6I3
EPRS1	P07814	EPS15	P42566	EPS8	Q12929
ERAP1	Q9NZ08	ERAP2	Q6P179	ERBB4	Q15303-1, Q15303-2, Q15303-3
ERBIN	Q96RT1	ERC1	Q96FL8	ERCC5	P28715
ERCC6	Q03468	ERCC8	Q13216	ERF	P50548
ERLEC1	Q96DZ1	ERLIN2	O94905	ERN1	O75460
ESCO2	Q56NI9	ESRRA	P11474	ESYT2	A0FGR8, A0FGR9
ETFA	P13804	ETNK2	Q9NVF9	EVA1A	Q9H8M9
EVC	P57679	EVC2	Q86UK5	EXOC1	Q9NV70
EXOC2	Q96KP1	EXOC4	Q96A65	EXOC5	O00471
EXOC6	Q8TAG9	EXOSC4	Q9NPD3	EXOSC6	Q5RKV6
EXTL2	Q9UBQ6	EXTL3	O43909	FAAP100	Q0VG06
FADD	Q13158	FAH	P16930	FAHD1	Q6P587
FAM114A2	Q9NRY5	FAM120B	Q96EK7	FAM20A	Q96MK3
FAM50A	Q14320	FAM53B	Q5JPI9	FAM98B	Q52LJ0
FANCD2	Q9BXW9	FANCI	Q9NVI1	FAR1	Q8WVX9
FARSB	Q9NSD9	FASN	P49327	FASTK	Q14296-4
FBN1	P35555	FBXL14	Q8N1E6	FBXL15	Q9H469
FBXL18	Q96ME1	FBXL19	Q6PCT2	FBXL3	Q9UKT7
FBXL5	Q9UKA1	FBXO10	Q9UK96	FBXO17	Q96EF6
FBXO27	Q8NI29, Q96EF6	FBXO30	Q8TB52	FBXO7	Q9Y3I1
FBXW5	Q969U6	FCHO1	O14526	FDX2	P10109, Q6P4F2
FEM1A	Q9BSK4	FEM1C	Q96JP0	FER	P16591
FERMT2	Q96AC1	FGD1	P98174	FGD4	Q96M96
FGF12	P61328	FGF2	P09038	FGF5	P12034-1
FIP1L1	Q6UN15	FITM2	A5D6W6, Q8N6M3	FKBP14	Q9NWM8
FKBP4	Q02790	FKBPL	Q9UIM3	FKTN	O75072
FLCN	Q8NFG4	FLNB	O75369	FLRT1	Q9NZU1
FLRT2	O43155	FN1	P02751	FNBP1L	Q5T0N5
FOLR3	P41439	FOSL1	P15407	FOXP1	Q9H334-8
FOXQ1	Q9C009	FPGT	O14772	FRAT2	O75474
FSTL1	Q12841	FTH1	P02794	FTL	P02792
FTO	Q9C0B1	FUCA1	P04066	FUND2	P46736
FXR1	P51114	FYN	P06241	FYTTD1	Q96QD9
FZD2	Q14332	FZD4	Q9ULV1	FZD6	O60353
FZD8	Q9H461	GABARAP	O95166	GALNS	P34059
GALNT1	Q10472	GALNT13	Q8IUC8	GALNT17	Q49A17, Q6IS24
GAPVD1	Q14C86	GAS1	P54826	GBA1	P04062
GBA2	Q9HCG7	GBE1	Q04446	GBP1	P32455, Q9H0R5
GBP2	P32456, Q9H0R5	GBP3	Q9H0R5	GCA	P28676
GCC2	Q8IWJ2	GCFC2	P16383	GCKR	Q14397
GCLM	P48507	GCNT1	Q02742	GDAP1	Q8TB36
GDF1	P27544	GDF5	P43026	GDNF	P39905
GEM	P55040	GEMIN4	P57678	GEMIN6	Q8WXD5
GEMIN7	Q9H840	GET3	O43681	GET4	Q7L5D6
GFM1	Q96RP9	GFM2	Q969S9	GFPT1	Q06210
GFRA1	P56159	GGA2	Q9UJY4	GGA3	Q9NZ52

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GID4	Q8IVV7	GINS3	Q9BRX5	GIT1	Q9Y2X7
GJA1	P17302	GJD3	Q8N144	GLA	P06280
GLB1	P16278	GLG1	Q92896	GLO1	Q04760
GLRX3	O76003	GM2A	P17900	GNA12	Q03113
GNAI3	P08754	GNAQ	P50148	GNB2	P62879
GNB4	Q9HAV0	GNG12	Q9UBI6	GNMT	Q14749
GNPAT	O15228	GNPDA1	P46926, Q8TDQ7	GNPDA2	Q8TDQ7
GOLGB1	Q14789	GOLIM4	O00461	GOPC	Q9HD26
GORASP1	Q9BQQ3	GPAT4	Q53EU6, Q86UL3	GPC4	O75487
GPC6	Q9Y625	GPCPD1	Q9NPB8	GPD1L	Q8N335
GPD2	P21695, P43304	GPFR1	Q99527	GPKOW	Q92917
GPNMB	Q14956	GPR143	P51810	GPR176	Q14439
GPS2	Q13227	GPSM1	Q86YR5	GPSM2	P81274
GPX4	P36969	GPX8	Q8TED1	GRB10	Q13322
GREB1	Q4ZG55	GRN	P28799	GRPEL2	Q8TAA5
GSK3A	P49841	GSS	P48637	GSTA4	O15217, P08263
GSTM1	P09488, P46439	GSTO1	P78417, Q9H4Y5	GSTZ1	O43708
GTF2A1	P52655	GTF2H3	Q13889	GTF3C1	Q12789
GTF3C2	Q8WUA4	GTF3C3	Q9Y5Q9	GTF3C5	Q9Y5Q8
GUCY1A2	P33402	GUCY1B1	Q02153	GYG1	P46976
H2AC25	Q7L7L0	H2AC6	Q93077	H2AX	P16104
H2AZ1	P0C0S5	H2BC11	P06899	H2BC12	O60814, Q93079, Q99877, Q99880
H2BC12L	P57053	H2BC5	P58876	H4C8	P62805
HABP4	Q5JVS0	HACD3	Q9P035	HACL1	Q9UJ83
HBEGF	Q99075	HBS1L	Q9Y450	HCFC1	P51610
HCFC2	Q9Y5Z7	HDAC1	Q13547	HDGF	P51858
HDHD3	P13716	HEBP2	Q9Y5Z4	HECTD1	Q9ULT8
HECTD2	Q5U5R9	HECTD3	Q5T447	HERC1	Q15751
HERC3	Q15034	HERC6	Q8IVU3, Q9UII4	HEXA	P06865
HEXB	P07686	HFE	Q30201	HHAT	Q5VTY9
HIBADH	P31937	HIBCH	Q6NVY1	HIF1A	Q16665
HIGD2A	Q9BW72	HIP1	O00291	HIRA	P54198
HIVEP3	Q5T1R4	HJURP	Q8NCD3	HKDC1	P52789, Q2TB90
HLA-A	P04439	HLA-H	P01893	HLCS	P50747
HLTF	Q14527	HMGB2	P26583	HMMR	O75330
HMOX1	P09601	HNMT	P50135	HNRNPA0	Q13151
HNRNPA1	P09651	HNRNPA2B1	P22626	HNRNPDL	O14979
HNRNPF	P52597	HNRNPH1	P31943, P55795	HOMER3	Q9NSC5
HPRT1	P00492	HPS1	Q92902	HPS4	Q9NQG7
HRAS	P01112	HS3ST3A1	Q9Y663	HS3ST3B1	Q9Y662, Q9Y663
HSCB	Q8IWL3	HSD17B12	Q53GQ0	HSD17B14	Q9BPX1
HSD17B2	P37059	HSD17B4	P51659	HSD17B8	Q92506
HSP90AA1	P07900	HSP90AB1	P08238	HSP90B1	P14625
HSPA13	P48723	HSPA4L	O95757	HSPA5	P11021
HSPA6	P17066	HSPB1	P04792	HSPD1	P10809
HSPG2	P98160	HTRA1	Q92743	HTRA2	O43464
HYAL2	Q12891	HYAL3	P38567	IARS1	P41252
ICAM3	P32942	ICE1	Q9Y2F5	ICE2	Q659A1
ICMT	O60725	ID1	P41134	ID3	Q02535

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ID4	P47928	IDE	P14735	IDH2	P48735
IDH3A	P50213	IDUA	P35475	IFI44	Q8TCB0-1
IFIH1	Q9BYX4	IFIT2	P09913	IFNGR1	P15260
IFT20	Q8IY31	IFT27	Q9BW83	IFT43	Q96FT9
IFT56	A0AVF1	IFT57	Q9NWB7	IFT70A	Q86WT1, Q8N4P2
IFT74	Q96LB3	IFT80	Q9P2H3	IFT81	Q8WYA0
IFT88	Q13099	IGF2BP3	O00425	IGF2R	P11717
IL17RE	Q8NFR9	IL1R1	Q9HB29	IL1RAP	Q9NPH3-1
IL1RN	P18510	IL33	O95760	IL6R	P08887, P08887-2
IL6ST	P40189	IL7R	P16871	ILF3	Q12906
ILK	Q13418	IMMT	Q16891	IMP3	Q9NV31
IMPA1	P29218	IMPA2	O14732	IMPACT	Q9P2X3
ING3	Q9NXR8	ING5	Q8WYH8	INPP1	P49441
INPP4A	Q96PE3	INPP5F	Q9Y2H2	INSIG2	Q9Y5U4
INTS5	Q6P9B9	INTS7	Q9NVH2	INTS8	Q75QN2
INTS9	Q9NV88	INTU	Q9ULD6	IP6K1	Q92551, Q96PC2
IPMK	Q8NFU5	IPO7	O95373	IPO8	O15397
IQCE	Q6IPM2	IQGAP1	P46940	IQGAP3	Q86VI3
IRAK1	P51617	IREB2	P48200	IRF3	Q92985
IRS1	P35568	IRS2	Q9Y4H2	ISCA2	Q86U28
ISCU	Q9H1K1-1	ITCH	Q96J02	ITGA4	P13612
ITGB1	P05556	ITGB3BP	Q13352	ITGB4	P16144
ITGB5	P18084	ITPKB	P27987	ITPR2	Q14571
ITPR3	Q14573	ITSN1	Q15811	ITSN2	Q9NZM3
JADE2	Q9NQC1	JAM3	Q9BX67	JARID2	Q92833
JMJD4	Q9H9V9	JOSD1	Q15040	JUN	P05412
JUP	P14923	KAT2A	Q92830	KAT2B	Q92831
KAT5	Q92993	KAT6B	Q8WYB5	KAT8	Q9H7Z6
KAZN	Q674X7	KCNB1	Q14721	KCNJ14	Q9UNX9
KCNN4	O15554	KCNQ3	O43525	KCNS3	Q9BQ31
KCTD13	Q8WZ19	KCTD3	Q9Y597	KDELR1	P24390
KDM1B	Q8NB78	KDM3A	Q9Y4C1	KDM5A	P29375
KDM6A	O15550	KDM6B	O15054	KDSR	Q06136
KHDRBS1	Q07666	KHDRBS3	O75525	KHSRP	Q92945
KIAA0319	Q5VV43	KIAA1671	Q9C0C2	KIDINS220	Q9ULH0
KIF11	P52732	KIF14	Q15058	KIF15	Q9NS87
KIF18A	Q8NI77	KIF1B	O60333	KIF1C	O43896
KIF20A	O95235	KIF20B	Q96Q89	KIF21A	Q7Z4S6
KIF23	Q02241	KIF2A	O00139, Q8N4N8	KIF3A	Q9Y496
KIF3B	O15066	KIF4A	O95239	KIF5B	P33176
KIF5C	Q6PKD1	KIF6	Q6ZMV9	KIF7	Q2M1P5
KIFAP3	Q92845	KIFC3	Q9BVG8-2	KIRREL1	Q96J84
KLC1	Q07866	KLF15	Q9UIH9	KLF16	Q9BXXK1
KLHDC3	Q9BQ90	KLHL20	Q9Y2M5	KLHL21	Q9UJP4
KLHL3	Q9UH77	KMO	O15229	KMT2A	Q03164
KMT2B	Q9UMN6	KMT2C	Q8NEZ4	KMT5B	Q4FZB7, Q86Y97
KNL1	Q8NG31	KNTC1	P50748	KPNA4	O00629
KRAS	P01116-1, P01116-2	KRBOX4	Q5JUW0	KREMEN1	Q9ULT6
KRT10	P13645	KRT18	P05783	KRT7	P08729

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
KRT8	P05787, P13647	KRTAP2-3	P0C7H8	KRTAP2-4	Q9BYR9, Q9BYU5
KTN1	Q86UP2	KYAT3	Q6YP21	LAGE3	Q14657
LAMA4	Q16363	LAMB1	P07942	LAMC2	Q13753
LAMP1	P11279	LAMTOR4	Q0VGL1	LAMTOR5	O43504
LARGE1	O95461	LARP1	Q6PKG0	LARS1	Q9P2J5
LARS2	Q15031	LATS1	O95835	LCLAT1	Q6UWP7
LCP1	P13796	LGALS3	P17931	LHFPL2	Q6ZUX7
LHX2	P50458	LIMK1	P53667	LIMS1	P48059
LIN54	Q6MZP7	LIN7C	Q9NUP9	LIPE	Q05469
LIPG	Q9Y5X9	LIPH	Q8WWY8	LMAN1	P49257
LMBRD1	Q9NUN5	LMO7	Q8WWI1	LNPEP	Q9UIQ6
LONP1	P36776	LOX	P28300	LOXL4	Q96JB6
LPCAT1	Q8NF37	LPGAT1	Q92604	LRFN3	Q9BTN0
LRFN4	Q6PJG9	LRRCL1	Q9BTT6	LRRCL4	Q15048
LRRCL49	Q8IUZ0	LRRCL8A	Q8IWT6	LRRCL8B	Q6P9F7
LRRCL8C	Q8TDW0	LRRCL8D	Q7L1W4	LRRCL8E	Q6NSJ5
LRRFIP1	Q32MZ4	LRRK2	Q5S007	LSM10	Q969L4
LSM11	P83369	LTB4R	Q15722, Q9NPC1	LTBP2	Q14767
LTBR	P36941	LTN1	O94822	LYN	P07948
LYPD6B	Q8NI32	LYPLA2	O95372	LYRM4	Q9HD34
M6PR	P20645	MACO1	Q8N5G2	MAD1L1	Q9Y6D9
MAF	O75444	MAF1	Q9H063	MAGEF1	Q96MG7
MAGT1	Q9H0U3	MALT1	Q9UDY8	MAML2	Q8IZL2
MAN1C1	Q9NR34	MAN2A1	Q16706	MANF	P55145
MAOA	P21397	MAP1B	P46821	MAP2K2	P36507
MAP2K3	P46734	MAP2K4	P45985	MAP2K7	O14733
MAP3K11	Q16584	MAP3K5	Q99683	MAP3K7	O43318
MAP4K4	O95819	MAPK10	P53779	MAPK12	P53778
MAPK13	O15264	MAPK6	Q16659	MAPK9	P45984
MAPKAPK2	P49137	MAPRE1	Q15691	MARCHF6	O60337
MASTL	Q96GX5	MAVS	Q7Z434	MAX	P61244
MBD3	O95983	MBD5	Q9P267	MBD6	Q96DN6
MBIP	Q9NS73	MBOAT2	Q6ZWT7	MBOAT7	Q96N66
MBP	P13727	MBTPS2	O43462	MCAT	Q8IVS2
MCOLN1	Q9GZU1	MCRIP1	C9JLW8	MCUB	Q9NWR8
MDGA1	Q8NFP4	MDM2	Q00987	MDM4	O15151
ME1	P48163	ME2	P23368	MED1	O95243
MED13	Q9UHV7	MED13L	Q71F56	MED14	O60244
MED18	Q9BUE0	MED23	Q9ULK4	MED24	O75448
MED4	Q9NPJ6	MEF2A	Q02078	MEN1	O00255
MET	P08581	METAP2	P50579	METTLL1	Q9UBP6
METTLL23	Q86XA0	METTLL3	Q86U44	MFN1	Q8IWA4
MGAT4B	Q9UBM8, Q9UQ53	MGLL	Q99685	MGMT	P16455
MGST3	O14880	MIA2	Q96PC5	MIA3	Q5JRA6
MIB1	Q86YT6	MIB2	Q96AX9	MICA	Q29983
MICAL1	Q8TDZ2	MICU1	Q9BPX6	MICU2	Q8IYU8
MICU3	Q86XE3, Q8IYU8	MID1IP1	Q92748, Q9NPA3	MIEF1	Q9NQG6
MIGA1	Q8NAN2	MIGA2	Q7L4E1	MINK1	Q8N4C8
MIS18BP1	Q6P0N0	MITF	O75030-9	MLH1	P40692

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MLH3	Q9UHC1	MLLT1	Q03111	MLLT3	P42568
MLST8	Q9BVC4	MLYCD	O95822-2	MMAB	Q96EY8
MMGT1	Q8N4V1	MMP14	P50281	MMUT	P22033
MOB1B	Q7L9L4	MOCs1	Q9NZB8-1, Q9NZB8-5	MOCs3	O95396
MOGS	Q13724	MOSPD2	Q8NHP6	MPC2	O95563
MPHOSPH10	O00566	MPHOSPH8	Q99549	MPP2	Q08050
MPST	P25325	MPZ	P25189	MR1	Q8N490-2
MRM3	Q9HC36	MRPL1	Q9BYD6	MRPL14	Q6P1L8
MRPL16	Q9NX20	MRPL17	Q9NRX2	MRPL34	Q9BQ48
MRPL36	Q9P0J6	MRPL41	Q8IXM3	MRPL44	Q9H9J2
MRPL49	Q13405	MRPL57	Q9BQC6	MRPL9	Q9BYD2
MRPS12	O15235	MRPS15	P82914	MRPS16	Q9Y3D3
MRPS17	Q9Y2R5	MRPS2	Q9Y399	MRPS24	Q96EL2
MRPS25	P82663	MRPS30	Q9NP92	MRPS34	P82930
MRRF	Q96E11	MSH3	P20585	MSRB3	Q8IXL7
MT-ATP6	P00846	MT-ATP8	P03928	MT-ND1	P03886
MT-ND2	P03891	MT-ND3	P03897	MT1X	P80297
MT2A	P02795	MTA1	Q13330	MTFMT	Q96DP5
MTHFD1L	Q6UB35	MTHFD2	P13995	MTHFR	P42898
MTMR4	Q9NYA4	MTMR6	Q9Y217	MTMR9	Q96QG7
MTPN	Q96Q27	MTR	Q99707	MTREX	P42285
MTX1	Q13505	MUL1	O15151, Q969V5	MYBL1	P10243
MYC	P01106	MYD88	Q99836	MYDGF	Q969H8
MYH10	P35580	MYL12A	P19105	MYL12B	O14950
MYLK	Q15746	MYO18A	Q92614	MYO3B	Q8WXR4
MYO5A	Q9Y4I1	MYO5B	Q9ULV0	MYO6	Q9UM54
MYO9A	B2RTY4	MYRIP	Q8NFW9	MYSM1	Q5VVJ2
MZT1	Q08AG7	NAA35	Q5VZE5	NAB1	P08865
NAB2	Q15742	NABP1	Q96AH0	NAGLU	P54802
NALCN	Q8IZF0	NAPA	Q9Y2A7	NAPG	Q99747
NAT8L	Q8N9F0	NBN	O60934	NBR1	Q14596
NCAPG	Q9BPX3	NCAPG2	Q86XI2	NCBP1	Q09161
NCEH1	Q6PIU2	NCF2	P19878	NCK1	O43639, P16333
NCK2	O43639	NCOA1	Q15788	NCOA2	Q15596
NCR3LG1	Q68D85	NCS1	P43080	NDC80	O14777
NDOR1	Q9UHB4	NDRG1	Q92597	NDUFA11	Q86Y39
NDUFA2	O43678	NDUFA5	Q16718	NDUFA8	P51970
NDUFAF2	Q8N183	NDUFAF5	Q5TEU4	NDUFB10	O96000
NDUFS4	O43181	NECTIN3	Q9NQS3	NEDD1	Q8NHV4
NEDD4	P46934, Q96PU5	NEDD4L	Q96PU5	NEGR1	Q7Z3B1
NEIL3	Q8TAT5	NEK1	Q96PY6	NEK2	P51955
NEK3	Q9UHY1	NEK7	Q8TDX7	NELFCD	Q8IXH7
NEU1	Q99519	NF1	P21359	NFASC	O94856
NFIB	O00712	NFKBIB	Q15653	NFU1	Q9UMS0
NGF	P01138	NGRN	Q9NPE2	NHEJ1	Q9H9Q4
NHLRC2	Q8NBF2	NHLRC3	Q5JS37	NHS	Q6T4R5
NID2	Q14112	NIPAL1	Q6NVV3	NIPAL2	Q9H841
NIPAL4	Q0D2K0	NIPBL	Q6KC79	NIPSNAP2	O75323
NLGN1	Q8N2Q7	NLN	Q9BYT8	NLRC5	Q86WI3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
NLRP1	Q9C000	NLRX1	Q86UT6	NMB	Q14956
NME2	P22392	NME4	O00746	NMI	Q13287
NMRK1	Q9NWW6	NMT2	O60551	NOC4L	Q9BVI4
NOG	Q13253	NOP10	Q9NPE3	NOP2	P46087
NOXO1	Q8NFA2	NPAS2	Q99743	NPC2	P61916
NPHP1	O15259	NPHP3	Q7Z494	NPIPB3	Q92617
NPPA	P01160	NPRL2	Q8WTW4	NR1D1	P20393
NR1D2	Q14995	NR1H2	P55055	NR2F6	P10588
NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	NR3C2	P08235-1, P08235-2, P08235-3, P08235-4	NRAS	P01111
NRBP1	Q9UHY1	NRG1	Q02297	NRG2	O14511
NRIP1	P48552	NRP1	O14786	NSDHL	Q15738
NSF	P46459	NSMAF	Q92636	NSMCE3	Q9UNF1
NSRP1	Q9H0G5	NSUN4	Q96CB9	NT5C3B	Q969T7
NT5E	P21589	NTHL1	P78549	NTN4	Q9HB63
NUBP1	P53384	NUBPL	Q8TB37	NUDT14	O95848
NUDT16	Q96DE0	NUDT21	O43809	NUF2	Q9BZD4
NUP133	Q8WUM0	NUP153	P49790	NUP54	Q7Z3B4
NUP58	Q9BVL2-1, Q9BVL2-2	NXT1	Q9UKK6	OAZ1	P54368
OBSL1	Q5VST9	OCRL	Q01968	OFD1	O75665
OLA1	Q9NTK5	OLAH	Q9NV23	OPA1	O60313
OPN3	Q9H1Y3	ORAI2	Q96SN7	ORC3	Q9UBD5
OSBPL11	Q9BXB4	OSBPL5	Q9H0X9	OSBPL6	Q9BZF3
OSBPL8	Q9BZF1	OSBPL9	Q96SU4	OVCA2	Q9BZG8
OXA1L	Q15070	OXTR	P30559	P2RX6	O15547
P2RY11	Q96G91	P2RY6	Q15077	P3H1	Q32P28
P3H2	Q8IVL5	P4HA1	P13674	PACS1	Q6VY07
PADI1	Q9ULC6	PADI3	Q9ULW8	PAF1	P28328
PAFAH1B1	P43034	PAFAH1B3	Q15102	PAK3	Q13153
PALS1	Q8N3R9	PAN3	Q58A45	PANK1	Q8TE04
PANX2	Q96RD6	PAPOLA	P51003	PAPPA2	Q9BXP8
PAPSS1	O43252	PAPSS2	O95340	PARD6A	Q9NPB6
PARD6B	Q9BYG5	PARP14	Q460N5	PARP4	Q9UKK3
PARP6	Q2NL67	PARP9	Q8IXQ6	PARS2	Q7L3T8
PATL1	Q86TB9	PAX8	Q06710	PBRM1	Q86U86
PBX1	P40424	PC	P11498	PCBP1	Q15365
PCBP4	P57723	PCCB	P05166	PCDH7	O60245
PCGF2	P35227	PCLO	Q9Y6V0	PCM1	Q15154
PCOLCE2	Q9UKZ9	PCSK7	Q16549	PCYOX1L	Q8NBM8
PCYT1A	P49585	PCYT1B	Q9Y5K3	PCYT2	Q99447
PDCD1LG2	Q9BQ51	PDCD4	Q53EL6	PDCD6IP	Q8WUM4
PDCD7	Q8N8D1	PDE1C	Q01064, Q14123	PDE2A	O00408
PDE5A	O76074	PDIA5	Q14554	PDIA6	Q15084
PDK1	O15530	PDK3	Q15120	PDLIM5	Q96HC4
PDLIM7	Q9NR12	PDP2	Q9P2J9	PDPN	Q86YL7
PDS5A	Q29RF7	PDXK	O00764	PEA15	Q15121
PEAK1	Q9H792	PELI3	Q8N2H9	PELP1	Q8IZL8
PER1	O15534	PER2	O15055	PET117	Q6UWS5
PEX1	O43933	PEX19	P40855	PEX3	P56589

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PEX5	P50542-1, P50542-2	PFDN2	Q9UHV9	PFKFB2	O60825
PFKFB3	Q16875	PFKFB4	Q16877	PFKL	P17858
PFKM	P08237	PFKP	Q01813	PGAP1	Q75T13
PGGT1B	P53609	PGM2L1	Q6PCE3	PGM3	O95394
PGP	A6NDG6	PGRMC1	O00264	PHACTR2	O75167
PHC3	Q8NDX5	PHF20L1	A8MW92	PHF5A	Q7RTV0
PHIP	Q8WWQ0	PHKB	Q93100	PHLPP2	O43747
PHYH	O14832	PI4K2A	Q9BTU6	PI4K2B	Q8TCG2
PIAS1	O75925	PIAS4	Q8N2W9	PICALM	Q13492
PIGA	P37287	PIGB	Q92521	PIGC	Q92535
PIGK	Q92643	PIGN	O95427	PIGQ	Q9BRB3-2
PIGV	Q9NUD9	PIK3C2A	O00443	PIK3C3	Q8NEB9
PIK3CB	P42338	PIK3R1	P27986	PIK3R2	O00459
PIKFYVE	Q9Y2I7	PIM1	P11309	PIP4K2A	P48426
PIP4K2B	P78356	PIP4K2C	Q8TBX8	PIP4P1	Q86T03
PIP5K1B	O14986	PIP5K1C	O60331	PIR	O00625
PISD	Q9UG56	PITPNB	P48739	PITPNM3	Q9BZ71
PJA2	O43164	PKN1	Q16512	PKN2	Q16513
PLA2G12A	Q9BZM1	PLA2G15	Q8NCC3	PLA2R1	Q13018
PLAAT3	P53816	PLAT	P00750	PLB1	Q3MJ16
PLCB1	Q9NQ66	PLCB4	Q15147	PLCG1	P19174
PLD1	Q13393	PLD6	Q8N2A8	PLEKHA1	Q9HB21
PLEKHA5	Q9HAU0	PLEKHG3	A1L390	PLIN2	Q99541
PLK2	Q9NYY3	PLOD1	Q02809	PLOD2	O00469
PLPP2	O43688	PLPP3	O14495	PLPP4	Q5VZY2
PLPPR3	Q6T4P5	PLXNA1	Q9UIW2	PLXNB1	O43157
PLXNB3	O43157	PLXNC1	O60486	PMEPA1	Q969W9
PMP22	Q9NR77	PMPCB	O75439	PNKD	Q8N490-2
PNN	Q9H307	PNP	P00491	PNPLA3	Q9NST1
PNPLA6	Q8IY17	PNRC2	Q9NPJ4	POLD1	P28340
POLD4	Q9HCU8	POLDIP3	Q9BY77	POLE4	Q9NR33
POLI	Q9UNA4	POLK	Q9UBT6	POLR1C	O15160
POLR1D	P0DPB5	POLR1F	Q3B726	POLR2B	P30876
POLR2H	P52434	POLR2J	P52435	POLR3GL	Q9BT43
POLR3K	Q9Y2Y1	POLRMT	O00411	POMT2	Q9UKY4
PON2	Q15165	POP7	O75817	POR	P16435
POT1	Q9NUX5	POU2F1	P14859	POU3F2	P20265
PPARGC1A	Q9UBK2	PPARGC1B	Q86YN6	PPFIA4	O75335
PPFIBP1	Q86W92	PPFIBP2	Q8ND30	PPHLN1	Q8NEY8
PPIF	P62937	PPIG	Q13427	PPIL3	Q9H2H8
PPIL4	Q8WUA2	PPIP5K1	Q6PFW1	PPIP5K2	O43314
PPM1F	P49593	PPP1CB	P62140	PPP1R12A	O14974
PPP1R15A	O75807	PPP1R16A	P24298	PPP1R21	Q9Y6Q2
PPP2R1B	P30154	PPP2R3B	Q9Y5P8	PPP2R5C	Q13362
PPP2R5E	Q16537	PPP3CA	Q08209	PPP4C	P60510
PPP4R2	Q9NY27	PPP6C	O00743	PPP6R1	Q9UPN7
PPP6R3	Q5H9R7	PPRC1	Q5VV67	PPWD1	Q96BP3
PRC1	O43663	PRCC	Q92733	PRDX2	P32119
PRDX4	Q13162	PREB	Q9HCU5	PRKAA1	Q13131

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PRKAA2	P54646	PRKAB1	Q9Y478	PRKAB2	O43741
PRKACB	P22694	PRKAG2	Q9UGJ0	PRKCD	Q05655
PRKCI	P41743	PRKD3	O94806	PRKDC	P78527
PRKN	O60260	PRMT6	Q96LA8	PRP4K	Q13523
PRPF18	Q99633	PRPF19	Q9UMS4	PRPF3	O43395
PRPF40A	O75400	PRR5	P85299	PSAP	Q8IWL1
PSEN2	P49810	PSIP1	O75475	PSMA1	P25786
PSMA3	P25788	PSMB4	P28070	PSMB9	P28065
PSMC1	P62191	PSMC2	P35998	PSMC6	P62333
PSMD1	Q99460	PSMD12	O00232	PSMD14	O00487
PSMD6	Q15008	PSMD9	O00233	PSME4	Q14997
PSMG3	Q9BT73	PSPH	P78330	PTBP1	P26599
PTCD3	Q96EY7	PTDSS2	Q9BVG9	PTEN	P60484
PTGIS	Q16647	PTK2	Q05397	PTK2B	Q14289
PTPN1	P18031	PTPN11	Q06124	PTPN12	Q05209
PTPN13	Q12923	PTPN23	Q9H3S7	PTPN4	P29074
PTPRD	P23468	PTPRJ	Q12913	PTRH2	Q9Y3E5
PUF60	Q9UHX1	PUS1	Q9Y606-1	PVR	P16234
PYGL	P06737	PYGO2	Q9BRQ0	QPRT	Q15274
RAB11FIP2	Q7L804	RAB13	P51153	RAB1B	Q9H0U4
RAB21	Q9UL25	RAB24	Q969Q5	RAB27A	P51159
RAB27B	O00194	RAB29	O14966	RAB30	Q15771
RAB32	Q13637	RAB38	P57729	RAB3GAP1	Q15042
RAB3GAP2	Q9H2M9	RAB3IL1	Q8TBN0	RAB5C	P51148
RAB5IF	Q9BUV8	RAB7A	P51149	RAB8B	Q92930
RAB9A	P51151	RABEP1	Q15276	RABGEF1	Q9UJ41
RABGGTB	P53611	RAC3	P60763	RAD21	O60216
RAD9A	Q99638	RALA	P11233	RALGAPA1	Q6GYQ0
RALGAPB	Q86X10	RANBP2	P49792	RANGRF	Q9HD47
RAP1GDS1	P52306	RAPGEF1	Q13905	RAPGEF2	Q9Y4G8
RARB	P10826	RARRES2	Q99969	RARS2	Q5T160
RASA1	P20936	RASA2	Q15283	RASGRF1	Q13972
RAX	O75569	RB1	P06400	RBKS	Q9H477
RBM22	Q9NW64	RBM23	Q14498	RBM25	P49756
RBM26	Q5T8P6	RBM27	Q9P2N5	RBM5	P52756
RBPJ	Q06330	RCAN3	Q9UKA8	RCC1	P18754
RCC1L	Q96I51	RCC2	Q9P258	RCOR1	Q9UKL0
RDX	P35241	REC8	O95072	RECK	O95980
REL	Q00653	REPIN1	Q9BS31	REST	Q13127
RETREG1	Q9H6L5	RFC1	P35251	RFFL	O75771
RGL1	Q9NZL6	RGL4	Q12967	RGN	O43493
RGS11	O94810	RGS19	P49795	RGS20	O76081
RGS9	O75916	RHEB	Q15382	RHOF	Q9HBH0
RHOG	P84095	RHOT1	Q8IXI2	RHPN2	Q8IUC4
RIC1	Q4ADV7	RICTOR	Q6R327	RIF1	Q5UIP0
RILP	Q96MT3	RIMKLA	Q8IXN7	RIMKLB	Q9ULI2
RIN3	Q8TB24	RINT1	Q6NUQ1	RIOK2	Q9BVS4
RIOK3	O14730	RIOX1	Q9H6W3	RLF	P51460
RLIM	Q9NVW2	RNASEK	Q6P5S7	RNF14	Q9UBS8

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RNF144A	P50876	RNF144B	Q7Z419	RNF182	Q8N6D2
RNF185	Q96GF1	RNF19A	Q9NV58	RNF20	Q5VTR2
RNF220	Q5VTB9	RNF5	Q96GF1, Q99942	RNF7	Q9UBF6
RNGTT	O60942	RNLS	Q5VYX0	ROBO1	Q9Y6N7
ROCK1	Q13464	ROCK2	O75116	RORA	P35398
RPAP2	Q8IXW5	RPE	Q2QD12, Q96AT9	RPGRIP1L	Q68CZ1
RPL13	P26373, P40429	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL27	P61353	RPL32	P62888, P62910
RPL35A	P18077, P42766	RPL37A	P18077, P61513	RPL38	P63173
RPL39L	Q96EH5	RPLP1	P05386	RPLP2	P05387
RPN1	Q13200	RPRD1A	Q96P16	RPS10	P46783
RPS11	P62280	RPS14	P62263	RPS15	P62841
RPS16	P62249	RPS17	P08708	RPS19	P39019
RPS19BP1	Q86WX3	RPS20	P60866	RPS21	P62841, P63220
RPS27	P42677, Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS5	P46782	RPS6KA3	P51812
RPS6KA6	Q9UK32	RPS6KB1	P23443	RPTOR	Q8N122
RPUSD4	Q96CM3	RRAGA	Q5VZM2, Q7L523	RRAGC	Q9HB90
RRAGD	Q9NQL2	RRAS	P10301	RRM1	P23921
RRM2	P31350	RRM2B	Q7LG56	RRN3	Q9NYV6
RRP1	P05386	RSF1	Q16629	RTF1	Q92541
RTN3	O95197	RUFY1	Q96T51	RWDD2B	Q9Y3V2
RXRB	P28702	S1PR3	Q99500	SACM1L	Q9NTJ5
SAE1	Q9UBE0	SAFB	Q15424	SAFB2	Q15424
SAMD8	Q96LT4	SAP130	Q15393	SAP30L	Q9HAJ7
SAR1B	Q9Y6B6	SARM1	Q6SZW1-1	SAT2	Q96QD8
SBF2	Q86WG5	SBSN	Q6UWP8	SCAMP1	O15126
SCAP	Q9NP31	SCFD1	Q8WVM8	SCLY	Q96I15
SCN3B	Q9NY72	SCN8A	Q9UQD0	SCN9A	Q15858
SCO2	O43819	SCOC	Q9UIL1	SCP2	P22307
SCRIB	Q14160	SDC3	O75056	SDCCAG8	Q86SQ7
SDE2	Q6IQ49	SDHAF1	A6NFY7	SDHAF2	Q9NX18
SDHB	P21912	SDSL	Q96GA7	SEC22A	Q96IW7
SEC23A	Q15436	SEC23IP	Q9Y6Y8	SEC24B	O95487
SEC31A	O94979	SEL1L	Q9UBV2	SEMA3A	Q14563
SEMA4A	Q9H3S1	SEMA4D	Q92854	SENP3	Q9H4L4
SEPHS2	Q99611	SEPSECS	Q9HD40	SEPTIN2	Q15019
SEPTIN7	Q16181	SERINC1	Q9NRX5	SERPINE1	P05121
SERPINE2	P07093	SETD1A	O15047, Q9UPS6	SETD1B	Q9UPS6
SETD2	Q9BYW2	SETD7	Q8WTS6	SETDB2	Q96T68
SF1	Q13285	SF3B1	O75533	SGCB	Q16585
SGF29	Q96ES7	SGK1	O00141	SGK3	Q96BR1
SGO2	Q562F6	SGTA	O43765	SH2B1	Q9NRF2-2
SH3BP4	Q9P0V3	SH3GL1	Q99961	SHC3	Q92529
SHOC2	Q9UQ13	SHPRH	Q149N8	SHTN1	A0MZ66
SIK1	O00567	SIKE1	Q9BRV8	SIN3A	Q96ST3
SIRT6	Q8N6T7	SKA2	Q8WVK7	SKI	Q5VTY9
SKIC3	Q6PGP7	SKIC8	Q9GZS3	SKIL	P12757
SLAMF7	Q9NQ25	SLC11A2	P49281	SLC12A4	Q9UP95

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SLC16A7	O60669	SLC17A5	Q9NRA2	SLC19A1	P41440
SLC19A2	O60779	SLC1A1	P43005	SLC20A1	Q8WUM9
SLC22A15	Q8IZD6	SLC22A3	O75751	SLC22A5	O76082
SLC25A13	Q9UJS0	SLC25A5	P05141, P12236	SLC26A1	Q9H2B4
SLC26A5	Q8TE54	SLC27A4	Q6P1M0	SLC29A1	Q99808
SLC29A2	Q14542	SLC35A3	Q9Y2D2	SLC35B2	Q8TB61
SLC35C1	Q96A29	SLC36A1	Q7Z2H8	SLC37A2	Q8TED4-2
SLC39A1	Q9NP94, Q9NY26	SLC39A10	Q9ULF5	SLC39A14	Q15043
SLC39A6	Q13433	SLC39A7	Q92504	SLC39A8	Q9C0K1
SLC3A1	Q07837	SLC3A2	P08195	SLC43A2	Q8N370
SLC44A2	Q8IWA5	SLC45A2	Q9UMX9	SLC45A3	Q96JT2
SLC46A1	Q96NT5	SLC4A2	P04920	SLC4A4	Q9Y6R1
SLC4A8	Q2Y0W8	SLC52A2	Q9HAB3	SLC5A10	A0PJK1
SLC66A1	Q6ZP29	SLC6A8	P48029	SLC6A9	P48067
SLC7A11	Q9UPY5	SLC7A5	Q01650	SLC7A8	Q9UHI5
SLC9A3	P48764	SLC9A7	Q96T83	SLC9B1	Q4ZJI4
SLCO4A1	Q96BD0	SLK	Q9H2G2	SLU7	O95391
SLX4	Q8IY92	SMAD5	Q99717	SMAD6	O43541
SMAD9	O15198	SMARCA2	P51531	SMARCA5	O60264
SMARCB1	Q12824	SMC2	O95347	SMC5	Q8IY18
SMC6	Q96SB8	SMG1	P62308	SMG5	Q9UPR3
SMIM20	Q8N5G0	SMO	Q9NWM0-3	SMPD2	O60906
SMUG1	Q53HV7	SMURF2	Q9HAU4	SMYD3	Q9H7B4, Q9NRG4
SNAP23	O00161	SNAPC3	Q92966	SNRNP48	Q6IEG0
SNRNP70	P08621	SNRPA	P09012	SNRPB	P14678
SNX5	Q9Y5X3	SNX6	Q9Y5X3	SOAT1	P35610
SOCS4	Q8WXH5	SOCS5	O75159	SOCS6	O14544
SORBS3	O60504	SORCS3	Q17R88	SORT1	Q99523
SOS1	Q07889	SOS2	Q07890	SOX13	Q9UN79
SOX9	P48436	SP100	P23497	SP3	Q9Y2H2
SPAST	Q9UBP0	SPATA13	Q96N96	SPC25	Q9HBM1
SPDL1	Q96EA4	SPHK1	Q9NYA1	SPHK2	Q9NRA0
SPIDR	Q14159	SPOCD1	Q6ZMY3	SPOPL	Q6IQ16
SPPL2A	Q8TCT8	SPR	P35270	SPRED2	Q7Z698
SPRY2	O43597	SPSB1	Q96BD6	SPSB2	Q99619
SPTAN1	Q13813	SPTBN2	O15020	SPTLC1	O15269
SPTLC3	Q9NUV7	SPX	Q15427	SRF	P11831
SRP54	Q05519	SRP72	O76094	SRPK1	Q96SB4
SRRT	Q9BXP5	SRSF11	Q05519	SS18	Q15532
SSNA1	O43805	SSR1	P43307	SSTR1	P30872
SSU72	Q9NP77	ST3GAL1	Q11201	ST3GAL4	Q11206
ST3GAL6	Q9Y274	ST6GAL1	P15907	ST6GALNAC4	Q9H4F1
ST7	Q9Y561	STAG2	Q8N3U4	STAM2	O75886
STAMBP	O95630	STARD13	Q9Y3M8	STARD8	Q92502
STAT1	P42224	STC2	O76061	STEAP3	Q658P3
STING1	Q86WV6	STN1	Q9Y6Q2	STOM	P27105
STON1	Q9Y6Q2	STRN	O43815	STT3B	Q8TCJ2
STUB1	Q9UNE7	STX10	O60499	STX16	O14662
STX17	P56962	STX3	Q13277	STX4	Q12846

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
STX5	Q13190	STXBP3	O00186	SUCLG2	Q96199
SUGT1	Q9Y2Z0	SULT1A1	P50225	SULT1A2	P50226
SULT4A1	Q9BR01	SUMO3	P55854	SUPT20H	Q8NEM7
SUPT3H	O75486	SUPT4H1	P63272	SUPT6H	Q7KZ85
SURF4	O15260	SYDE1	Q6ZW31	SYNE1	Q8NF91
SYNGAP1	Q96PV0, Q9UJF2	SYNJ1	O43426	SYNJ2	O15056
SYS1	Q8N2H4	SYT1	P21579, Q8NBV8	TAB2	Q9NYJ8
TAB3	Q8N5C8	TAF1	P21675, Q8IZX4	TAF1D	Q9H5J8
TAF2	Q6P1X5	TAF4B	Q92750	TAF6	P49848
TAF6L	Q9Y6J9	TAF9B	Q16594, Q9HBM6	TANK	Q92844
TAOK1	Q7L7X3	TAOK3	Q9H2K8	TAPBP	O15533
TARBP2	Q15633	TASOR	Q9UK61	TATDN2	Q93075
TAX1BP1	Q86VP1	TBC1D13	Q9NVG8	TBC1D16	Q8TBP0
TBC1D20	Q96BZ9	TBC1D7	Q9P0N9	TBC1D8B	Q0IIM8
TBCA	O75347	TBK1	Q9UHD2	TBRG4	Q969Z0
TCEA1	P23193	TCF12	Q99081	TCF7	P36402
TCIRG1	Q13488	TCP1	P17987	TEAD1	P28347
TECR	Q9NZ01	TEN1	Q86WV5	TENT4B	Q8NDF8
TERF1	P54274	TES	Q9NZU5	TET2	Q6N021
TEX264	Q9Y6I9	TFB1M	Q8WVM0	TFDP1	Q14186
TFPI	P10646	TGFA	P01135	TGFB2	P61812
TGFBR1	P36897	TGFBR3	Q03167	TGIF1	Q15583
TGM1	P22735	THBS1	P07996	THOC3	Q96J01
THOC6	Q86W42	THOP1	P52888	THRAP3	Q9Y2W1
THSD4	Q6ZMP0	THUMPD1	Q9NXG2	TIA1	P31483
TIAM2	Q8IVF5	TICAM2	Q86XR7	TIMM10	P62072
TIMM13	Q9Y5L4	TIMM22	Q9Y584	TIMM44	O43615
TJP1	Q07157	TKT	P29401	TLE1	Q04724
TLR1	Q15399, Q9Y2C9	TLR3	O15455	TLR6	Q9Y2C9
TMED10	P49755	TMED3	Q9Y3Q3	TMED5	Q9Y3A6
TMED7	Q9Y3B3	TMEM11	P17152	TMEM115	Q12893
TMEM126B	Q8IUX1	TMEM132A	Q24JP5	TMEM177	Q53S58
TMEM186	Q96B77	TMEM216	Q9P0N5	TMEM223	A0PJW6
TMEM30A	Q9NV96	TMEM87A	Q8NBN3	TMF1	P82094
TMOD2	Q9NZR1	TMOD3	Q9NYL9	TMSB4X	P62328
TMX3	Q96JJ7	TNC	P24821	TNFAIP8	O95379
TNFRSF14	Q92956	TNIP1	Q15025	TNIP2	Q8NFZ5
TNK2	Q07912	TNKS	O95271, Q9H2K2	TNKS2	Q9H2K2
TNPO1	Q92973	TNRC6A	Q8NDV7	TNRC6B	Q9UPQ9
TOLLIP	Q9H0E2	TOMM70	O94826	TOP2A	P11388
TOP2B	Q02880	TOP3A	Q13472	TOR1AIP1	Q5JTV8
TOR1B	O14657	TOR4A	Q9NXH8	TP53	P04637
TPCN1	Q9ULQ1	TPD52	Q9Y2Q9	TPD52L1	Q16890
TPGS1	Q6ZTW0	TPM1	P09493	TPP1	Q96AP0
TPP2	P29144	TPR	P12270	TPRN	Q4KMQ1
TPST2	O60704	TPX2	Q9ULW0	TRAF7	Q6Q0C0
TRAK1	Q9UPV9	TRAPPC11	Q7Z392	TRAPPC13	A5PLN9
TRAPPC2L	Q9UL33	TRAPPC3	O43617	TRAPPC5	Q8IUR0
TRAPPC6B	O75865, Q86SZ2	TRAPPC8	Q9Y2L5	TRAPPC9	Q96Q05

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TRDMT1	O14717	TRIAP1	O43715	TRIM2	Q9C040
TRIM25	Q14258	TRIM27	P14373	TRIM28	Q13263
TRIM3	P53365	TRIM32	Q13049	TRIM36	Q9NQ86
TRIM41	Q8WV44	TRIM47	Q8WV44	TRIO	O75962
TRIP11	Q15643	TRIP12	Q14669	TRIP4	Q15650
TRMT12	Q53H54	TRMT13	Q9NUP7	TRMU	O75648
TRNT1	Q96Q11	TRPC1	P48995	TRUB2	O95900
TSC22D3	Q99576	TSC22D4	Q99576	TSEN54	Q7Z6J9
TSNAX	Q99598	TSPAN14	Q8NG11	TSPAN15	O95858
TSR3	Q9UJK0	TST	Q15477	TTC21B	Q7Z4L5
TTLL12	Q14166	TTLL3	A6PVC2, Q9Y4R7	TTLL7	Q6ZT98
TTN	Q8WZ42	TTYH3	Q9C0H2	TUBA1A	Q71U36
TUBA4A	P68366	TUBB2A	Q13885, Q9BVA1	TUBB2B	Q13885, Q9BVA1
TUBB4A	P04350	TUBGCP5	Q96RT8	TULP4	Q9NRJ4
TUT4	Q5TAX3	TWF1	Q12792	TXLNA	P40222
TXNDC11	Q6PKC3	TXNIP	Q9H3M7	TXNL4A	P83876
TYSND1	Q2T9J0	TYW1	Q9NV66	U2SURP	O15042
UACA	Q9BZF9	UAP1	Q16222	UBA2	Q9UBT2
UBA3	Q8TBC4	UBA5	Q9GZZ9	UBA52	P62979, P62987
UBA6	A0AVT1	UBA7	P22314, P41226	UBAC1	Q9BSL1
UBE2D3	P61077	UBE2D4	Q9Y2X8	UBE2I	P63279
UBE2J2	Q8N2K1	UBE2K	P61086	UBE2Q1	Q7Z7E8
UBE2Q2	Q8WVN8	UBE2W	Q96B02	UBE3A	Q05086
UBE4A	Q14139	UBL4A	P11441	UBN1	Q9NPG3
UBOX5	O94941	UBQLN1	Q9UMX0	UBQLN2	Q9UHD9
UBR1	Q8I WV7	UBR2	Q8I WV7, Q8I WV8	UBR4	Q5T4S7
UBXN7	O94888	UCHL3	P15374	UCHL5	Q9Y5K5
UCK1	Q9HA47	UCN	P55089	UCP2	P55851, P55916
UFL1	O94874	UGCG	Q16739	UGGT1	Q9NYU2
UGGT2	Q9NYU1	UGP2	Q16851	UHMK1	Q8TAS1
UIMC1	Q96RL1	ULK3	Q6PHR2	UNC119B	A6NIH7
UNC13B	O14795	UNK	Q9H9P5	UPF1	Q92900
UPF2	Q9HAU5	UPP1	Q16831	UQCC3	Q6UW78
UQCC5	Q8WVI0	UQCC6	Q69YU5	UQCR11	O14957
UQCRC1	P31930	UQCRC2	P22695	UQCRFS1	P47985
USF1	Q8NFU3	USF2	Q15853	USO1	O60763
USP11	P51784	USP12	O75317, Q86T82	USP13	Q92995
USP14	P54578	USP15	Q9Y4E8	USP16	Q9Y5T5
USP19	O94966	USP2	O75604	USP21	Q9UK80
USP24	Q9UPU5	USP25	Q9UHP3	USP28	Q96RU2
USP30	Q70CQ3	USP33	Q8TEY7	USP34	Q70CQ2
USP44	Q70CQ1, Q9H0E7	USP47	Q96K76	USP48	Q86UV5
USP8	P40818	USP9X	Q93008	UTP14C	Q5TAP6
UTRN	P46939	UTY	O14607	UVRAG	Q9P2Y5
UXT	Q9UBK9	VAC14	Q08AM6	VAMP2	P63027
VAMP4	O75379	VAMP7	P51809	VANGL1	Q8TAA9
VASH1	Q7L8A9	VAT1	Q99536	VBP1	P61758
VCL	P18206	VDAC3	Q9Y277	VEGFB	P49765
VLDLR	P98155	VPS11	Q9H270	VPS18	Q9P253

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
VPS29	Q9UBQ0	VPS33A	Q96AX1	VPS36	Q86VN1
VPS37B	A5D8V6, Q9H9H4	VPS45	Q9NRW7	VPS4B	O75351
VPS54	Q9P1Q0	VTA1	Q9NP79	WAPL	Q7Z5K2
WASF1	Q92558	WDCP	Q9H6R7	WDR11	Q8WWQ0, Q9BZH6
WDR18	Q9BV38	WDR24	Q96S15	WDR35	Q9P2L0
WDR4	P57081	WDR48	Q8TAF3	WDR6	Q9NNW5
WDR70	Q9NW82	WDR75	Q8IWA0	WDR81	Q562E7
WDR82	Q6UXN9	WDR83	Q9BRX9	WDR91	A4D1P6
WHAMM	Q8TF30	WHRN	Q9P202	WLS	Q5T9L3
WNT2B	Q93097	WNT5A	P41221	WRAP53	Q9BUR4
WRN	Q14191	WSB1	Q9Y6I7	WWC1	Q8IX03
WWP1	Q9H0M0	WWP2	O00308	WWTR1	Q9GZV5
XBP1	P17861-2	XDH	Q06278	XPA	P23025
XPO1	O14980	XRN1	Q8IZH2	YBX1	P67809
YEATS2	Q9ULM3	YES1	P07947	YME1L1	Q96TA2
YRDC	Q86U90	YTHDC1	Q96MU7	YTHDC2	Q9H6S0
ZBED1	O96006	ZBTB17	Q13105	ZC3H18	Q86VM9
ZC3H3	Q8IXZ2	ZC3H6	Q8N5P1	ZC3HAV1	Q7Z2W4
ZCCHC8	Q6NZY4	ZDHHC2	Q9UIJ5	ZDHHC20	Q5W0Z9
ZDHHC21	Q8IVQ6	ZDHHC7	Q9NXF8	ZDHHC8	Q9ULC8
ZEB1	P37275	ZFP28	Q8NHY6	ZFP90	Q8TF47
ZKSCAN1	P17029	ZMIZ1	Q92520	ZMYM2	Q9UBW7
ZNF10	P21506	ZNF101	Q8IZC7	ZNF12	P17014
ZNF160	Q9HCG1	ZNF175	Q9Y473	ZNF189	O75820
ZNF195	O14628	ZNF202	O95125	ZNF205	O95201
ZNF217	O75362	ZNF23	P17027	ZNF233	Q9NZL3
ZNF25	P17030	ZNF253	O75346	ZNF263	O14978
ZNF266	Q14584	ZNF268	Q96BR6	ZNF324B	Q6AW86
ZNF33B	Q06730	ZNF354C	Q86Y25	ZNF37A	P17032
ZNF398	Q9BS31	ZNF41	P51814	ZNF426	Q9BUY5
ZNF429	Q86V71	ZNF432	O94892	ZNF433	Q8N7K0
ZNF441	Q8N8Z8	ZNF484	Q5JVG2, Q8N8J6	ZNF506	Q5JVG8
ZNF528	Q3MIS6	ZNF529	Q6P280, Q8NDQ6	ZNF546	Q86UE3
ZNF558	Q96NG5	ZNF561	Q8N587	ZNF584	Q8IVC4
ZNF596	Q8TC21	ZNF598	Q86UK7	ZNF616	Q08AN1
ZNF621	Q6ZSS3	ZNF626	Q68DY1	ZNF628	P51786
ZNF655	Q8N720	ZNF668	Q96K58	ZNF678	Q5SXM1
ZNF681	Q96N22	ZNF689	Q96CS4	ZNF70	Q9UC06
ZNF708	P17019	ZNF721	Q8TF20	ZNF737	O75373
ZNF74	Q16587	ZNF770	Q6IQ21	ZNF771	Q7L3S4
ZNF772	Q68DY9	ZNF785	A8K8V0	ZNF786	Q8N393
ZNF791	Q3KP31	ZNF793	Q6ZN11	ZNF92	Q03936
ZSCAN22	Q68DY9	ZSWIM8	A7E2V4	ZW10	O43264
Input	ChEBI Id				
CIT	16947				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ABCA1	ENSG00000165029	ABCC3	ENSG00000108846	ABCF2	ENSG000000033050
ACADM	ENSG00000117054	AGT	ENSG00000135744	AKT1	ENSG00000142208
ANKRD1	ENSG00000148677	ANXA3	ENST00000264908	ANXA4	ENST00000394295
APAF1	ENSG00000120868	ARC	ENSG00000198576	ARFGAP1	ENSG00000101199
ARL4C	ENSG00000188042, ENST00000339728	ASAH1	ENSG00000104763	ASNS	ENSG00000070669
ATM	ENSG00000149311, ENST00000278616	ATP6V0A1	ENSG00000033627	ATP6V0B	ENSG00000117410
ATP6V0C	ENSG00000185883	ATP6V0D1	ENSG00000159720	ATP6V0E1	ENSG00000113732
ATP6V1A	ENSG00000114573	ATP6V1B2	ENSG00000147416	ATP6V1C1	ENSG00000155097
ATP6V1E1	ENSG00000131100	ATP6V1F	ENSG00000128524	ATP6V1G1	ENSG00000136888
ATP6V1H	ENSG00000047249	AXIN1	ENSG00000103126	BACH1	ENSG00000156273
BAX	ENSG00000087088	BCL2L1	ENSG00000171552	BCL2L11	ENSG00000153094
BDP1	ENSG00000072135	BHLHE41	ENSG00000123095	BMP4	ENSG00000125378
BNIP3L	ENSG00000104765	BOLA2B	ENSG00000183336	BRWD1	ENSG00000185658
CA9	ENSG00000107159	CAPZA1	ENSG00000116489	CAT	ENSG00000121691
CBX5	ENSG00000094916	CCN1	ENSG00000145386	CCN2	ENSG00000118523
CCNA2	ENSG00000145386	CCNB2	ENSG00000157456	CDC25B	ENSG00000101224
CDH2	ENSG00000170558	CDK5	ENSG00000164885	CEBPB	ENSG00000172216
CHD1	ENSG00000153922, ENST00000284049.7	CITED2	ENSG00000164442	CLOCK	ENSG00000134852
CNN2	ENSG00000064666	CNOT6L	ENST00000264903	CPEB1	ENST00000563519
CPS1	ENSG00000021826	CPT1A	ENSG00000110090	CREB1	ENSG00000118260
CRY2	ENSG00000121671	CSF1	ENSG00000184371	CTSD	ENSG00000117984
CXXC1	ENSG00000154832	CYP1A1	ENSG00000140465	DBP	ENSG00000105516
DDB2	ENSG00000134574	DDIT4	ENSG00000168209	DICER1	ENSG00000100697
DDIT4	ENSG00000134574	DLGAP5	ENSG00000126787	DNAJB9	ENSG00000128590
DKK1	ENSG00000107984	EDEM1	ENSG00000134109	EEPD1	ENSG00000122547
DNAJC3	ENSG00000102580	EGR2	ENSG00000122877	EHF	ENST00000257831
EGF	ENSG00000138798	ELL2	ENST00000237853	ELOVL2	ENSG000000012660
EIF2AK2	ENSG00000055332	ENPP2	ENST00000427067	EPAS1	ENSG00000116016
ELOVL5	ENSG000000012660	ERBB4	ENSG00000178568	ESRRA	ENSG00000173153
EPHA4	ENSG00000116106	EXTL3	ENSG000000012232	FANCD2	ENSG00000144554, ENST00000419585
EXTL2	ENSG00000162694	FASN	ENSG00000169710	FGF2	ENSG00000138685
FANCI	ENSG00000140525, ENST00000310775	FKBP4	ENSG00000004478	FN1	ENSG00000115414
FKBP14	ENSG00000106080	FTL	ENST00000331825	GBP1	ENSG00000117228
FTH1	ENST00000273550	GBP3	ENSG00000117226	GCLM	ENSG00000023909
GBP2	ENSG00000162645	GEM	ENSG00000164949	GFPT1	ENSG00000198380
GDNF	ENSG00000168621	GPRC5A	ENST000000014914	GRB10	ENSG00000106070
GPR143	ENSG00000101850	GSK3A	ENSG00000105723	GSTO1	ENSG00000148834
GREB1	ENSG00000196208, ENST00000381486.6	HIF1A	ENSG00000100644	HLA-A	ENSG00000206503
HDGF	ENSG00000143321	HMOX1	ENSG00000100292	HNRNPA2B1	ENSG00000122566
HLA-H	ENSG00000206341	HNRNPF	ENSG00000169813	HSP90AA1	ENSG00000080824
HNRNPDL	ENSG00000152795	HSPA5	ENSG00000044574	HSPA6	ENSG00000173110
HSP90B1	ENSG00000166598	HSPD1	ENSG00000144381	HTRA2	ENSG00000115317
HSPB1	ENSG00000106211	ID3	ENSG00000117318	ID4	ENSG00000172201
ID1	ENSG00000125968	IL1R1	ENSG00000115594	IL1RN	ENSG00000136689
IFIT2	ENSG00000119922	IRAK1	ENSG00000184216	IRF3	ENSG00000126456
IL6R	ENSG00000160712				

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
ITCH	ENSG00000078747	ITGA4	ENSG00000115232	ITGB1	ENSG00000150093
KDM1B	ENST00000297792	KDM5A	ENST00000399788	KDM6B	ENSG00000132510
KLF15	ENSG00000163884	KLHDC3	ENSG00000124702	KRT18	ENST00000388835
KRT7	ENST00000331817	KRT8	ENST00000293308	LCP1	ENSG00000136167
LGALS3	ENSG00000131981	LIPE	ENSG00000079435	LONP1	ENSG00000196365
MAOA	ENSG00000189221	MAP4K4	ENSG00000071054	MAPK6	ENSG00000069956
MDM2	ENSG00000135679	ME1	ENSG00000065833	MED13	ENST00000397786
MET	ENSG00000105976	MGLL	ENSG00000074416	MINK1	ENSG00000141503
MITF	ENSG00000187098	MLH1	ENSG00000076242	MPZ	ENSG00000158887
MT-ATP6	ENST00000361899	MT-ATP8	ENST00000361851	MT-ND1	ENST00000361390
MT-ND2	ENST00000361453	MT-ND3	ENST00000361227	MT2A	ENSG00000125148
MYC	ENST00000613283	MYDGF	ENSG00000074842	MYO5A	ENSG00000197535
NAB2	ENSG00000166886	NDRG1	ENSG00000104419	NOP2	ENSG00000111641
NPAS2	ENSG00000170485	NPPA	ENSG00000175206	NR1D1	ENSG00000126368
PAX8	ENSG00000125618	PBX1	ENSG00000185630	PCBP4	ENSG00000090097
PDCD4	ENSG00000150593	PDIA5	ENSG00000065485	PDIA6	ENSG00000143870
PDPN	ENSG00000162493	PER1	ENSG00000179094	PER2	ENSG00000132326
PIK3R1	ENSG00000145675	PIM1	ENSG00000137193	PLIN2	ENSG00000147872
PLK2	ENSG00000145632	PMP22	ENSG00000109099	POLRMT	ENSG00000099821
POU2F1	ENSG00000143190	POU3F2	ENST00000328345	PPARGC1A	ENSG00000109819
PPP1R15A	ENSG00000087074, ENST00000200453	PREB	ENSG00000138073	PSAP	ENSG00000122852, ENSG00000185303
PTEN	ENST00000371953	PTPN1	ENSG00000196396	PTPN11	ENSG00000179295
PTPN12	ENSG00000127947	PTPN13	ENSG00000163629	PTPN23	ENSG00000076201
PTPN4	ENSG00000088179	RAB27A	ENSG00000069974	RALA	ENSG00000006451
RGL1	ENSG00000143344	RNU1-1	ENSG00000206652	ROBO1	ENSG00000169855
RORA	ENSG00000069667	RRM2	ENSG00000171848	RRM2B	ENSG00000048392
SCO2	ENSG00000130489	SCP2	ENSG00000175215	SEC31A	ENSG00000138674
SERPINE1	ENSG00000106366	SGK1	ENST00000237305	SLC4A4	ENST00000264485
SLC7A11	ENSG00000151012	SMAD6	ENSG00000137834	SOCS4	ENSG00000180008
SOX9	ENST00000245479	SP100	ENSG00000067066	SRF	ENSG00000112658
SSR1	ENSG00000124783	STAT1	ENSG00000115415	STEAP3	ENSG00000115107
STT3B	ENSG00000163527	TATDN2	ENSG00000157014	TCP1	ENSG00000120438
TET2	ENST00000265149	TFB1M	ENSG00000029639	TGFA	ENSG00000163235
TGFBR3	ENSG00000069702, ENST00000212355	THBS1	ENSG00000137801	TIAM2	ENSG00000146426
TJP1	ENSG00000104067	TKT	ENSG00000163931	TOP2A	ENSG00000131747
TP53	ENSG00000141510	TPP1	ENSG00000166340	TPST2	ENSG00000128294
TRIAP1	ENSG00000170855	TRIM2	ENSG00000109654	TRIM25	ENSG00000121060
TRIM3	ENSG00000110171	TUT4	ENSG00000134744	UXT	ENSG00000126756
WWC1	ENSG00000113645	XBP1	ENST00000216037, ENST00000344347	ZBTB17	ENSG00000116809
ZEB1	ENSG00000148516				

Interactors (3188)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ABCA1	O95477	P02647	ABCD3	P28288	P40855
ABCF3	Q9NUQ8	P14373	ABHD15	Q6UXT9	P21673
ABHD3	Q8WU67	P04201	ABHD6	Q9BV23	Q9P296
ABI1	Q8IZP0	Q07889	ABI2	Q9NYB9-2	Q13177

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ABLM1	O14639	Q9H257	ABLM3	O94929-2	Q53FT3
ABRAXAS1	Q6UWZ7	P53611	ACAA2	P42765	P0DTC3
ACAD11	Q709F0	Q14493	ACADM	P11310	P0DTC5
ACBD6	Q9BR61	P12931	ACCS	Q96QU6	P14927
ACOX2	Q99424	P63172	ACSL3	O95573	Q8NEZ3, Q96RY7
ACSL4	O60488	Q9H2J7	ACTMAP	Q5BKX5-3	Q9NQ94
ACTN2	P35609	Q92934	ACTR10	Q9NZ32	Q8N5V2
ACTR1B	P42025	Q8N5V2	ACTR2	P61160	Q14247
ACTR3	P61157	O00401	ACTR5	Q9H9F9	O96019
ACTR6	Q9GZN1	Q15274	ACY1	Q03154	P55735
ADAM10	O14672	P12931	ADAM9	Q13443	Q99962
ADAMTS1	Q9UHI8	P08833	ADAMTSL3	P82987	Q7Z3S9
ADARB1	P78563-4	P19525	ADAT3	Q96EY9	Q8WUP2
ADCY9	O60503	P15514	ADGRG3	Q86Y34	Q99437
ADI1	Q9BV57	P50281	ADIPOR1	Q96A54	P25103
ADIPOR2	Q86V24	Q9P0P0	ADORA2B	P29275	P29274
ADSS2	P30520	Q8NB12	AEN	Q8WTP8-2, Q8WTP8	Q13685
AFDN	P55196	P12931	AFF4	Q9UHB7	P55199
AFG2B	Q9BVQ7	P25445	AFG3L2	Q9Y4W6	P42858
AGAP3	Q96P47	P61981	AGFG1	P52594	P51809
AGFG2	O95081	Q9NQZ5	AGT	P01019	P11021
AHCTF1	Q8WYP5	P06748	AHCYL1	O43865	P23526
AHDC1	Q5TGY3	Q13185	AHNAK	Q09666	P18827
AIDA	Q96BJ3	P25963	AIFM1	O95831	P62937
AIG1	Q9NVV5-2	O75396	AIMP2	Q13155	P03508
AK3	Q9UIJ7	P02649	AK4	P27144	P13501
AKAP1	Q92667	P13861	AKAP11	Q9UKA4	P51530
AKAP7	Q9P0M2	P17612	AKAP8	O43823	P13861
AKAP9	Q99996-2	Q9BTA9	AKR1A1	P14550	O76024
AKR1B10	O60218	Q13085	AKT1	P31749	P54278
AKT1S1	Q96B36	P31946	AKT2	P31751	P31749
AKT3	Q9Y243	P07900	ALDH3B1	P43353	P35813
ALG1	Q9BT22	P13569	ALG11	Q2TAA5	O43914
ALG13	Q9NP73-4	Q01974	ALG2	O75340	Q8WUM4
ALKBH8	Q96BT7-2	P55081	ALMS1	Q8TCU4	Q8IWZ6
ALOX12	P18054	P02545	ALOXE3	Q9BYJ1	Q8NFP9
AMBRA1	Q9C0C7, Q9C0C7-3	O75385	AMDHD2	Q9Y303-2	P55212
AMIGO1	Q86WK6	Q9H2H9	AMPD2	Q01433	P18827
ANAPC2	Q9UJX6	O60566, Q12834	ANGPTL2	Q9UKU9	P11021
ANGPTL7	O43827	P17987	ANK2	Q01484	Q9P2E9
ANK3	Q12955	Q15796	ANKIB1	Q9P2G1	P22736
ANKMY2	Q8IV38	Q15185	ANKRD1	Q9CR42	P19338
ANKRD17	O75179	P55075	ANKRD28	O15084	P53350
ANKRD40	Q6AI12	P23526	ANKRD44	Q8N8A2	P46531
ANKS1A	Q49AR9	P50458	ANKS1B	Q7Z6G8-3	P61586
ANKS6	Q68DC2	O15155	ANLN	Q9NQW6	P07948
ANOS1	P23352	P11362	ANP32E	Q9BTT0	P52294
ANTKMT	Q9BQD7	O14964	ANXA3	P12429	P13569
ANXA4	P09525	P55212	ANXA5	P08758	Q96CV9

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ANXA6	P08133	P10636-8	ANXA8	P13928	Q00403
AP1G1	P22892	Q14677	AP1S2	P56377	P56539
AP2B1	P63010	P29353	AP3B1	O00203	P0DTC4
AP3M1	Q9Y2T2	Q9UBH0	APAF1	O14727	P99999
APBB2	Q92870-2	Q969S8	APC	P25054	P49841
APEH	P13798	P13798	APEX1	P27695	Q09472
API5	Q9BZZ5	Q13838	APLF	Q8IW19	Q86WJ1
APLP1	P51693	P05067-4	APOL6	Q9BWW8	Q9UHA4
APOOL	Q6UXV4	P46059	APPBP2	Q92624	P23526
APPL1	Q9UKG1	O14908	AQR	O60306	O75152
ARB2A	Q8WUF8	P63208	ARC	O60936	Q14790
ARF5	P84085	Q9UNA3	ARFGAP1	Q9NP61	P17302
ARFGAP2	Q8N6H7	P05155	ARFGAP3	Q9NP61	P17302
ARFGEF1	Q9Y6D6	Q9NRM0	ARFGEF3	Q5TH69	P15907
ARFIP1	P53367	O60906	ARFIP2	P53365	Q92730
ARHGAP12	Q8IWW6	P60709	ARHGAP21	Q5T5U3	P29692
ARHGAP31	Q2M1Z3	Q9Y490	ARHGAP33	O14559	P06241
ARHGAP42	A6NI28	P46108	ARHGAP45	Q92619	Q08379
ARHGEF11	O15085	O43157	ARHGEF12	Q9NZN5	Q9UM54
ARHGEF26	Q96DR7	P31327	ARHGEF3	Q9NR81	P14373
ARHGEF39	Q8N4T4	Q2T9J0	ARHGEF4	Q9NR80-3	P25054
ARHGEF6	Q15052	Q13177	ARID2	Q68CP9	P52294
ARID4A	P29374	Q13547	ARID4B	Q4LE39-3	Q96JC9
ARID5A	Q03989	P40937	ARID5B	Q14865	Q92769, Q13547
ARIH1	Q9Y4X5	Q9BYZ6	ARK2N	Q96B23	P63165
ARL2	P36404	Q9BTW9	ARL3	P36405	P36404
ARL4C	P56559	P52306	ARL6IP5	O75915	P25103
ARMC6	Q6NXE6	O14975	ARMC8	Q8IUR7	Q6VN20, Q96S59
ARMC9	Q7Z3E5	Q8N183	ARMCX1	Q9P291	Q92917, Q9UBB9
ARMCX5	Q6P1M9	P09917	ARMH4	Q86TY3	Q9NPF0
ARPP19	P56211	P05067	ARRB1	P49407	P12931
ARRDC1	Q8N5I2	Q96PU5	ARRDC3	Q96B67	P51668
ARRDC4	Q8NCT1	Q96PU5	ARSB	P15848	P01185
ARSI	Q5FYB1	P11021	ASAH1	Q13510, Q13510-2	Q13285
ASAP1	Q9ULH1	P12931	ASAP2	O43150	P19174
ASB13	Q8W XK3	Q93034	ASB3	Q9Y575	Q93034
ASCC3	Q8N3C0	P63244	ASF1A	Q9Y294	P49736
ASH1L	Q9NR48	P05067	ASIC2	Q16515	Q9NRD5
ASIC3	Q9UHC3	Q14160	ASMTL	O95671	Q96CV9
ASPH	Q12797	Q9NWR8, Q9H4I9, Q8NE86	ASPM	Q8IZT6	P11802
ASPSR1	Q9BZE9	P55072	ATE1	O95260	P01185
ATF6	P18850	Q8NH Y2	ATF7IP	Q6VMQ6	Q15047
ATG101	Q9BSB4	O75385	ATG16L2	Q8NAA4	Q9H1Y0
ATG2A	Q2TAZ0	Q9GZQ8	ATG2B	Q96BY7	Q13309
ATG4A	Q8WYN0	Q9GZQ8	ATG9A	Q7Z3C6	Q96QE2
ATL3	Q6DD88	Q92911	ATM	Q13315	Q9NZQ7
ATN1	Q86V38	P23588	ATOSA	Q32MH5	P14373
ATP11B	A0A0C4DG94	A6NI15	ATP13A1	Q9HD20	Q96FT7
ATP1A1	P05023	P12931	ATP1B1	P05026	P21926

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATP1B3	P54709	P27824	ATP2B4	P23634	P01258
ATP2C1	P98194	P80370	ATP5F1D	P30049	P23786
ATP6AP1	Q15904	P27449, P61421	ATP6V0A1	Q93050	Q15904
ATP6V0A2	Q9Y487	Q15904	ATP6V0B	Q99437	P22732
ATP6V0C	P27449	Q15904	ATP6V0D1	P61421	Q15904
ATP6V0E1	O15342	P05090	ATP6V1A	P38606	Q9UI12
ATP6V1B2	P21281	Q9UI12	ATP6V1C1	P21283	P21281, P38606
ATP6V1C2	Q8NEY4	O75348, P21283, P21281, P36543, P38606	ATP6V1E1	P36543	Q9UI12
ATP6V1F	Q16864	Q9UI12	ATP6V1G1	O75348	P0C6X7
ATP6V1H	Q9UI12	P42772	ATP9A	O75110	Q9NWB7
ATRX	P46100	P84243, Q9UER7	ATXN1	P54253	Q6VY07
ATXN10	Q9UBB4	Q12891	ATXN7	O15265	Q8N2S1
ATXN7L2	Q5T6C5	P43897	ATXN7L3	Q14CW9	Q92831
AUP1	Q9Y679	P27824	AVPI1	Q5T686	A6NIH7
AXIN1	O15169	Q93008	AZI2	Q9QYP6	Q13114
B3GALT6	Q96L58	P40855	B3GLCT	Q6Y288	P13671
B4GALT3	O60512	P27824	B4GALT5	O43286	P16410
B4GAT1	O43505	P17302	BABAM1	Q9NWW8	P38398
BACE2	Q9Y5Z0	P10997	BACH1	O14867	O75330
BAD	Q92934	Q07817	BAG6	P46379-2	P10451
BAIAP2	Q9UQB8-6	Q13322	BARD1	Q99728	Q05048
BASP1	P80723	P10636-8	BAX	Q07812	P55957
BAZ1A	Q9NRL2	O60264	BAZ2B	Q9UIF8	Q96GX9
BBS1	Q8NFJ9	P68104	BBS10	Q8TAM1	Q8IWZ6
BBS4	Q96RK4	Q99814	BBX	Q8WY36	Q13547
BCAN	Q96GW7	P07948	BCAP29	Q9UHQ4	P51572
BCAS2	O75934	Q02548	BCDIN3D	Q7Z5W3	Q9NPC8
BCKDHA	P12694	P06746	BCKDHB	P21953	P11182
BCL10	O95999	O95999, Q9UDY8	BCL2L1	Q07817	Q9C000
BCL2L11	O43521	Q92934	BCL7A	Q4VC05	Q9H8M2
BCL7B	Q9BQE9	Q969G3, Q12824	BCL9L	Q67FY2	P35222
BCLAF1	Q9NYF8	O75152	BDH1	Q02338	P46059
BDP1	P46678	Q9UL45	BEND3	Q5T5X7	Q13547
BEX1	Q9HBH7	Q9NRD5	BEX2	Q9BXY8	P19883
BHLHE41	Q99PV5, Q9C0J9	Q16665	BICD1	Q96G01	P49841
BIRC2	Q13490	Q13546, Q9Y572	BIRC3	Q13489	Q13546, Q9Y572
BIRC6	Q9NR09	P53350	BIVM	Q86UB2	Q13114
BLOC1S3	Q6QNY0	Q9UL45, Q96EV8, P78537	BLOC1S4	Q9NUP1	Q9UL45, Q96EV8, P78537
BLOC1S6	Q9UL45	Q9NUP1, Q9UL45, Q96EV8, P78537, Q6QNY0	BLTP2	Q14667	P60709
BLTP3A	Q6BDS2	Q9Y3M2	BLTP3B	A0JNW5	P31946
BLZF1	Q9H2G9	Q9P0N9	BMERB1	Q96MC5	Q8WZA2
BMP1	P13497	O60216	BMP4	P12644	Q9UMX1
BMPR1A	P36894	P12644	BMS1	Q14692	P62753
BNIP2	Q12982	Q9NRZ7	BNIP3	Q12983	O60313
BNIP3L	O60238	Q9NQ11	BOD1	Q96IK1-2	P68104
BOD1L1	Q8NFC6	P21741	BOLA1	Q9Y3E2	Q86SX6
BOLA2B	Q9H3K6	P19320	BRAF	P15056	Q12967

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BRCC3	P46736	P38398	BRD2	P25440	Q13761
BRI3	O95415	P26572	BRINP1	O60477	P09651
BRIP1	Q9BX63	P54278	BROX	Q5VW32	P23193
BRPF1	P55201	P62805	BRPF3	Q9ULD4-2	P11021
BRWD1	Q9NSI6-4	P07550	BSG	P35613	O15228
BTAF1	O14981	P46059	BTBD2	Q9BX70	P40692
BTBD6	Q96KE9	Q9UER7	BUB1	O43683	P51530
BUB1B	O60566	Q13257	BYSL	Q13895	Q96MT3
C11orf54	Q9H0W9-3	Q13153	C11orf65	Q8NCR3	P12830
C12orf57	Q99622	O14907	C14orf119	Q9NWQ9	P61587
C16orf87	Q6PH81	Q9UM11	C19orf25	Q9UFG5	Q12981, Q9NZ43, Q9P2W9, O75396
C1QTNF5	Q9BXJ0	P43080	C1S	P09871	P05155
C1orf226	A1L170	P53350	C2CD5	Q86YS7	P24941
C2orf49	Q9BVC5	O76024	C3orf38	Q5JPI3	P11142
C4orf33	Q8N1A6	P55212	C5orf22	Q49AR2	P13725
C6orf141	Q5SZD1	Q9NYZ3	C6orf47	O95873	Q13257
C9orf85	Q96MD7	P46736, Q9NWW8	CA9	Q16790	P21291
CAB39	Q9Y376	Q15831	CABLES1	Q8TDN4	P80370
CACFD1	Q9UGQ2	Q96T60	CACTIN	Q8WUQ7	P67870
CAD	O76075	O00273	CALCOCO2	Q13137	P29353
CALD1	Q05682	Q13813	CALM2	P62161	P52888
CAMKK1	Q8N5S9	Q9Y265, Q9Y230	CAMSAP2	Q08AD1	P29692
CAND1	Q86VP6	Q93034, Q9UBF6	CANT1	Q8WVQ1	Q96JA1
CANX	P27824	Q15904	CAP2	P40123	P23528
CAPN10	Q9HC96	Q16595	CAPN15	O75808	Q9NWF9
CAPN7	Q9Y6W3	Q9Y6D9	CAPRIN2	Q6IMN6	O75197
CAPZA1	P47753, P13127	Q9Y5K6	CARD11	Q9BXL7	O95999, Q9UDY8
CARD19	Q96LW7	O95999	CASK	O14936	Q13009
CASKIN1	Q8VHK2, Q8WXD9	O14936	CASP3	P42574	O60551
CASP4	P49662	P51878	CAST	O15446	O15160, O95602, P19388, P61218
CAT	P04040	Q2T9J0	CAV2	P51636	Q15392
CAVIN1	Q6NZI2	Q9Y6D9	CAVIN2	O95810	Q6NZI2
CBARP	Q8N350	P36873	CBLB	Q13191	Q13882
CBR1	P16152	P00533	CBR3	O75828	P16152
CBX5	P45973	Q8N9R8	CBX6	O95503	P11940
CBX8	Q9HC52	Q96RU7	CCAR2	Q8N163	P48431
CCDC106	Q9BWC9	P40429	CCDC107	Q8WV48	Q12981, Q9P2W9
CCDC112	Q8NEF3-2	Q9Y2T1	CCDC124	Q96CT7	P06748
CCDC125	Q86Z20	Q9HAQ2	CCDC127	Q96BQ5	P41208
CCDC136	Q96JN2-2	O75604	CCDC137	Q6PK04	P78563
CCDC14	Q49A88-3	P48730	CCDC186	Q7Z3E2	P0DTC6
CCDC24	Q8N4L8	O14561	CCDC25	Q86WR0	Q9H2D1
CCDC40	Q4G0X9	Q9HBG6	CCDC43	Q96MW1, Q86WV7	Q96T60
CCDC47	Q96A33	Q92985	CCDC50	Q8IVM0	Q13546
CCDC51	Q96ER9	P17900	CCDC6	Q16204	Q13164
CCDC85A	Q96PX6	P51955	CCDC88A	Q3V6T2	Q13257
CCDC90B	Q9GZT6	Q92993	CCDC97	Q96F63	P18846
CCN1	P20248	Q13415	CCN2	P29279	P98164

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCN3	P48745	O43559	CCN5	O76076	P49639
CCNA2	P20248	Q13415	CCNB2	O95067	P06493
CCNC	P24863	Q15648	CCNE2	O96020	Q08999
CCNH	P51946	Q15669	CCNT2	O60583	Q9UHB7
CCP110	O43303	Q8IW35	CCSER2	Q9H7U1	P61981
CCT4	P50991	Q13882	CCT6A	P40227	Q8NCC3
CCT8	P50990	Q13882	CD109	Q6YHK3	P23946
CD200	P41217	Q8TD46	CD2AP	Q9Y5K6	Q8WUM4
CD2BP2	O95400	P36873	CD46	P15529	P78504
CD63	P08962	P05556	CD9	P21926	P05556
CDAN1	Q8IWIY9	Q9Y294	CDC14B	O60729	P43246
CDC16	Q13042, Q8R349	P51955	CDC25B	P30305	P31946
CDC26	Q99JP4, Q8NHZ8	P51955	CDC27	P30260	P51955
CDC34	P49427	Q13616	CDC37L1	Q7L3B6	P07900
CDC40	O60508	P62306, P14678, P62304	CDC42BPA	Q5VT25	P60953
CDC42EP2	O14613	P60953	CDC73	Q6P1J9	Q9P210, Q12996, Q95639, Q10570
CDCA2	Q69YH5	P36873	CDH2	P19022	O60716
CDIP1	Q9H305	O43561-2	CDIPT	O14735	Q8N9R8
CDK11A	Q9UQ88-1	O96017	CDK11B	P21127	Q13153
CDK17	Q00537	P06493	CDK19	Q9BWU1	Q15648
CDK2AP1	O14519	P52294	CDK2AP2	O75956	Q92769
CDK4	P11802	P42772	CDK5	Q03114	Q16620
CDK5RAP2	Q96SN8	P17612	CDK6	Q00534	P42772
CDK7	P50613	P28715	CDKN2AIP	Q9NXV6	O94913, Q9H0D6
CDKN3	Q16667, Q00526	P06493	CDYL	Q9Y232	Q13257
CDYL2	Q8N8U2	Q9UBU9	CEBPB	P17676	P35638
CELSR2	Q9HCU4	Q92838	CENPB	P07199	Q9P031
GENPC	Q03188	P06748	CENPE	Q02224	Q9Y6D9
CENPU	Q71F23-1, Q71F23	P53350	CEP112	Q8N8E3	P31946
CEP126	Q9P2H0	O43809	CEP170	Q5SW79	Q92845
CEP170P1	Q96L14	A1L4H1	CEP19	Q96LK0	P55072
CEP192	Q8TEP8	O14965	CEP250	Q9BV73	Q96CV9
CEP290	O15078	Q8IW35, O43303	CEP295	Q9C0D2	Q9NV70
CEP350	Q5VT06	P35221	CEP55	Q53EZ4	Q8WUM4
CEP57	Q86XR8	Q96PC5	CEP57L1	Q8IYX8-2, Q6P2R3	P09917
CEP70	Q8NHQ1	P43897	CEP83	Q9Y592-2	P18848
CERK	Q8TCT0	O75496	CERS4	Q9HA82	Q9UKG4
CERS6	Q6ZMG9-2	O75396	CERT1	Q9Y5P4	O00186
CFAP299	Q6V702	Q16236	CFAP36	Q96G28	P36404
CFI	P05156-PRO_0000027569, P05156-PRO_0000027570, P05156	P08603	CFLAR	O15519, O15519-1	Q14790
CGNL1	Q0VF96	P31946	CHCHD1	Q96BP2	P04618
CHCHD3	Q9NX63	Q96QF0	CHCHD4	Q8N4Q1	O43768
CHD1	O14646	Q96KQ7	CHD1L	Q86WJ1	Q9NTX7
CHD8	Q9HCK8	P09651	CHD9	Q3L8U1-3	Q07869
CHDH	Q8NE62	O95861	CHERP	Q8IWX8	P21802
CHI3L1	P36222	P27797	CHIC2	Q9UKJ5	O96015

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CHM	P24386	Q92696	CHMP1A	Q9HD42	P40818
CHMP4B	Q9H444	Q96FZ7, Q8WUX9, Q9Y3E7	CHMP5	Q9NZZ3	Q9Y3E7
CHMP7	Q8WUX9	Q9H444	CHPF2	Q9P2E5	P0DTC8
CHPT1	Q8WUD6	Q9Y2B1	CHST12	Q9NRB3	Q9UI95
CHST15	Q7LFX5	P27824	CHTOP	Q9Y3Y2	Q14739
CIAO1	O76071	P51530	CIAO2A	Q9H5X1	O75582
CIAO2B	Q9Y3D0	P16284	CIAO3	Q9H6Q4	P27540
CIAPIN1	Q6FI81	Q9H3K6	CIC	Q96RK0	Q13257
CILK1	Q9UPZ9	Q13451	CIR1	Q86X95	Q07955, Q01130
CIRBP	Q14011	Q92973	CISD2	Q8N5K1	O15155
CISD3	P0C7P0	Q6PJG9	CIT	O14578	O14907
CITED2	Q99967	Q09472	CIZ1	Q9ULV3	Q9P0V3
CKAP2	Q8WWK9	Q13257	CKAP5	A2AGT5, Q14008	Q9Y6A5
CKS2	P33552	P06493	CLASP1	Q7Z460	P35613
CLASP2	O75122	P49841	CLCC1	Q96S66	P59632
CLCF1	Q9UBD9, Q9UBD9-PRO_0000015616	P26992, O75462	CLCN7	P51798	Q96CA5
CLDN15	Q96FX9	Q8IY26	CLHC1	Q8NHS4	Q96ES7
CLIC4	Q9Y696	O15247	CLINT1	Q14677	O00443
CLIP1	P30622	P0DTC6	CLIP3	Q96DZ5	O76024
CLOCK	O08785	P36873	CLP1	Q92989	Q9UPV0
CLPP	Q16740	P54252	CLSTN1	O94985	Q01523
CLTA	P09496	Q9Y6A5	CLTC	Q00610	Q9NYZ3
CLUAP1	Q96AJ1	Q9NQC8, Q9BW83, Q9P2H3, Q9NWB7, Q9UG01, Q13099	CLUH	O75153	Q96PE7
CMAS	Q8NFW8	P48059	CMPK1	P30085	Q6DKK2
CMTM7	Q96FZ5	Q15109	CNN2	Q99439	P11142
CNNM1	Q9NRU3	Q12974	CNNM3	Q8NE01	Q12974
CNOT1	A5YKK6	Q8NDV7	CNOT2	Q9NZN8	Q9UBE8
CNOT3	O75175	P55072	CNOT6L	Q96LI5	P09237
CNOT8	Q9UFF9	P09237	CNOT9	Q92600	P09237
CNP	P09543	Q12824	CNPY2	Q9Y2B0	Q8WY64
CNPY3	Q9BT09	Q14160	CNTN1	Q12860	Q92823
COA1	Q9GZY4	Q9BQ95	COA6	Q5JTJ3	Q9NPB3
COA7	Q96BR5	P29279	COASY	Q13057	P23443
COBLL1	Q53SF7	P31946	COG3	Q96JB2	Q9NZQ7
COG5	Q9UP83	P35613	COG6	Q9Y2V7	O95999
COL4A5	P29400-2	Q8TB40	COL5A1	P20908	P01033
COL5A2	P05997	Q6ISS4	COL8A1	P27658	O43559
COLQ	Q9Y215	P22303	COMMD4	Q9H0A8	Q14186, O75461
COMT	P21964	P30518	COP1	Q8NHY2	Q96RU7
COPA	P53621	Q8WUJ3	COPB1	P53618	P56945
COPB2	P35606	P19838	COPG2	Q9UBF2	P27824, P04233
COPRS	Q9NQ92	P62805	COPS2	P61201	Q15831
COPS4	Q9BT78	Q15831	COQ9	O75208	P09601
CORO1B	Q9BR76	Q9NWB7	COX4I1	P13073	P11182
COX5A	P20674	P00403	COX6B1	P14854	P00403
COX7C	P15954	P00403	CPEB1	Q9BZB8	Q92797
CPNE7	Q9UBL6-2	P14373	CPS1	P31327	O95861

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CPSF1	Q10570	Q92797, Q9C0J8, Q6UN15, O95639	CPSF2	Q9P2I0	Q92797, Q9C0J8, Q6UN15
CPSF6	Q16630-2, Q16630-1, Q16630	O43809, Q9P2I0, Q9C0J8, Q6UN15, O95639, Q7Z6E9	CPSF7	Q8N684-3, Q8N684	O43809
CPT1B	A2RRE8	P42858	CRABP2	P29373	Q9HAW8
CRBN	Q96SW2	P48729	CREB1	P16220	P18846
CREB5	Q02930-3	P19883	CRELD1	Q96HD1-2	P19404
CRELD2	Q6UXH1-2	P12830	CRMP1	Q14194	P07900
CROT	Q9UKG9	P16671	CRTAP	O75718	O95076
CRTC1	Q6UUUV9	P31946	CRTC2	Q53ET0	Q9UMX1
CRY2	Q49AN0	Q9H0D6	CRYBB3	P26998	P05814
CRYL1	Q9Y2S2	O00443	CRYZL1	O95825	Q13572
CSDE1	O75534	P0DTD2	CSF1	P09603	P07333
CSGALNACT2	Q8N6G5	Q9GZU1	CSK	P41240	Q68CZ2
CSNK1A1	P48729	O75330	CSNK1G2	P78368	P18827
CSNK1G3	Q9Y6M4	P78368	CSPP1	Q1MSJ5-1	Q08379
CSRNP1	Q96S65	P36873	CSRP3	P50461	P42858
CSTB	P04080	P13569	CSTF3	Q12996	Q6P1J9
CSTPP1	Q9H6J7-2	P02649	CTBP1	Q13363	P48552
CTBS	Q01459	P59665	CTCF	P49711	P40429
CTDNEP1	O95476	Q9Y5M8	CTDSPL2	Q05D32	P52294
CTNNA1	P35221	P25963	CTNNAL1	Q9UBT7	P11532
CTNND1	O60716	Q12913	CTR9	Q6PD62	O00267
CTSA	P10619	P61586	CTSB	P07858	P40692
CTSD	P07339	Q8NBJ7	CTSH	P09668	P07711
CTSK	P43235	P07711	CTSL	P07711	P43235
CTSO	P43235	P07711	CTTNBP2NL	Q9P2B4	P12273
CTU1	Q7Z7A3	P50552	CUL2	Q13617	Q15843
CUL5	Q93034	Q9Y314	CUTA	O60888	P22736
CWC27	Q6UX04	P0DTC4	CXADR	P97792-1	P19793
CXXC1	Q9P0U4	P60484	CXorf38	Q8TB03	Q02548
CYB561A3	Q8NBI2	Q9NUQ2	CYBC1	Q9BQA9	O15155
CYC1	P08574	P40855	CYP1A1	P04798	Q15813
CYSTM1	Q9H1C7	Q6ZMH5	DAP3	P51398	Q14197
DAPK3	O43293	Q9BQ90	DARS1	P14868	P05412
DAZAP1	Q96EP5	P13569	DAZAP2	Q15038	Q9HAU4
DBNL	Q9UJU6	P11142	DBP	Q10586	P35638
DBT	P11182	Q96E11	DCBLD2	Q96PD2	P46109
DCP1A	Q9NPI6	Q92900	DCP2	Q8IU60	Q92900
DCPS	Q96C86	Q96C86	DCTN4	Q9UJW0	Q14203
DCTN6	O00399	Q9Y5N6	DDB1	Q16531	Q15843
DDB2	Q92466	Q9NYZ3	DDI2	Q5TDH0-2	P14136
DDIT4	Q9NX09	P16284	DDIT4L	Q96D03	Q9Y316
DDR1	Q08345-1	P29353	DDR2	Q16832	P29353
DDX1	Q92499-1	P04618	DDX23	Q9BUQ8	P62316, P62306, P14678, P62304
DDX28	Q9NUL7	Q9NYY8	DDX39A	O00148	P82979
DDX39B	Q13838	P82979, Q9Y3Y2, O75152	DDX42	Q86XP3	O75533
DDX5	P17844	P09651	DDX50	Q9BQ39	Q15646
DDX6	P26196	Q92900	DEAF1	O75398	P49841

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DECR1	Q16698	Q95997	DEF8	Q6ZN54-2	Q14957
DEK	P35659	Q9UER7	DENND4A	Q7Z401	P18433
DENND4C	Q5VZ89	P31946	DERA	Q9Y315	Q9Y3C6
DEXI	O95424	Q9Y5Y5	DFFA	O00273	O76075
DFFB	O76075	O00273	DGAT1	O75907	P61073
DGCR2	P98153	Q8TC07	DGKA	P23743	P12931
DGKQ	P52824	P0DPK4	DGLUCY	Q7Z3D6	P25788
DHCR24	Q15392	Q9H1C4	DHRS13	Q6UX07	P32971
DHRS7	Q9Y394	Q9NTJ5	DHX29	Q7Z478	P06748
DHX30	Q7L2E3	Q99558	DHX34	Q14147	Q16659
DHX36	Q9H2U1	Q16629	DHX37	Q8IY37	P06748
DHX38	Q92620	Q8WUA2, Q92917, O60231	DHX40	Q8IX18	P52294
DHX57	Q6P158	Q9Y4Z2	DHX58	Q96C10	Q9UKV8, Q9UPY3
DIAPH3	Q9NSV4	Q6UWR7	DICER1	Q9UPY3, Q9UPY3- 1	Q9UKV8
DIS3	Q9Y2L1	P05120	DKK1	O94907	O75197
DLD	P09622	P11182	DLGAP5	Q15398	P61769
DMD	P11532	P43657	DMTN	Q08495	P31946
DMWD	G5E9A7	Q8WUX9	DMXL1	Q9Y485	P21281
DMXL2	Q8TDJ6	O75695	DNAAF2	Q9NVR5	Q8NAV1
DNAJA1	P31689	Q9GZX7	DNAJB14	Q8TBM8	P13569
DNAJB2	P25686	P13569	DNAJB4	Q9UDY4	Q9Y266
DNAJB5	O75953	P25685	DNAJB9	Q9UBS3	Q13217
DNAJC10	Q8IXB1	Q9UBX5	DNAJC3	Q91YW3	P11021
DNAJC6	O75061	P10636-8	DNAJC7	Q99615	Q9BQ52
DNAJC8	O75937	P02545	DND1	Q8IYX4	Q96JH7
DNM1L	O00429	P62937	DOCK1	Q14185	P46109
DOCK10	Q96BY6	P42357	DOCK5	Q9H7D0	P46109
DOCK7	Q96N67	Q9UM54	DOCK9	Q9BZ29	P84022
DOHH	Q9BU89	Q9NQZ5	DOK7	Q18PE1	Q9UKG1
DOLK	Q9UPQ8	Q9NZQ7	DOT1L	Q8TEK3	P42568, Q03111
DPAGT1	Q9H3H5	P42858	DPH1	Q9BZG8	O75386
DPH5	Q9H2P9	P0DTD2	DPM3	Q9P2X0-2	O75396
DPP8	Q6V1X1	P14678	DRAP1	Q14919	Q00403
DSC3	Q02487	Q14126	DSG2	Q14126	P13569
DST	Q03001	Q96CV9	DTNBP1	Q96EV8	A1L3X0
DTX3	Q8N9I9	Q13882	DUSP10	Q9Y6W6	P60709
DUSP23	Q9BVJ7	Q8TBC4	DUSP5	Q16690	P28482
DVL1	O14640	Q9Y4D1	DVL2	O14641, Q60838	P53350
DXO	O77932	P16284	DYNC1H1	Q14204	P43034
DYNC1I1	O14576-2	Q9GZQ8	DYNC1I2	Q13409	P43034
DYNC1LI2	O43237	P13569	DYNC2I1	Q8WVS4	Q9Y266
DYNC2LI1	Q8TCX1	Q14203	DYNLL1	P63167	O75330
DYNLRB1	Q9NP97	P10636-8	DZIP3	Q86Y13	P51668
E2F4	Q16254	P11802	EAPP	Q56P03	P62306, P14678, P62304
ECD	O95905	Q15185	ECHDC2	Q86YB7	O75380
ECI2	O75521	Q16659	ECPAS	Q5VYK3	Q9NRM0
ECSIT	Q9BQ95	Q9H244	EDC3	Q96F86	P46109
EDC4	Q6P2E9	Q8IZH2	EDEM1	Q92611	Q9UMX1

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EDEM3	Q2HXL6	P30101	EDRF1	Q3B7T1	O14832
EEF1D	P29692-2, P29692	P26641, P29692	EEIG1	Q5T9C2-3	P56945
EEPD1	Q7L9B9	P18827	EFEMP1	Q12805	Q9H2F3
EFNA1	P20827	P54764	EFNB2	P52799	P48163
EGF	P01133	P00533	EGFL7	Q9UHF1	Q9UBX5
EGLN2	Q96KS0	Q16665	EGR2	A0A0S2Z3U4	Q08379
EHBP1	Q8NDI1	Q9NWB7	EHD3	Q9NZN3	Q9H1K0
EHHADH	Q08426	P60709	EHMT2	Q96KQ7	Q13185
EI24	O14681	Q12982	EIF1AY	O14602	P47813
EIF2A	P05198	P19525	EIF2AK2	P19525	O75152
EIF2B4	Q9UI10	P01019	EIF2B5	Q13144	P21579
EIF2S3	P41091	P38936	EIF3A	Q14152	P47813
EIF3E	P60228	Q99814	EIF3J	O75822	Q04637
EIF3M	Q7L2H7	O00232	EIF4A2	Q14240	Q04637, P06730
EIF5	P55010	P47813	EIF5B	O60841	P14921
ELF2	Q15723	Q01196	ELK1	P19419	P63279
ELL2	O00472	P07550	ELOB	Q15370	Q16665
ELOVL2	Q9NYP7	O14653	ELOVL5	Q9NYP7	O14653
ELOVL7	A1L3X0	Q96EV8	ELP1	O95163	Q15306
EMB	Q6PCB8	P17931	EMC2	Q15006	O43868
EMC6	Q9BV81	Q8N4V1	EMC8	O43402	P27824
EML2	A0A0C4DGQ7	P40937	EMP1	P54849	Q9Y5Q5
ENAH	Q03173	Q9UQB8	ENC1	O14682	Q13501
ENDOD1	O94919	Q9NRZ7	ENDOG	Q14249	P11142
ENTR1	Q96C92	Q9HC29	ENTREP3	P81408	Q30201
EOGT	Q5NDL2-3	Q9UKG4	EPAS1	Q99814	P40818
EPB41L1	Q9H4G0	Q9ULK5	EPB41L2	O43491	P42568, Q03111
EPB41L3	Q9Y2J2	P49841	EPHA4	P54764	P05177
EPHB1	P54762	Q14451	EPHB2	P29323-3	O14832
EPHX1	P07099	Q9Y5M8	EPN1	Q9Y6I3	P42566
EPRS1	P07814	Q8IWL3	EPS15	P42566	Q9P0V3
EPS8	Q12929	P49419	ERAP1	Q9NZ08	Q15392
ERBB4	Q15303	P29353	ERBIN	Q96RT1	P36873
ERC1	Q8IUD2	P01222	ERCC5	P28715	P50613
ERCC6	Q03468	P18827	ERCC8	Q13216-1	Q03468
ERF	P50548	P28482	ERGIC1	Q969X5	P04601
ERGIC3	Q9Y282	Q96AA3	ERLEC1	Q96DZ1	Q5K4L6
ERLIN2	O94905	P11021	ERN1	O75460-1	P11021
ERP29	P30040	Q9Y315	ERVK3-1	Q96A10	O95429
ESCO2	Q56NI9	Q08379	ESRRA	P11474	Q9UBK2
ESYT2	A0FGR8	P61981	ETFA	P13804	Q8IWL3
ETFRF1	Q6IPR1	O14561	ETNK2	Q9NVF9	Q16623
ETV1	P50549	Q8NHY2	EVI2B	P34910-2	P22732
EVI5	O60447	P53350	EVI5L	Q96CN4	P43115
EXOC1	Q9NV70	P18848	EXOC2	Q96KP1	Q13304
EXOC4	Q96A65	P04601	EXOC5	O00471	Q13882
EXOC6	Q8TAG9	P08173	EXOSC4	Q9NPD3	P42285
EXOSC6	Q5RKV6	Q13901, Q9NQT4	EXTL2	Q9UBQ6	Q6ZMH5
EXTL3	O43909	Q6ZMH5	F8A1	P23610	P42858
F8A2	P23610	P42858	F8A3	P23610	P42858

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FAAP100	Q0VG06	P99999	FADD	Q13158	O95243
FADS3	Q9Y5Q0	Q8N5I4	FAH	P16930	P00533
FAHD1	Q6P587-2	Q12982	FAM111A	Q96PZ2	P04637
FAM114A2	Q9NRY5	Q969Z0	FAM118B	Q9BPY3	Q8N307
FAM120B	Q96EK7	Q12933	FAM162A	Q96A26	P08238
FAM168B	A1KXE4-2	O75593	FAM171B	Q6P995	P21802
FAM177A1	Q8N128	O14975	FAM184A	Q6P9G8	Q08379
FAM199X	Q6PEV8	P49674	FAM210A	Q96ND0	Q9Y366
FAM241A	Q8N8J7	O14786	FAM53B	Q14153	P31946
FAM81A	Q8TBF8	Q9UBB9	FAM83G	A6ND36	P48729
FAM89B	Q9QUI1	P12755	FAM8A1	Q9UBU6	Q9NPF0
FAM98B	Q52LJ0	P10636-8	FANCD2	Q9BXW9	O00255
FANCI	Q9NVI1	Q9Y5B0	FAR1	Q8WVX9	P13569
FARSB	Q9NSD9	P10636-8	FASN	P49327	Q16665
FAT1	Q9QXA3	Q14160	FAXDC2	Q96IV6	Q12846
FBH1	Q8NFBZ0	P63208	FBN1	P35555	P15502
FBRSL1	Q9HCM7	P68400, P67870	FBXL14	Q8N1E6	P35813
FBXL15	Q9H469	Q9HAU4	FBXL18	Q96D16	P61978
FBXL19	Q6PCT2-2	Q6ZPD8	FBXL3	Q9UKT7	Q9UJ68
FBXL5	Q9UKA1	Q92905	FBXO10	Q9UK96	O43924
FBXO17	Q96EF6	O60551	FBXO27	Q8NI29	P63208
FBXO3	Q9UK99	P04608	FBXO30	Q8TB52	Q9UM54
FBXO38	Q6PIJ6	Q93009	FBXO7	Q9Y3I1	Q9P0P0
FBXW5	Q969U6	P63208, Q13616	FCHO1	O14526	Q13882
FEM1A	Q9BSK4	P09110	FEM1C	Q96JP0	Q93034
FER	P16591	P08238	FERMT2	Q96AC1	P35222
FERRY3	Q9NQ89	P55212	FEZ2	Q9UHY8	P54252
FGD4	Q96M96	P17252	FGD6	Q6ZV73	P31946
FGF2	P09038	O15530	FHIP2A	Q5W0V3	Q13304
FICD	Q9BVA6	P42858	FIP1L1	Q6UN15-4, Q6UN15	Q9P2I0, Q9C0J8, O95639, Q10570
FITM2	Q8N6M3	Q9H244	FKBP14	Q9NWM8	P01222
FKBP4	Q02790	Q9UKV8	FKBP1	Q9UIM3	P49674
FKTN	O75072	Q9Y2B1	FLNB	O75369	Q13233
FLRT1	Q9NZU1	P42336	FMC1	Q96HJ9	Q9NUJ1
FMR1	Q06787	Q04637	FN1	P02751	P05556
FNBP1L	Q5T0N5	Q9Y4D1	FNDC3A	Q9Y2H6	O43511
FNDC3B	Q53EP0	Q9NPD5	FOSL1	P15407	P15336
FOSL2	P15408	Q8NHY2	FOXF2	Q12947	Q07954
FOXN2	P32314	Q9Y297	FOXP1	Q9H334	Q9BZS1
FOXQ1	Q9C009	Q04726	FOXRED2	Q8IWF2	P27824
FRAT2	O75474	P49841	FRMD6	Q96NE9, Q96NE9-2	P40692
FRYL	O94915	P46531	FSD1	Q9BTV5	P30740
FSTL1	Q12841	P08571	FTH1	P02794	Q92574
FTL	P02792	Q96L34, O43303	FTO	Q9C0B1-2	Q92989
FUBP1	Q96AE4	P61981	FUCA1	P04066	Q8NBK3
FUNDC2	Q9BWH2	Q8IVP5	FXR1	P51114	Q13416
FYN	P06241	P12931	FYTDD1	Q96QD9	P82979, O75152
FZD2	Q14332	P43119	FZD4	Q9ULV1	P42858
FZD8	Q9H461	P56704	G2E3	Q7L622	O76024

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GABARAP	O95166	O75385	GABRE	P78334	O60427
GALNS	P34059	Q8NBK3	GAPVD1	Q14C86	P49674
GAS2L1	Q99501	Q96B97	GATD1	Q8WUU5	Q13547
GBA1	P04062	P51159	GBP1	P32455	P32455
GBP2	P32456	P32455	GBP3	Q9H0R5	P32455
GCA	P28676	P28070	GCC2	Q8IWJ2	P18509
GCFC2	P16383	P62304	GCKR	Q14397	P35557
GCLM	P48507	P10746	GCNT1	Q02742	Q6DKK2
GDAP1	Q8TB36	O15392	GDAP2	Q9NXN4	Q9Y3M2
GDF5	P43026	P29373	GDNF	P39905- PRO_0000034005	Q92673
GEM	P55040	Q9BUH8	GEMIN4	P57678	O75604
GEMIN6	Q8WXD5	P62316, P62306, P14678, P62308, P62314, P62304	GEMIN7	Q9H840	P62304
GET3	O43681	P10451	GET4	Q7L5D6	Q9HB29
GFM1	Q96RP9	P03508	GFM2	Q969S9	P14373
GFPT1	Q06210	P13569	GFRA1	P56159-2, Q62997	Q92673
GGA2	Q9UJY4	Q9Y316	GGA3	Q9NZ52	Q6VY07
GID4	P38263	Q5SW96	GIGYF1	O75420	Q86UK7
GINM1	Q9NU53	Q14118	GINS3	Q9BRX5	P49736
GIT1	Q9Y2X7	Q13177	GJA1	P17302, P08050	Q07157
GJD3	Q8N144	Q07157	GLA	Q9UHE5	O15342
GLB1	P16278	P61586	GLG1	Q92896	O15530
GLIS3	Q8NEA6-2	Q96RU7	GLO1	Q04760	P13569
GLRX3	O76003	Q16790	GM2A	P17900	P62736
GMEB1	Q9Y692	P18827	GMFB	P60983	Q86X55
GNAI3	P08754	P13569	GNAQ	P50148	P49407
GNB1L	Q9BYB4	P53611	GNB2	P62879	Q13371
GNB4	Q9HAV0	P18827	GNG12	Q9UBI6	Q14192
GNMT	Q14749	P24311	GNPAT	O15228	Q6UWR7
GNPDA1	P46926	Q9Y303	GNPDA2	Q8TDQ7	Q9Y303
GOLGA6L9	A6NEM1	P43897	GOLGB1	Q14789	Q9Y3A6
GOLIM4	O00461	Q9Y2C2	GOLT1B	Q9Y3E0	O95197
GOPC	Q9HD26	P60484	GORAB	Q5T7V8	O75396
GORASP1	Q9BQQ3	P30622	GPANK1	O95872	Q13190
GPAT4	Q86UL3	O60443	GPATCH2	Q9NW75-2	Q9C005
GPATCH2L	Q9NWQ4-1	Q96RU7	GPBP1	Q86WP2	Q9UBU9
GPC4	O75487	O15530	GPC6	Q9Y625	P32249
GPD1L	Q8N335	P13501	GPD2	P43304	Q8WTR4
GPKOW	Q92917	P23193	GPN2	Q9H9Y4	P62875
GPNMB	Q14956-1, Q14956	P00533	GPR108	Q9NPR9	Q9H8X2
GPR143	P51810	Q16655, P14679	GPR155	Q7Z3F1	P25090
GPR160	Q9UJ42	P17252	GPR3	P46089	P05067-4
GPRASP3	Q6PI77	P48552	GPRC5A	Q8NFJ5	P13569
GPRC5C	Q9NQ84	Q9H015	GPS2	Q13227	P18848
GPSM1	Q86YR5-3	O14503	GPX8	Q8TED1	Q92730
GRAMD4	Q6IC98	Q96QG7	GRB10	Q13322-4	P12931
GRN	P28799	Q13569	GRPEL2	Q8TAA5	P11182
GSK3A	P49840	P49841	GSTA4	O15217	O95995
GSTO1	P78417	P04792	GSTZ1	O43708	Q12933

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GTF2A1	P52655	P20226	GTF2H3	Q13889	P50613
GTF2IRD1	Q9UHL9	Q6VMQ6	GTF3C1	Q12789	Q15185
GTF3C3	Q9Y5Q9	P53350	GTF3C5	Q9Y5Q8	P53350
GUCD1	Q96NT3-2	Q13163	GUCY1A2	P33402	Q14160
GUF1	Q8N442	P01375	GXYLT1	Q4G148	P27824
GZF1	Q9H116	P78563	H1-10	Q92522	P06748
H2AC25	Q7L7L0	P19404	H2AX	P16104	Q13315
H2AZ1	P0C0S5	P62805	H2BC11	P06899	P10636-8
H2BC12	O60814	Q12824	H2BC5	P58876	Q01831
H4C8	P62805	P0DTC4	HABP4	Q5JVS0	Q12873
HACD3	Q9P035	P0DTC4	HBEGF	Q99075	P00533
HBS1L	Q9Y450	Q9BRX2	HCFC1	P51610	P08047
HCFC2	Q9Y5Z7	Q92560	HDAC1	Q13547	Q9UJW3
HDGF	P51858	P78527	HDHD3	Q9BSH5	O75593
HEATR5A	Q86XA9	O75688	HECTD1	Q9ULT8	O75496
HECTD3	Q5T447	Q6NZI2	HEG1	Q9ULI3	P17931
HELZ	P42694	Q8IWL3	HERC1	Q15751	O95395
HERC3	Q15034	Q9UNU6	HES6	Q96HZ4	Q14469
HEXB	P07686	P06865	HFE	Q30201	O14672
HHAT	Q5VTY9	O00560	HIBADH	P31937	O75396
HID1	Q8IV36	Q9BZL1	HIF1A	Q16665	P40818
HIGD2A	Q9BW72	P40855	HINFP	Q9BQA5	O14561
HIP1	O00291	Q86US8	HIRA	P54198	P84243
HIVEP1	P15822	P84022	HIVEP2	P31629	P63208
HIVEP3	Q5T1R4	P61981	HJURP	Q8NCD3	P62805
HKDC1	Q2TB90	P19367	HLA-A	P04439	Q16288
HLA-H	P01893	P11142	HLCS	P50747	Q9NUX5
HLTF	Q14527	Q9NTX7	HMBX1	Q6NT76	Q9HAQ2
HMG20A	Q9NP66	O95863	HMGB2	P26583	P48729
HMGXB4	Q9UGU5	Q9H2G4	HMMR	O75330	Q9ULW0
HMOX1	P09601	Q9NUX5	HNRNPA0	Q13151	Q92973
HNRNPA1	P09651	Q8IZH2	HNRNPA1L2	Q32P51	P09651
HNRNPA2B1	P22626	P09651	HNRNPDL	O14979	Q00839
HNRNPF	P52597	O43347	HNRNPH1	P31943	P09651
HOMER3	Q9NSC5	P50458	HOMEZ	Q8IX15-3	P61956
HOOK1	Q9UJC3	Q9P0N9	HOOK3	Q86VS8	Q9NWB7
HPF1	Q05164	O43791	HPRT1	P00492	Q9H1K1
HPS1	Q92902	Q9C0C7	HRAS	P01112	Q12967
HS1BP3	Q53T59	P06396	HSBP1L1	C9JCN9	Q9UBN6
HSCB	Q8IWL3	P23526	HSD17B12	Q53GQ0	P04049
HSD17B14	Q9BPX1	P29372	HSD17B4	P51660	P42858
HSD17B8	Q92506	Q96EN8	HSP90AA1	P07900	P12931
HSP90AB1	P08238	P12931	HSP90B1	P14625	P11021
HSPA13	P48723	P01375	HSPA4L	O95757	P11021
HSPA5	P11021	P01222	HSPA6	P17066	Q99558
HSPB1	P04792	Q9BRX2	HSPD1	P10809	Q16822
HSPG2	P98160- PRO_0000391621, P98160- PRO_0000391622, P98160	P35968	HTRA1	Q92743	P78504

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HTRA2	O43464	Q8WXH5	HYAL2	Q12891	P21217
HYAL3	O43820	O75553	HYCC1	Q9BYI3	Q9UJH6
HYCC2	Q8IXS8	P14373	IARS1	P41252	P05412
IBTK	Q9P2D0	Q06187	ICA1	Q05084	Q9NRD5
ICA1L	Q8NDH6-2	Q9C005	ICAM3	P32942	P11686
ICE1	Q9Y2F5	P55199	ICE2	Q659A1	P55199
ICMT	O60725	O00186	ID1	P41134	Q03135
ID3	Q02535	P51149	ID4	P47928	Q9Y4Z2
IDE	P14735-1	P10997	IDH2	P48735	P46059
IDH3A	P50213	P13569	IER5	Q5VY09	Q00613
IFIH1	Q9BYX4	P05161	IFIT2	P09913	Q86WV6
IFNE	Q86WN2	Q8WXG9	IFNGR1	P15260	Q9NPF0
IFRD2	Q12894	Q9Y6K5	IFT20	Q8IY31-3	Q13286
IFT27	Q9BW83	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099	IFT43	Q96FT9, Q96FT9-2	Q7Z4L5, Q8NEZ3, Q96RY7, Q9HBG6, Q9P2L0
IFT56	A0AVF1	Q9NQC8, Q9Y547, Q9BW83, Q9P2H3, Q9NWB7, Q9Y366, Q9UG01, Q96LB3, Q9H7X7, Q8WYA0, Q13099	IFT57	Q9NWB7	Q12981, Q9P2W9
IFT70A	Q86WT1	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099	IFT74	Q96LB3	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q9P2H3, Q9UG01, Q13099
IFT80	Q9P2H3	Q9Y266	IFT81	Q8WYA0	Q9NQC8, A0AVF1, Q9Y547, Q9BW83, Q9P2H3, Q9NWB7, Q9Y366, Q9UG01, Q96LB3, Q9H7X7, Q13099
IFT88	Q13099	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q9BW83, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099	IGF2BP3	O00425	Q9H074
IGF2R	P11717	Q9NZ52	IHO1	Q8IYA8	P61421
IKZF5	Q9H5V7	P61024	IL1R1	P14778	Q9Y4K3
IL1RAP	Q9NPH3	P01584	IL1RN	P18510	P14778
IL6R	P08887-PRO_0000450730, P08887	P05231	IL6ST	P40189	P05231
IL7R	P16871	P27449	ILF3	Q12906	P40429
ILK	Q13418	Q8NF37	ILRUN	Q9H6K1	P11142
IMMT	Q16891	P54252	IMP3	O00425	Q9H074
IMPA1	P29218-3	P42858	IMPA2	O14732	Q8TBC4
IMPACT	Q9P2X3	P20618	ING3	Q9NXR8	P52294

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ING5	Q8WYH8	Q00994	INPP1	A0A0S2Z3X1	P20393
INPP5F	Q9Y2H2	P31946	INSIG2	Q9Y5U4	Q8TB40
INTS5	Q6P9B9	P08173	INTS7	Q9NVH2	P67775
INTS8	Q75QN2	P67775	INTS9	Q9NV88	P52294
IP6K1	Q92551	P21802	IPO11	Q9UI26	Q9NRM0
IPO13	O94829	P47813	IPO7	O95373	Q9NZQ7
IPO8	O15397	Q9UKV8	IQCE	Q6IPM2	P25788
IQGAP1	P46940	P60709	IQGAP3	Q86VI3	P40855
IRAK1	P51617	Q92985	IREB2	P48200	P43353
IRF2BP1	Q8IU81	Q14160	IRF2BP2	Q7Z5L9	Q8N6T7
IRF2BPL	Q9H1B7	P14316	IRF3	Q14653	Q96EN8
IRS1	P35568	P11940	IRS2	Q9Y4H2	P27986
ISCA2	Q86U28	Q9H4P4	ISCU	Q9H1K1	Q8IWL3
ISOC1	Q96CN7	P04004	ITCH	Q96J02	P08151
ITFG1	Q8TB96	P11686	ITGA4	P13612	P05556
ITGB1	P05556	P08648	ITGB3BP	Q13352	Q969Y2
ITGB4	P16144	P48509	ITGB5	P18084	P04004
ITPKB	P27987	Q8N9R8	ITPR2	Q14571	Q9H244
ITPR3	Q14573	Q92560	ITPRIP	Q8IWB1	Q96FT7
ITSN1	Q15811	Q07889	ITSN2	Q9NZM3	Q9UBC2
IVNS1ABP	Q9Y6Y0	P53350	JADE2	Q9NQC1-2	P01100
JAM3	Q9BX67	Q9BX67	JARID2	Q92833-1, Q92833	Q15022
JAZF1	Q86VZ6	Q9Y265	JOSD1	Q15040	P14373
JPH1	Q9HDC5	Q5SY16	JUN	P05412	P29692
JUP	P14923	P01160	KANK2	Q63ZY3	Q15669
KAT2A	Q92830	Q01094	KAT2B	Q92831	Q16665
KAT5	Q92993	O94844	KAT8	Q9D1P2	P51610
KATNA1	O75449	Q9Y4X5	KATNAL1	Q9BW62	Q14061
KATNBL1	Q9H079	Q9UBU9	KAZN	Q674X7-2	O75604
KBTBD4	Q9NVX7	P11142	KCNJ14	Q99712	P63252
KCNN4	O15554	P61073	KCNS3	Q9BQ31	Q13370
KCNT2	Q6UVM3	Q14160	KCTD13	Q8WZ19	P46108
KCTD16	Q68DU8	P05067-4	KCTD20	Q7Z5Y7	P36873
KCTD3	Q9Y597	P26045	KCTD9	Q7L273	O14786
KDELR1	P24390	Q92796	KDM1B	Q8NB78	Q92560
KDM3A	Q9Y4C1	O75582	KDM5A	P29375	P06400
KDM6A	O15550	O43474	KDM6B	Q5NCY0	Q9BVI0
KDSR	Q06136	Q9H5K3	KHDRBS1	Q07666	P12931
KHDRBS3	O75525	P40938	KHSRP	Q92945	P05412
KIAA0232	Q92628	Q8N9F0	KIAA0825	Q8IV33	Q96CV9
KIAA1143	Q96AT1	Q9H5Z1, Q8WUD4, Q9BRR8, O60508, O94906	KIAA1217	Q5T5P2-6	Q13882
KIAA1671	Q9BY89	Q9Y3R4	KIAA1958	Q8N8K9	O14543
KIDINS220	Q9EQG6	P46109	KIF14	Q15058	Q9Y3A6
KIF18A	Q8NI77	P36873	KIF1B	O60333-2	Q9BRX2
KIF1C	O43896	P31946	KIF20A	O95235	Q13257
KIF20B	Q96Q89	P52294	KIF21A	Q7Z4S6	Q14678
KIF23	E9Q5G3, Q02241-2, Q02241	Q9H0H5	KIF2A	O00139	Q5S007
KIF3A	Q9Y496	Q5T8I9	KIF3B	O15066	Q9Y496, Q92845

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
KIF5B	P33176	O14977	KIF5C	O60282	P68400, P67870
KIF7	Q2M1P5	P10070	KIFAP3	Q92845	Q13177
KIFC3	Q9BVG8-5	Q99728	KIRREL1	Q96J84	Q07157
KIZ	Q2M2Z5	Q96NL6	KLC1	Q07866-2, Q07866	Q12982
KLF10	Q13118	P08047	KLF15	Q9UIH9	P12931
KLF16	Q9BXX1	Q12857	KLF2	Q9Y5W3	Q9HCE7
KLHDC3	Q9BQ90	Q15843	KLHL20	Q9Y2M5	O14832
KLHL21	Q9UJP4	P23025	KLHL3	Q9UH77	P06396
KLHL7	Q8IXQ5	Q8N4J0	KMO	O15229-2	P42858
KMT2A	Q03164	O00255	KMT2B	Q9UMN6	O00255
KMT2C	Q8NEZ4	Q14686, Q15291, P61964, Q9UBL3	KMT2E	Q8IZD2	P12273
KMT5B	Q4FZB7	Q9H2G4	KNL1	Q8NG31-2, Q8NG31	O60566
KNOP1	Q1ED39	Q96CW1	KNTC1	P50748	P46059
KPNA4	O00629	Q9BQB4	KRAS	P01116	P61586
KRBOX4	Q5JUW0	Q13263	KRI1	Q8N9T8	P40429
KRIT1	O00522	Q15185	KRT10	P13645	P19320
KRT18	P05783	Q9NUX5	KRT7	P08729	P09429
KRT8	P05787	Q9Y6K9	KRTAP2-3	P0C7H8	Q8WUK0
KRTAP2-4	Q9BYR9	P49638	KTI12	Q96EK9	Q9HC97
KTN1	Q86UP2-3	P20645	LAGE3	Q14657	Q96S44
LAMA4	Q16363	P04637	LAMB1	P07942	P17931
LAMC2	Q13753	Q70UQ0	LAMP1	P11279	Q15836
LAMTOR4	Q0VGL1	O15392	LAMTOR5	O43504	P68104
LANCL1	O43813	Q9Y315	LARGE1	O95461	Q6VN20
LARP1	Q6PKG0	Q9H074	LARP4	Q6P4E2	Q02548
LARS1	Q9P2J5	O00560	LARS2	Q15031	O95861
LATS1	O95835	P06493	LCLAT1	Q6UWP7	Q92504
LCOR	Q96JN0, Q96JN0-2	Q13363	LCP1	P13796	Q8NG68
LEPROT	O15243	Q9UBD6	LGALS3	P17931	Q9NPD5
LGALSL	Q3ZCW2	Q9Y6D9	LHFPL2	Q6ZUX7	Q14162
LHX2	P50458	O00712, Q12857	LIMCH1	Q9UPQ0-1	Q96JA1
LIMK1	P53667	P23528	LIMS1	P48059	Q9Y3P9
LIN54	Q6MZIP7	P52747	LIN7C	Q9NUP9	Q8N3R9
LIPG	Q9Y5X9	Q9Y3A6	LIPH	Q8WWY8	O75197
LLGL1	Q15334	Q05513	LMAN1	P49257	P04578
LMBRD2	Q68DH5	P22732	LMO7	Q8WWI1-3	Q96JA1
LNPEP	Q9UIQ6	P27824	LNPK	Q9C0E8-2	O60664
LONP1	P36776	Q4G176	LONRF3	Q496Y0	P35232
LOX	P28300	Q9UBX5, P15502	LOXL4	Q96JB6	O60341
LPCAT1	Q8NF37	O14975	LPGAT1	Q92604	P22732
LPP	Q93052	P05120	LRBA	P50851	P16410
LRFN3	Q9BTN0	P13385	LRFN4	Q6PJG9	P23468, Q13332
LRPAP1	P30533	P98155	LRRC1	Q9BTT6	P36873
LRRC14	Q15048	Q92905	LRRC20	Q8TCA0	P60709
LRRC40	Q9H9A6	Q96JA1	LRRC58	Q96CX6	Q93034
LRRC61	Q9BV99	Q16512	LRRC8A	Q8IWT6	O43914
LRRC8B	Q6P9F7	Q13370	LRRC8C	Q8TDW0	P25090
LRRC8E	Q6NSJ5	P43119	LRRFIP1	Q32MZ4	Q96LB3

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LRRFIP2	Q9Y608	P13569	LRRIQ3	A6PVS8	O43639
LRRK2	Q5S007	P13639	LRWD1	Q9UFC0	Q9Y5N6
LSM10	Q969L4	P62306	LSM11	P83369	P62306, P62304
LSM14B	Q9BX40	P10636-8	LTBP2	Q14767	Q9UBX5
LTBR	P36941	Q9NPF0	LTN1	O94822	Q9Y5B0
LUC7L	Q9NQ29-3	Q13247, Q16629	LXN	Q9BS40	Q9UQK1
LYN	P07948	P12931	LYPLA2	O95372	Q9Y496
LYRM4	Q9HD34	Q9H1K1	LZTS1	Q9Y250	P30307
LZTS2	Q9BRK4	Q9Y316	M6PR	P20645	Q9NZ52
MACF1	Q9UPN3	P03372	MACO1	Q8N5G2	Q96FE5
MAD1L1	Q9Y6D9	Q13257, Q9Y6D9	MAD2L1BP	Q15013	O43511
MAF	O75444	P18848	MAF1	Q9H063	Q9UJK0
MAGI3	Q5TCQ9	Q9NTX7	MAGT1	Q9H0U3	P27824
MALT1	Q9UDY8	O95999, Q9UDY8	MAN1C1	Q9NR34	O95363
MAN2A1	Q16706	Q03405	MAP1A	P78559	Q9GZQ8
MAP1B	P46821	Q9GZQ8	MAP1LC3B2	A6NCE7	Q9NT62
MAP1S	Q66K74	O14543	MAP2K2	P36507	P07900
MAP2K3	P46734	Q16539	MAP2K4	P45985	Q13233
MAP2K7	O14733	O43318	MAP3K11	Q16584	P27986
MAP3K2	Q9Y2U5	Q13163	MAP3K20	Q9NYL2	O75582
MAP3K5	Q99683	P25445	MAP3K7	O43318	Q9Y4K3
MAP3K7CL	P57077	Q13227	MAP3K9	P80192	P17252
MAP4	P27816	P10636-8	MAP4K2	Q12851	Q9NUX5
MAP4K3	Q924I2	Q99962	MAP4K4	O95819	Q9H0R5
MAP7	Q14244	Q9Y3M2	MAP7D3	Q8IWC1	Q9Y3M2
MAPK10	P53779	P49407	MAPK12	P53778	Q14160
MAPK13	O15264	P31749	MAPK6	Q16659	P02787
MAPK9	P45984	P45983	MAPKAPK2	P49137	Q16539
MAPKBP1	O60336	O43597	MAPRE1	Q15691	P30622
MARCHF5	Q9NX47	Q12981	MARCHF7	Q9H992	Q8IWW6
MARCKSL1	P49006	O60551	MARF1	Q9Y4F3	P61964
MARK1	Q9P0L2	P0DTD2	MARK2	Q7KZI7	P0DTD2
MAST2	Q6P0Q8	P27037	MATR3	P43243	P09651
MAVS	Q7Z434	Q13568	MAX	P61244	P29692
MAZ	P56270	P43146	MBD3	O95983-2	Q9Y5J6
MBD6	Q96DN6	O43639	MBIP	Q9NS73-5	P52306
MBNL1	Q86VM6	Q08117	MBNL3	Q9NUK0	O75553
MBOAT7	Q96N66	Q86WV6	MBP	P13727	Q12946
MBTD1	Q05BQ5	P62805	MCAT	Q8IVS2	Q00839
MCC	P23508	P51955	MCOLN1	Q9GZU1	Q9GZU1
MCRIP2	Q9BUT9	O15234	MCUB	Q9NWR8	Q15047
MCUR1	Q96AQ8	P23786	MDM2	Q00987	Q15648
MDM4	O15151	P62837	ME1	P48163	P48163
ME2	P23368	P10809	MED1	Q15648	P53350
MED13	Q9UHV7	Q15648	MED13L	Q71F56	Q15648
MED14	O60244	Q15648	MED18	Q9BUE0	Q15648
MED23	Q9ULK4	Q15648	MED24	O75448	Q15648
MED4	Q9NPJ6	Q15648	MEF2A	Q02078-6	Q13164
MEN1	O00255-2, O00255	Q03164	MEPCE	Q7L2J0	Q9Y3E7
MET	P08581	P12931	METAP2	P50579	Q93034

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
METRNL	Q641Q3	Q01523	METTL13	Q8N6R0	Q12933
METTL15	A6NJ78	P46059	METTL16	Q86W50	P53701
METTL17	Q9H7H0	Q9UBX5	METTL27	Q8N6F8	Q9NQ94
METTL3	Q86U44	Q08J23	MFHAS1	Q9Y4C4	Q9Y496
MFN1	Q8IWA4	Q16611	MFSD5	Q6N075	O75396
MFSD6	Q6ZSS7	P29033	MFSD8	Q8NHS3	Q9H244
MGAT4B	Q9UQ53	Q96CV9	MGLL	Q99685	P07550
MGST3	O14880	O75396	MIA2	Q96PC5	Q9HC35
MIA3	Q5JRA6	Q92504, Q6ZMH5	MIB1	Q86YT6	Q9UHD2
MIB2	Q96AX9	Q9UHD2	MICA	Q29983	Q15836
MICALL1	Q8N3F8	P31946	MICU1	Q9BPX6	Q9H4I9, Q9BPX6, Q8NE86
MICU2	Q8IYU8	Q9H4I9, Q9BPX6, Q8NE86	MID1IP1	Q9NPA3	P16473
MIEF1	Q9NQG6	P53365, O60664	MIEN1	Q9BRT3	O95231
MIER1	Q8N108-16	Q15109	MIER2	Q8N344	Q92769, Q13547
MIER3	Q7Z3K6	Q92769, Q13547	MIGA2	Q7L4E1	O14975
MILR1	Q7Z6M3	Q92538	MINDY1	Q8N5J2-3	P55212
MINDY3	Q9H8M7	P78549	MINK1	Q8N4C8	Q00613
MIPEP	Q99797	Q9BQP7	MIS18BP1	Q6P0N0	P56539
MITF	O75030	P19484	MKI67	P46013	P53667
MKRN2	Q9H000	P51668	MLH1	P40692	P54278
MLH3	Q12083	O14641	MLLT1	Q03111	P04608
MLLT3	P42568	Q8TEK3	MLST8	A0A0A0MR05	P99999
MLYCD	O95822-2	P52756	MMAB	Q96EY8	P11182
MMGT1	Q8N4V1	O75396	MMP14	P50281	P53667
MMUT	P22033	P42858	MNT	Q99583	O43639
MOB1B	Q7L9L4	Q13188, Q13043	MOB4	Q9QYW3	P42566
MOCS1	Q9NZB8	Q9BQ90	MOCS3	O95396	Q9H5K3
MOGS	Q13724	P27824	MON2	Q7Z3U7	P21453
MORC3	Q14149	Q9NWF9	MORC4	Q8TE76	P40763
MOSPD2	Q8NHP6	P40855	MOXD1	Q6UVY6	O76003
MPC2	O95563	P02654	MPHOSPH10	O00566	P62753
MPHOSPH8	Q99549	Q15047	MPLKIP	Q8TAP9	Q9ULX6
MPP2	Q08050	Q13416	MPST	P25325	O96006
MR1	Q95460-1	P61769	MRM3	Q9HC36	Q96T88
MRPL1	Q9BYD6	P67809	MRPL16	Q9NX20	P67809
MRPL17	Q9NRX2	P67809	MRPL34	Q9BQ48	O14975
MRPL41	Q8IXM3	Q14197	MRPL44	Q9H9J2	Q15047
MRPL49	Q13405	P67809	MRPL57	Q9BQC6	P62241
MRPL9	Q9BYD2	P21673	MRPS12	O15235	P46199
MRPS15	P82914	P62753	MRPS16	Q9Y3D3	Q9H9Z2
MRPS17	Q9Y2R5	Q92845	MRPS2	Q9Y399	Q00839
MRPS24	Q96EL2	Q9P031	MRPS25	P82663	Q92845
MRPS30	Q9NP92	Q14197	MRPS34	P82930	Q9Y2I6
MRRF	Q96E11	P11182	MRTFB	Q9ULH7	P11831
MSANTD3	Q96H12	P06396	MSH3	P20585	P43246
MSRB3	Q8IXL7	Q9UKN5	MT-ATP8	P03928	P25090
MT-ND1	P03886	Q9P0J0	MT-ND2	P03891	O75380
MT2A	P02795	P09429	MTA1	Q13330	Q16665

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MTCH1	Q9NZJ7	P0DTC6	MTFP1	Q9UDX5	Q96CV9
MTG1	Q9BT17	Q9NRD5	MTHFD1L	Q6UB35	Q96GW9
MTHFD2	P13995	P49137	MTHFR	P42898	A0PJW6
MTMR4	Q9NYA4	P09917	MTMR6	Q9Y217	Q9Y216, Q96QG7
MTMR9	Q96QG7	Q13614	MTNAP1	Q9BSJ5	Q8N2W9
MTPN	P58546	P47756	MTR	Q99707	Q9Y4U1
MTREX	P42285	P52298, Q09161	MTSS2	Q765P7	P60709
MTX1	Q13505	Q05940	MUL1	Q969V5	P40855
MYADM	Q96S97	Q9Y624	MYC	P01106	P17480
MYD88	Q99836	Q15109	MYDGF	Q969H8	Q14554
MYEF2	Q9P2K5-2	Q9UGI0	MYH10	P35580	Q14164
MYL12A	P19105	P35579	MYL12B	O14950	P60709
MYO18A	Q92614	P19105	MYO1B	O43795	Q5S007
MYO3B	Q8WXR4	P08238	MYO5A	Q9Y4I1	P03372
MYO5B	Q9ULV0	P40855	MYO5C	Q9NQX4	P40855
MYO6	Q9UM54	P98082	MYO9A	B2RTY4	Q8TC07
MYPOP	Q86VE0	Q92845	MYRIP	Q8NFW9	P51159
MZT1	Q08AG7	Q6ZMU5	NAA15	Q9BXJ9	Q8WVK2
NAA40	Q86UY6	P49736	NAA60	Q9H7X0	P55212
NAB2	Q15742	P40937	NABP1	Q96AH0	Q5VWX1
NAGLU	P54802	P27824	NAP1L5	Q96NT1	P18848
NAPA	P54920	Q12981, Q9NZ43, Q9P2W9, O75396	NAPG	Q99747	Q9P2W9, O75396
NAT8L	Q8N9F0	P40855	NBN	O60934	Q13315
NBR1	Q14596	P40818	NCAPG	Q9BPX3	P04201
NCBP1	Q09161	Q86U42	NCDN	Q9UBB6	Q15185
NCEH1	Q6PIU2	P27824	NCF2	P19878	Q92845
NCK1	P16333	Q07889	NCK2	O43639	Q05209
NCOA1	Q15788	Q13569	NCOA2	Q15596	O00482-1
NCOA5	Q9HCD5	P36873	NCOA7	Q8NI08	Q9GZQ8
NCR3LG1	Q68D85	P47813	NCS1	P62166	Q9UKN5
NDC80	O14777	Q8IY31, Q8WYA0	NDOR1	Q9UHB4	Q9UBE8
NDRG1	Q92597	Q9BYZ2	NDRG4	Q9ULP0-2	O60906
NDUFA11	Q86Y39	O75489	NDUFA2	O43678	Q9P0J0
NDUFA5	Q16718	Q9P0J0	NDUFA8	P51970	Q9P0J0
NDUFAF2	Q8N183	O75396	NDUFAF5	Q5TEU4	P01185
NDUFB10	O96000	P05455	NDUFS4	O43181	P19526
NECTIN3	Q9NQS3-1, Q9NQS3	P15151, Q15223	NEDD1	Q8NHV4	Q8IW35, O43303
NEDD4	P46934-4	Q9HAU4	NEDD4L	Q96PU5-5	O15105
NEGR1	Q7Z3B1	P61916	NEIL3	Q8TAT5	Q15185
NEK1	Q96PY6	P04004	NEK2	P51955	Q8NG66
NEK3	P51956	P32971	NEK7	Q8TDX7	Q9ULW0
NELFCD	Q8IXH7	Q8WX92	NEPRO	Q6NW34	P62753
NEU1	Q99519	Q8NBQ5	NF1	P21359	P05067
NFIB	O00712	P50458	NFKBIB	Q15653	P19838
NFU1	Q9UMS0	P21741	NGF	P01138-PRO_0000019600, P01138	P08138
NGRN	Q9NPE2	P09651	NHEJ1	Q9H9Q4, Q9H9Q4-1	Q13426
NHERF2	Q15599	Q9BXI6	NHLRC2	Q8NBF2-2	P04792

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NHSL1	Q5SYE7	P16333	NHSL3	Q9P206	Q9UQB8
NID2	Q8IV28	P10745	NIF3L1	Q9GZT8	P42772
NIN	Q8N4C6	P49841	NINJ1	Q92982	Q16585
NIPAL1	Q6NVV3	Q09328	NIPAL4	Q0D2K0	P21964
NIPBL	Q6KC79	P45973	NIPSNAP2	O75323	Q05513
NIPSNAP3A	Q9UFN0	P10809	NKTR	P30414	Q15696
NLN	Q9BYT8	Q5SW96	NLRC5	Q86WI3	O15111, O14920
NLRP1	Q9C000	P29466	NLRX1	Q86UT6	P49411
NMB	Q14956-1, Q14956	P00533	NME2	P19804	Q05193
NME4	O00746	Q9NPB3	NME5	P56597	P42858
NMI	Q13287	Q96QG7	NMT2	O60551	Q9BY76
NOC3L	Q8WTT2	P19525	NOC4L	Q9BVI4	O00410
NOP10	Q9NPE3	Q9NPB3	NOP16	Q9Y3C1	O00410
NOP2	P46087	P04733	NPAS2	Q99743	O43639
NPAT	Q14207	P16104	NPC2	P61916	P00751
NPHP1	C9J082	P01019	NPHP3	Q7Z494	A6NIH7
NPIPB3	Q92617	P59634	NPPA	P01160	Q6GTS8
NPPB	P16860	P25788	NR1D1	P20393	P06727
NR1D2	Q6NSM0	Q8WY64	NR1H2	Q60644	Q96EB6
NR2F6	P10588	P83916	NR3C1	P06536	Q15788
NR3C2	P08235	Q9UBK2	NRAS	P01111	P41440
NRBP1	Q9UHY1	P51530	NRDC	O43847	P04637
NRG1	Q02297-10	P21854	NRIP1	P48552	Q9Y6K1
NRP1	P97333	P08648	NRROS	Q86YC3	P11021
NSF	P46459	P59635	NSMAF	Q92636	P60709
NSMCE3	Q9CPR8	P08138	NSMF	Q9EPI6	P27361
NSRP1	Q9H0G5	Q8WUA2	NSUN3	Q9H649	Q6PJG9
NSUN4	Q96CB9	O95861	NT5E	P21589	P36404
NTMT1	Q9BV86	P18754	NTN4	Q9HB63	P15056
NTPCR	Q9BSD7	Q8NEZ3, Q96RY7	NUBP1	P53384	Q9Y5Y2
NUBPL	Q8TB37	Q9Y5Y2	NUCB2	P80303	P24522
NUCKS1	Q9H1E3	Q12824	NUDT14	O95848	O95848
NUDT16	Q96DE0	P27930	NUDT21	O43809	O43809, Q16630-1
NUDT6	P53370	P46199	NUF2	Q9BZD4	Q9Y3P9
NUFIP2	Q7Z417	Q9UBX5	NUP133	Q8WUM0	P55735
NUP153	P49790	Q9UPY3	NUP54	Q7Z3B4	Q8IY31
NUP58	Q9BVL2	Q8WUX9	NUSAP1	Q9BXS6	Q14974
NXPE3	Q969Y0	Q6IN84	NXT1	Q9UKK6	Q9Y3Y2
OAF	Q86UD1	P56559	OARD1	Q9Y530	P09874
OAZ1	P54368	O14977	OBSL1	O75147	Q9UBX5
OCIAD1	Q9NX40	Q9Y2C2	OCRL	Q01968	Q9NYZ3
ODAD2	Q5T2S8	P43354	ODR4	Q5SWX8	Q96FT7
OFD1	O75665	Q4AC94	OGFOD3	Q6PK18	P27824
OGFRL1	Q5TC84	P11802	OLA1	Q9NTK5	P04591
OLFM2	O95897	O14641	OPA1	O60313	Q5S007
ORC3	Q9UBD5	Q13415	OSBPL11	Q9BXB4	P09917
OSBPL5	Q9H0X9	P54802	OSBPL6	Q9BZF3	P53350
OSBPL8	Q9BZF1	O15530	OSBPL9	Q96SU4	Q92685
OTUD4	Q01804	P04004	OXA1L	Q2M1J6	P17252

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OXLD1	Q5BKU9	Q4G176	OXNAD1	Q96HP4	P10809
OXR1	Q8N573-5	Q14145	P2RY6	Q15077	Q14974
P3H1	Q32P28	Q70UQ0	P3H2	Q8IVL5	O95076
P4HA1	P13674	Q6UN15	PACS1	Q6VY07	Q9NZ52
PACS2	Q86VP3	P63104	PADI1	Q9ULC6	Q9UGK3
PADI3	Q9ULW8	Q9Y496	PAF1	Q8N7H5	P23193
PAFAH1B1	P43034	P07900	PAFAH1B3	Q15102	Q8WUX9
PAK3	O75914	O14907	PALD1	Q9ULE6	Q92985
PALS1	Q8N3R9, Q8N3R9-1	P59637	PAN3	Q58A45	P11940
PANX2	Q495U3	P06396	PAPSS1	O43252	P52294
PAPSS2	O95340	Q96CV9	PAQR5	Q9NXX6	O95470
PARD6A	Q9NPB6	Q05513	PARD6B	Q9BYG5	Q05513
PARP14	Q460N5	O43914	PATL1	Q86TB9	Q8IZH2
PAX8	Q06710	O00712, Q12857	PBK	Q96KB5	Q14160
PBLD	P30039	O75791	PBRM1	Q86U86	P46531
PBX1	P40424	Q12948	PBX3	Q96AL5	O60906
PBX4	Q9BYU1	Q00403	PC	P11498	P48163
PCBP1	Q15365	P11940	PCBP4	P57723	P11142
PCDH7	O60245	P34925	PCDHGC3	Q9UN70	P12830
PCED1A	Q9H1Q7	Q7Z3S9	PCGF1	Q9BSM1	Q03111
PCGF2	P35227	Q93009	PCM1	Q15154	Q8IW35, O43303
PCMTD1	Q96MG8	Q93034, Q9UBF6	PCNX2	A6NKB5-5	Q8WTS1
PCOTH	Q58A44	P25788	PCSK7	Q16549	P17612
PCYOX1L	Q8NBM8	P27824	PCYT1A	P49585	Q8N0U8
PCYT1B	Q9Y5K3-3	Q8N0U8	PCYT2	Q99447	O15357
PDCD1LG2	Q9BQ51	Q9NZQ7	PDCD2	Q16342	O60260
PDCD4	Q53EL6	P02649	PDCD6IP	Q8WUM4	P12931
PDCD7	Q8N8D1	Q15696	PDE2A	O00408	O00170
PDF	Q99988	Q13185	PDIA5	Q14554	Q99538
PDIA6	Q15084	P11021	PDK1	Q15118	P00403
PDK3	Q15120	P10515, Q15119	PDLIM4	P50479	P12814
PDLIM5	Q96HC4	O75386	PDLIM7	Q9NR12	Q96QF0
PDS5A	Q29RF7	O60216, Q7Z5K2, Q96FF9	PDXDC1	Q6P996	P13569
PDZD2	O15018	P59637	PEA15	Q15121	P28482
PEAK1	Q9H792	P46109	PELI3	Q8N2H9-4	P54652
PELP1	Q8IZL8	P19838	PER1	O15534	P49674
PER2	O15055	P49674	PEX1	O43933	O75925
PEX19	P40855	Q9NZQ7	PEX3	P56589	Q9Y5Y5, P40855
PEX5	P50542	O14734	PFDN2	Q9UHV9	Q13164
PFKFB2	O60825	P31946	PFKFB3	Q16875	P31946
PFKFB4	Q16877	P31946	PFKL	P17858	Q969Y2
PFKM	P08237	Q8IUQ4	PFKP	Q01813	Q8NEZ3, Q9HBG6
PGAP4	Q9BRR3	Q6ZMH5	PGAP6	Q9HCN3	P0DPK4
PGGT1B	P53609	Q9NQ87	PGRMC1	O00264	P05177
PHACTR2	O75167	P60709	PHC3	Q8NDX5	O15294
PHF11	Q9UIL8	O00560	PHF20L1	A8MW92	P62805
PHF23	Q9BUL5	P99999	PHF5A	Q7RTV0	O76024
PHKB	Q93100	Q8WUJ3	PHLDB2	Q86SQ0	Q93009
PHLPP2	Q6ZVD8	Q8TAF3, O94782	PHYH	O14832	O75925
PI4K2A	Q9BTU6	Q96J02	PI4K2B	Q8TCG2	Q03405

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PIAS1	O75925	O00628, O14832	PIAS4	Q8N2W9	O60551
PICALM	Q13492-2, Q13492	Q9NYZ3	PID1	Q7Z2X4	Q8N9N5
PIGB	Q92521	P19526	PIGN	O95427	P43119
PIGQ	Q9BRB3	Q9P0P0	PIGV	Q9NUD9	P50454
PIH1D2	Q8WWB5	O60469	PIK3C2A	O00443	O95630, Q96FJ0
PIK3C3	Q8NEB9	Q9P2Y5	PIK3CB	P42338	Q13485
PIK3R1	P27986-2, P27986	P12931	PIK3R2	O00459	P03496
PIM1	P11309-2	Q9UNQ0	PINX1	Q96BK5	O14746
PIP4K2A	P48426	P15907	PIP4K2B	P78356	Q15831
PIP4K2C	Q8TBX8	Q15831	PIP4P1	Q86T03	P05919
PIP5K1C	O60331	P53611	PIR	O00625	P02675
PISD	Q9UG56	Q8NEZ3, Q96RY7	PITPNB	P48739	P55212
PJA2	O43164	P13569	PKIB	Q9C010-2	P17612
PKIG	Q9Y2B9	P17612	PKN1	Q16512	P26641
PKN2	Q16513	O76075	PLA2G12A	Q9BZM1	O76024
PLAA	Q9Y263	P55072	PLAGL2	Q9UPG8	O94955
PLAT	P00750	Q04721	PLB1	Q6P1J6-2	P17252
PLCG1	P19174	P16234	PLD1	Q13393	P23528
PLD6	Q8N2A8	P40855	PLEKHA1	Q9HB21	Q9P0V3
PLEKHA5	Q9HAU0	Q9NWB1	PLEKHA7	Q6IQ23-2	Q15669
PLEKHG3	A1L390	P31946	PLEKHM1	Q9Y4G2	P51149
PLEKHM3	Q6ZWE6	P63104	PLEKHO1	Q53GL0	Q9HCE7
PLIN2	Q99541	Q8WTS1	PLK2	Q9NYY3	Q13177
PLOD1	Q02809	P55317	PLOD2	O00469	Q9Y230
PLPP2	O43688	Q7Z3S9	PLPP3	O14495	O60716
PLPP4	Q5VZY2	P17302	PLSCR3	Q9NRY6	Q15915
PLXNA1	Q9UIW2	Q01523, P59665	PLXNB1	O43157	P52198, Q92730, P61587
PLXNB3	Q9ULL4	P08581	PLXNC1	O60486	O75326
PMP22	A0A6Q8PF08	Q969S8	PMPCB	O75439	P01160
PNKD	Q8N490-3	Q13153	PNMA1	Q8ND90	Q96L34
PNMA2	Q9UL42	Q96L34	PNN	Q9H307	Q7Z6E9
PNP	P00491	P17612	PNPLA6	Q8IY17	Q8TDQ0
PNRC2	Q9NPJ4	Q92900	POC5	Q8NA72, Q8NA72-3	P41208
POGK	Q9P215	P08238	POGLUT3	Q7Z4H8	P0DTC8
POLD1	P28340	Q13485	POLD4	Q9HCU8	P12004
POLDIP2	Q9Y2S7	P13196	POLDIP3	Q8WUT1	Q969Y2
POLI	Q9UNA4	Q15843	POLR1C	O15160	P23025
POLR1D	P0DPB6	O15160, Q9GZS1, O95602, O15446, Q9H9Y6, P19388, P61218, P62875	POLR1F	Q3B726	Q9NYV6
POLR2B	P30876	Q9Y5B0	POLR2H	P52434	Q8IXW5
POLR2J	P52435	Q9Y5B0	POLR3GL	Q9BT43	O15160, P19388, P61218, P62875
POLR3K	Q9Y2Y1	O15160, P19388, P61218, P62875, P52434	POLRMT	O00411	Q00059
POMT2	Q9UKY4	P37268	PON2	Q15165	Q12891
POP7	O75817	Q96S44	POT1	Q9NUX5	P50747
POU2F1	P14859	P14859	POU3F2	P20265	P56693
PPARGC1A	Q9UBK2	P37231	PPFIBP1	Q86W92	P31946

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PPFIBP2	Q8ND30	P46108	PPHLN1	Q8NEY8	P84103
PPIC	P45877	Q8N9N5	PPIF	P30405	P53365
PPIG	Q13427	P82979	PPIL3	Q9H2H8	P02545
PPIL4	Q8WUA2	O60563	PPIP5K1	Q6PFW1	P61981
PPIP5K2	O43314	Q13257	PPM1G	O15355	O96017
PPM1H	Q9ULR3	P31946	PPP1CB	P62140	Q9UQK1
PPP1R12A	Q10728	P35240	PPP1R12C	Q9BZL4	Q9UI12
PPP1R15A	O75807	P62136	PPP1R16A	Q96I34	Q92696, P53611
PPP1R21	Q6ZMI0-5	P99999	PPP1R37	O75864	O75593
PPP1R3B	Q86XI6	P36873	PPP1R3E	Q9H7J1	P31946
PPP1R3F	Q6ZSY5	P61964	PPP1R9A	Q9ULJ8	Q9UM54
PPP1R9B	Q96SB3	P36873	PPP2R1B	P30154	Q9BRV8
PPP2R2B	Q00005	O15530	PPP2R3A	Q06190	Q9NYZ3
PPP2R3B	Q9Y5P8	Q9HC29	PPP2R3C	Q969Q6	O95197
PPP2R5C	Q13362-3, Q13362-2, Q13362-1	O96017	PPP2R5E	Q16537	O96017
PPP3CA	Q08209	P49841	PPP4C	P60510	O00267
PPP4R2	Q9NY27	Q9Y570	PPP4R3B	Q5MIZ7	Q92620
PPP6C	O00743	O43318	PPP6R1	Q9UPN7	Q14694
PPP6R3	Q5H9R7	O14965	PPWD1	Q96BP3	P55735
PRC1	O43663	P36941	PRCC	Q92733	P38646
PRDX2	P32119	P10599	PRDX4	Q13162	P18428
PREB	Q9HCU5	P27824	PRIMPOL	Q96LW4	O95714
PRKAA1	Q13131	O75385	PRKAA2	P54646	O75385
PRKAB1	Q9Y478	O75385	PRKAB2	O43741	P01185
PRKACB	P22694	P13861	PRKAG2	Q9UGJ0	Q13501
PRKCD	Q05655	P29353	PRKCI	P41743	Q8N5V2
PRKD3	O94806	P63027	PRKDC	P78527	P46531
PRKN	O60260-5	O94844	PRMT6	Q96LA8	Q9Y365
PRP4K	Q13523	P32121	PRPF18	Q99633	Q9BUH8
PRPF19	Q9UMS4	O75533	PRPF3	O43395	Q13107
PRPF40A	O75400-2	Q86US8	PRR11	Q96HE9	Q15361
PRR14	Q9BWN1	Q13185	PRR14L	Q5THK1	Q15185
PRR19	A6NJB7-2	Q96RU7	PRR5	P85299-2	O14964
PRTFDC1	Q9NRG1	Q8NFF5	PRXL2A	Q9BRX8	Q9UNA3
PSAP	Q8IWL2	Q9UGM3	PSD3	Q9NYI0	P63104
PSD4	Q8NDX1-2	Q5SW96	PSEN2	P49810	P30626
PSIP1	O75475	Q03164	PSMA1	P25786	P57764
PSMA3	P25788	P27449	PSMB4	P28070	Q9Y5K5
PSMB9	P28065	Q15788	PSMC1	P62191	O00487
PSMC2	P35998	O60216	PSMC6	P62333	P46109
PSMD1	Q99460	O00487	PSMD12	O00232	O00487
PSMD14	O35593, O00487	Q9Y5K5, Q16186	PSMD6	Q15008	O00487
PSMD9	O00233	P08887-2	PSME4	Q14997	P60896
PSMG3	Q9BT73	Q9BQB4	PTAR1	Q7Z6K3	O15498
PTBP1	P26599	Q9NZI8	PTBP3	O95758	Q92900
PTCD2	Q8WV60	P04004	PTCD3	Q96EY7	Q16629
PTDSS2	Q9BVG9	P35414	PTEN	P60484	P60484
PTK2	P34152, Q05397	P12931	PTK2B	Q14289	P12931
PTPN1	P18031	P12931	PTPN11	Q06124	Q8WU20

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PTPN12	Q05209	P29353	PTPN13	Q12923	P59637
PTPN23	Q9H3S7	P12931	PTPN4	P29074	Q9BY67
PTPRD	P23468	O14965	PTPRH	Q9HD43	P10912
PTPRJ	Q62884	P12931	PTRH2	Q9Y3E5	Q9H2J7
PTRHD1	Q6GMV3	Q9H244	PTTG1IP	P53801	Q16585
PUF60	Q9UHX1	P05455	PUSL1	Q8N0Z8	P0DTC8
PVR	P15151	O43929	PWWP2B	Q6NUJ5	Q13547
PYGL	P06737	Q13451	QPCTL	Q9NXS2	P40855
QPRT	Q15274	O00443	QSOX2	Q6ZRP7	Q9NRD5
R3HDM1	Q15032	P31946	RAB11FIP2	Q7L804	P31946
RAB13	P51153	P0DTC3	RAB1B	Q9H0U4	Q92696
RAB21	P35282	P17301	RAB24	Q969Q5	Q14353
RAB27A	P51159	Q8IYJ3	RAB27B	O00194	Q8IYJ3
RAB29	O14966	Q92696	RAB30	Q15771	Q92696
RAB32	Q13637	Q92696	RAB38	P57729	Q5S007
RAB3GAP1	Q15042	P09917	RAB3GAP2	Q9H2M9	P13569
RAB3IL1	Q8TBN0	Q96QF0	RAB5C	P51148	Q9UH99
RAB5IF	Q9BUV8	O60704	RAB7A	P51149	O75348, P21281, P61421
RAB8B	Q92930	Q9UK39	RAB9A	P51151	Q92696, P53611
RABEP1	Q15276	Q9NZ52	RABGAP1L	Q5R372	Q9Y3P9
RABGEF1	Q9UJ41	Q9H0F6	RABGGTB	P53611	O15498
RABIF	P47224	Q9NZU5	RABL2B	Q9UNT1-2	P06396
RABL3	Q5HYI8	Q9NRZ7	RABL6	Q3YEC7-3	P17252
RAC3	P60763	P52565	RAD21	O60216	Q7Z5K2, Q96FF9
RAD9A	Q99638	P40937	RADX	Q6NSI4	P40692
RAI14	Q9P0K7	P53350	RALA	P11233	P13569
RALGAPA1	Q6GYQ0	P63104	RALGPS2	Q86X27	P31946
RANBP2	P49792, Q9ERU9	P46060, P18754	RANBP3	Q9H6Z4	P35813
RANGRF	Q9HD47	Q8N307	RAP1GDS1	P52306	P17480
RAPGEF1	Q13905	P46109	RAPGEF2	Q9Y4G8	P46937
RAPH1	Q70E73	Q99962	RARB	P10826-2	Q9UBK2
RARS2	Q5T160	P21462	RASA1	P20936, P50904	Q13480
RASEF	Q8IZ41	P13569	RASL10B	Q96S79	Q9H3L0
RAVER1	Q8IY67	O75791	RAX	O75569	P03496
RB1	P06400	O96017, O14757	RBAK	Q9NYW8	O95995
RBKS	Q9H477	Q9H477	RBM15B	Q8NDT2	Q86VM9
RBM19	Q9Y4C8	P16104	RBM22	Q9NW64	P62306, P62304
RBM23	Q86U06	Q9Y5V3	RBM24	Q9BX46	P05120
RBM25	P49756	Q86VM9	RBM26	Q5T8P6	Q9Y4Z2
RBM3	P98179	Q9Y6M1	RBM41	Q96IZ5	Q9Y2I6
RBM45	Q8IUH3	Q9Y316	RBM5	P52756	P27797
RBMS1	P29558	P84103	RBPJ	Q06330-6, Q06330	P46531
RBPM5	Q93062	Q53HV7	RCAN3	Q9UKA8	Q13322
RCC1	P18754	O14980	RCC1L	Q96I51	O00425
RCN2	Q14257	Q9Y4K3	RCOR1	Q9UKL0	O95863
RDX	P35241	P52565	REC8	O95072	Q9UNE7
RECK	Q6P9E2	Q96LC9	RECQL	P46063	P52292
REEP3	Q6NUK4	P31946	REL	Q04864-2	P30622
RELCH	Q9P260	P21709	RELT	Q969Z4	Q8NFM7

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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REST	Q8VIG1	Q96MT3	RETREG1	Q9H6L5-1	Q9GZQ8
RETREG2	Q8NC44	Q9GZQ8	REXO4	Q9GZR2	P40429
REXO5	Q96IC2	Q15185	RFC1	P35251	P35250, P40938, P40937, P35249
RFFL	Q8WZ73	P29353	RFX1	P22670	Q13547
RFX7	Q2KHR2	P31946	RGPD2	P0DJD1	P25103
RGPD3	A6NKT7	Q96PC5	RGPD5	Q99666	P61769
RGPD6	Q99666	P61769	RGS19	P49795	Q12837
RGS20	O76081-6	Q8WUK0	RHBDD1	Q8TEB9	O75396
RHBDD3	Q9Y3P4	Q6UWB1	RHBDL2	Q9NX52-3	Q969S0
RHEB	Q15382-PRO_0000082708	O43924	RHOF	Q9HBH0	P52306
RHOG	P84095	P52565	RHOT1	Q8IXI2	O60260
RHPN2	Q8IUC4	P61981	RIC1	Q4ADV7	P20340
RICTOR	Q6R327	P22891	RIF1	Q9Y581	P98164
RILP	Q96NA2	P51149	RILPL1	Q5EBL4-3	Q86T03
RIMKLA	Q8IXN7	Q96EN8	RIN3	Q8TB24	P46109
RINT1	Q6NUQ1	Q12981, Q9NZ43, Q9P2W9, O75396	RIOK2	Q9BVS4	Q13895
RIOK3	O14730	Q92851	RITA1	Q96K30	Q9NPI6
RLF	Q13129	Q13185	RLIM	Q9NVW2	Q9H0E2
RMDN1	Q96DB5	P05181	RMDN2	Q96LZ7	Q9Y4Z2
RMRP	EBI-9875627	O14746	RNASEH2C	Q8TDP1	P11142
RNASEK	Q6P5S7	Q5QGT7	RNF10	Q8N5U6	P51668
RNF14	Q9UBS8	P51668	RNF141	Q8WVD5	Q9H4B4
RNF144A	P50876	Q96AA3	RNF144B	Q7Z419	Q07812
RNF145	Q96MT1	P0DTC5	RNF166	Q96A37	P36941
RNF170	Q96K19	Q13370	RNF182	Q8N6D2	P51668
RNF185	Q96GF1	P51668	RNF19A	Q9NV58	P00441
RNF20	Q5VTR2	O96017	RNF220	Q9BWG1	Q96CV9
RNF26	Q9BY78	P62837	RNF5	Q99942	P51668
RNF7	Q9UBF6	Q8WXH5	RNGTT	O60942	P52294
RNLS	Q5VYX0	P23634	RO60	P10155	P05455
ROBO1	Q9Y6N7	O75044	ROCK1	Q13464	P61586
ROCK2	O75116	O15530	ROGDI	Q9GZN7	P21281
RORA	P35398-4	Q9BZS1	RPAP1	Q9BWH6	Q9NUX5
RPAP2	Q8IXW5	Q15648	RPAP3	Q9H6T3	P36873
RPGR	Q92834-2, Q92834	O43924	RPGRIP1L	Q68CZ1-2, Q68CZ1	O75161
RPL13	P26373	Q15361	RPL18	Q07020	P40429
RPL18A	Q02543	Q15361	RPL19	P84098	P40429
RPL27	P61353	Q99558	RPL32	P62910	Q99558
RPL35A	P18077	P17252	RPL37A	P61513	P17612
RPL38	P63173	P63244	RPLP1	P05386	O43809
RPLP2	P05387	P62753	RPN1	P04843	Q14165
RPP25L	Q8N5L8	Q9UPY3	RPRD1A	Q96P16-3	Q9UBU9
RPRM	Q9NS64	O60493	RPS10	P46783	Q9H2U1
RPS11	P62280	Q04637	RPS14	P62263	Q15361
RPS15	P62841	P62937	RPS16	P62249	Q8IZH2
RPS17	P08708	P63244	RPS19	P39019	P11309
RPS19BP1	Q86WX3	P42357	RPS20	P60866	Q15843

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
RPS21	P63220	P63244	RPS27	P42677	Q00987
RPS28	P62857	P16104	RPS29	P62273	P63244
RPS3	P23396	Q15843	RPS5	P46782	P16104
RPS6KA3	P51812	Q12849	RPS6KA6	Q9UK32	P27361
RPS6KB1	P23443-2	P08151	RPTOR	Q8N122	O75385
RPUSD2	Q8IZ73	Q9H2A2	RPUSD4	Q96CM3	Q00613
RRAGA	Q7L523	P61916	RRAGC	Q9HB90	P17931
RRAGD	Q9NQL2	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4	RRAS	P10301	Q9UBH0
RREB1	Q92766	O15379	RRM1	P23921	P31350
RRM2	P31350	Q9UM11	RRM2B	Q7LG56	Q13315
RRN3	Q9NYV6	Q8NFM7	RRP1	P56182	Q13724
RRS1	Q15050	P40429	RSBN1	Q5VWQ0	Q9BYW2
RSF1	Q96T23	P09429	RSPRY1	Q96DX4	Q14435
RSRC2	Q7L4I2-2	Q08379	RTF1	Q92541	P49407
RTL10	Q7L3V2	P21796	RTL6	Q6ICC9	Q9P2W9
RTL9	Q8NET4	Q00613	RTN3	O95197-3	Q9NTJ5
RUFY1	Q96T51	Q96QF0	RUFY2	Q8WXA3	Q9Y639
RWDD2B	P57060	Q96HC4	RXRB	P28702	P37231
S100A13	Q99584	Q92845	S100A16	Q96FQ6	Q15915
S100A2	P29034	P35579	S100A4	P26447	P15514
S100A6	P06703	P35579	S1PR3	Q99500	P21964
SACM1L	Q9NTJ5	Q9H244	SACS	Q9NZJ4	Q6UWR7
SAE1	Q9UBE0	Q06710	SAFB	Q15424	P09651
SAFB2	Q14151	O75152	SAMD1	Q6SPF0	Q13309
SAP130	Q15393	P53350	SAP30L	Q9HAJ7	Q92769, Q13547
SAR1B	Q9Y6B6	Q9Y5M8	SARM1	Q6SZW1	Q99836
SAT2	Q96F10	P21673	SBF2	Q86WG5	Q13614
SBSN	Q6UWP8	P15498	SCAF11	Q99590	Q07955
SCAF8	Q9UPN6	Q08379	SCAMP1	O15126	Q969Z0
SCAMP2	O15127	P06576	SCAMP3	O14828	P80370
SCAND1	P57086	Q6ZPD8	SCAP	Q9NP31	O14543
SCAPER	Q9BY12	P20248	SCFD1	Q8WVM8	P16234
SCN3B	Q9NY72	P21359	SCOC	Q9UIL1-3	Q9BRV8
SCP2	P22307	Q03135	SCRIB	Q14160	Q13177
SCRN3	Q0VDG4	Q8NB12	SCX	Q7RTU7	O43639
SCYL1	Q96KG9	O94979	SCYL2	Q6P3W7	O14640
SDC3	A0A0S2Z4U3	P63252	SDCCAG8	Q86SQ7	P61088
SDF4	Q9BRK5	P50548	SDHAF1	A6NIFY7	Q8IWL3
SDHAF2	Q9NX18	P31040	SDHB	P21912	Q8IWL3
SEC22A	Q96IW7	P35232	SEC23A	Q15436	O94979, P55735
SEC23IP	Q9Y6Y8	Q9Y547	SEC24B	O95487	Q15436
SEC31A	O94979	Q15436, P55735	SEC62	Q99442	O43914
SEC63	Q9UGP8	Q8N6T3	SEL1L	Q9UBV2	Q16288
SELENOO	Q9BVL4	Q6IN84	SEMA3C	Q99985	P18827
SEMA4A	Q9H3S1	Q9Y4D7	SEMA4C	Q9C0C4	O14908
SEMA4D	Q92854	Q9ULL4	SENP3	Q9H4L4	Q8IZL8
SENP7	Q9BQF6	Q13185	SEPSECS	Q9HD40	P04591
SEPTIN10	Q9P0V9	Q15019	SEPTIN11	Q9NVA2	Q15019

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SEPTIN2	Q15019-2	Q9UNE7	SEPTIN7	Q16181	Q15019
SEPTIN8	Q92599	Q15019	SERINC1	Q9NRX5	Q06710
SERPINE1	P05121	Q8NBQ5	SESTD1	Q86VW0	Q15561
SETD1A	O15047	P51610	SETD1B	Q9UPS6	Q9UER7
SETD2	Q9BYW2	P00740	SETD7	Q8WTS6	P26358
SETX	Q7Z333	P63279	SF1	Q13285	Q9NPJ4
SF3B1	O75533	O43809	SFT2D2	O95562	Q8NBQ5
SGCB	Q16585	P01375	SGF29	Q96ES7	Q6VY07
SGK1	O00141	P49815	SGK3	Q96BR1	Q96J02
SGO2	Q562F6	Q13257	SGTA	O43765	P59635
SGTB	Q96EQ0	P10451	SH2B1	Q9NRF2	O60674
SH3BGR1	O75368	P00533	SH3BP4	Q9P0V3	O14908
SH3BP5L	Q7L8J4	Q9NWX6	SH3D21	A4FU49	P00533
SH3GL1	Q99961	Q8WUM4	SH3GLB1	Q9Y371	P02647
SH3RF2	Q8TEC5	Q9ULX6	SHC3	Q92529	Q86U10
SHCBP1	Q8NEM2	P05771	SHOC2	Q9UQ13	Q96RT1
SHPRH	Q149N8-1	P61088	SHTN1	A0MZ66	P31946
SIK1	P57059	Q9Y248, Q9BRT9	SIKE1	Q9BRV8	P67775
SIN3A	Q96ST3	P12755	SIRT6	Q8N6T7	Q04206
SIRT7	Q9NRC8	Q16656	SIX3	O95343	Q92570
SKA2	Q8WVK7	Q9NWB7	SKI	P12755	P36941
SKIC3	Q6PGP7	P06748	SKIC8	Q9GZS3	Q16143
SKIL	P12757	P23025	SLAMF7	Q8BHK6	P06241
SLC10A7	Q0GE19	P04114	SLC11A2	P49281-3	P00403
SLC12A4	Q9UP95	P20340	SLC16A13	Q7RTY0	Q96BA8
SLC16A7	O60669	P09601	SLC19A1	P41440	Q15836
SLC19A2	O60779	Q92911	SLC1A1	P43005	Q6IAN0
SLC20A1	Q8WUM9	Q9UM47	SLC22A15	Q8IZD6	Q86TM6
SLC22A5	O76082	Q9H015	SLC25A13	Q9UJS0	Q9Y5J6
SLC25A23	Q9BV35	Q8N695	SLC25A24	Q6NUK1	P13569
SLC25A40	Q8TBP6	P02452	SLC25A5	P05141	P12236
SLC27A4	Q6P1M0	P40855	SLC29A1	Q99808	P43119
SLC30A6	Q6NXT4	Q8TAD4	SLC30A7	Q8NEW0	Q9Y2C2
SLC30A9	Q6PML9	Q8TC07	SLC35B2	Q8TB61	Q96BA8
SLC35C1	Q96A29	P05090	SLC35E1	Q96K37	Q9Y366
SLC35F6	Q8N357	P38606	SLC36A1	Q7Z2H8	P05067
SLC38A10	Q9HBR0	Q30201	SLC38A7	Q9NVC3	P62495
SLC39A1	Q9NY26	P21964	SLC39A10	Q9ULF5	Q6ZMH5
SLC39A13	Q96H72	O15173	SLC39A14	Q15043	Q9NRZ7
SLC39A6	Q13433	Q6ZMH5	SLC39A7	Q92504	Q9UBV7, O75063
SLC39A8	Q9C0K1	Q09328	SLC3A2	P08195	P27824
SLC43A2	Q8N370	P0DPK4	SLC43A3	Q8NBI5	P55072
SLC44A2	Q8IWA5	O75695	SLC4A1AP	Q9BWU0	Q15696
SLC6A8	P48029	P43119	SLC7A11	Q9UPY5	P16070
SLC7A8	Q9UHI5	O60906	SLMAP	Q14BN4	Q9BRV8
SLU7	O95391	O75152	SLX4	Q8IY92	P53350
SLX4IP	Q5VYV7	P53350	SMAD5	Q99717	Q9HAU4
SMAD6	O43541-2, O35182	O95630	SMAD9	O15198, O15198-2	Q13485
SMARCA2	P51531	Q8NFD5	SMARCA5	O60264	P62805

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SMARCAD1	Q9H4L7	P43246	SMARCB1	Q12824	Q92925, Q8NFD5, P51531, O96019, O75177, Q15532
SMC2	O95347	P01106	SMC5	Q8IY18	Q9H668
SMC6	Q96SB8	Q9Y547, Q9BW83	SMCHD1	A6NHR9	Q96GD4
SMCO4	Q9NRQ5	P20645	SMG1	Q96Q15	Q92900
SMG5	Q9UPR3	Q92540	SMIM26	A0A096LP01	P06493
SMPD2	O60906	P49675	SMUG1	Q53HV7-2	Q8IVS8
SMURF2	Q9HAU4	Q9HAU4	SMYD3	Q9H7B4	Q16512
SNAP23	O00161	Q15836	SNAP47	Q5SQN1	Q9P2W9, O75396
SNAPC3	Q92966	O76024	SNRNP48	Q6IEG0	Q15696
SNRNP70	P08621	Q09161	SNRPA	P09012	O75152
SNRPB	P14678-2	P52747	SNX10	Q9Y5X0	P17252
SNX13	Q9Y5W8	P61073	SNX14	Q9Y5W7	P61073
SNX29	Q8TEQ0	Q13077	SNX30	Q8WV41	Q05193
SNX33	Q8WV41	Q05193	SNX5	Q9Y5X3	Q9Y639
SNX6	Q9UNH7	Q5T8I9	SNX8	Q9Y5X2	P16435
SOAT1	P35610	O75326	SOCs4	Q8WXH5	Q8N3R9
SOCs5	O75159	Q93034, Q9UBF6	SOCs6	O14544	P46937
SON	P18583	O75533	SORBS3	O60504	Q13177
SORT1	Q99523	P02787	SOS1	Q62245	P62993-1
SOS2	Q07890	P19174	SOX12	O60248	O00712, Q12857
SOX13	Q9UN79	Q12933	SOX9	P48436	Q12857
SP100	P23497	Q9Y6K1	SP140	Q13342	Q16236
SP3	Q02447	P03372	SP6	Q3SY56	P17612
SPACA9	Q96E40	O00471	SPAG4	Q9NPE6	O14653
SPAG5	Q96R06	Q15648	SPAST	Q9UBP0, Q9UBP0-1	Q9UF56
SPATA13	Q96N96-6	P51570	SPATA2L	Q8IUW3	P02649
SPATS2	Q86XZ4	P36941	SPC25	Q9HBM1	P42566
SPDL1	Q96EA4	P25963	SPEG	Q15772	P31946
SPG11	Q96JI7	Q05193	SPG21	Q9NZD8	P42772
SPHK1	Q9NYA1-2	Q7Z4W1	SPHK2	Q9NRA0	Q96CV9
SPIDR	Q14159	Q06609	SPIN1	Q9H2V7	Q07817
SPNS1	Q9H2V7	Q07817	SPOPL	Q6IQ16	Q99836
SPPL2A	Q8TCT8	O76024	SPPL3	Q8TCT6	Q9H015
SPRED2	Q7Z698	P27797	SPRING1	Q9H741	P40855
SPRY2	O43597	P22681	SPRY4	Q8WW59	P24534
SPRYD4	Q8WW59	P24534	SPRYD7	Q5W111	Q86UT5
SPSB1	Q96BD6	Q13153	SPSB2	Q99619	Q15369, Q15370, Q93034, Q9UBF6
SPTAN1	Q13813	Q00597, P40692	SPTLC1	O15269-2	Q8WXH5
SPTY2D1	Q68D10	P78563	SPX	Q9BT56	Q6IN84
SRCAP	Q6ZRS2	Q13257	SREK1IP1	Q8N9Q2	Q9NRZ7
SRF	P11831	Q969V6	SRP54	P61011	P11142
SRP72	O76094	P48431	SRPK1	Q9GSB4	Q9Y3Y2
SRRT	Q9BXP5	P09651	SRSF11	Q05519	Q92844
SS18	Q15532	P15336	SS18L2	Q9UHA2	Q9H8M2
SSBP2	P81877	Q9UBH0	SSBP4	Q9BWG4	Q9UBH0
SSNA1	O43805	Q9P0N9	SSR1	P43307	P27824
SSU72	Q9NP77	O60216	SSX2IP	EBI-11742485	Q96CJ1

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ST3GAL1	Q11201	Q9ULF5	ST7	Q9Y561	Q9NPI5
STAG2	Q8N3U4, Q8N3U4-2	O60216, Q7Z5K2, Q96FF9	STAM2	O75886	O95630, Q92783
STAMBP	O95630	Q9Y3E7	STAT1	P42224	P23458
STC2	O76061	P0DTC8	STIL	Q15468	Q9HC77
STING1	Q86WV6	Q96N66	STK39	Q9UEW8	P43119
STN1	Q9H668	Q9UBD5, Q13416	STOM	P27105	O43916
STOX2	Q9P2F5	P49674	STRN	O55106, O43815	Q9BRV8
STRN3	Q13033-2, Q13033	Q9BRV8	STT3B	Q8TCJ2	Q14165
STUB1	Q9UNE7	P08138	STX10	O60499-2	Q12846
STX11	O75558	O15498	STX12	Q86Y82	Q12981, Q9NZ43, Q9P2W9
STX16	O14662-5, O14662	Q12846	STX17	P56962	O00161
STX2	P32856-2	O75396	STX3	Q13277	P51809
STX4	Q12846	Q9NZ43, O75396	STX5	Q13190	Q9P2W9, O75396
STXBP3	O00186	Q12846	STXBP5	Q5T5C0	Q13277
STXBP6	Q8NFX7	Q13277	STYK1	Q6J9G0	P49841
SUCLG2	Q96I99	Q8IWL3	SUCO	Q9UBS9	P07237
SUGT1	Q9Y2Z0	Q8IW35	SULT4A1	Q9BR01	Q9NUX5
SUMO3	P55854	P63279	SUPT20H	Q8NEM7	Q16539
SUPT3H	O75486	Q92831	SUPT4H1	P63272	O00267
SUPT6H	Q7KZ85	O00267	SURF4	O15260	Q86WV6
SVIL	O46385	P35579	SYDE1	Q6ZW31	P62258
SYNE1	Q8NF91-1, Q8NF91	Q9UH99, O94901	SYNGAP1	Q96PV0	Q6VMQ6
SYNJ1	Q62910, O43426	Q99962	SYNJ2	O15056	P31946
SYNJ2BP	P57105	Q9NR61	SYS1	Q8N2H4	Q8NBQ5
SYT1	P21579	P98155	SYT17	Q9BSW7	P02787
SYTL3	Q4VX76	P51159	TAB2	Q9NYJ8	P52747
TAB3	Q8N5C8	O43318	TACC1	O75410, O75410-7	Q9Y316
TAF1	P21675	P20226	TAF1D	Q9H5J8	P06746
TAF2	Q6P1X5	P35249	TAF6	P49848	P20226
TAF6L	Q9Y6J9	P31948	TAF9B	Q9HBM6	P35249
TANGO2	Q6ICL3	Q99962	TANK	Q92844	P53350
TAOK1	Q7L7X3	P61981	TAOK2	Q9UL54	P36941
TAOK3	Q9H2K8	O75460	TAPBP	O15533	Q30201
TARBP2	Q15633	P03496	TASOR	Q9UK61	Q99549
TATDN2	Q93075	Q03405	TAX1BP1	Q86VP1	Q8TC07
TBC1D13	Q9NVG8	O60664	TBC1D20	Q96BZ9	P55056
TBC1D2B	Q9UPU7	P09917	TBC1D30	Q9Y2I9-2	Q9UGI0
TBC1D5	Q92609	Q9BZP6	TBC1D7	Q9P0N9	P23025
TBC1D9B	Q66K14	P21462	TBK1	Q9UHD2	P59596
TBKBP1	A7MCY6	Q9UKV8	TBRG4	Q969Z0	Q9UBN6
TCAF1	Q9Y4C2-2	P52747	TCAF2	A6NFBQ2	P80370
TCEA1	P23193	Q8WVC0	TCEAL1	Q15170	O14757
TCEAL4	Q96EI5	Q93009	TCEAL8	Q8IYN2	O76024
TCEAL9	Q9UHQ7	Q6VN20, Q96S59	TCF12	Q99081	P30626
TCF7	P36402	P35222	TCIRG1	Q13488	Q15904
TCP1	P17987	Q13882	TDRD7	Q8NHU6	P42771
TEAD1	P28347	Q06609	TECPRI	Q7Z6L1	Q9H1Y0, Q9NT62, O94817
TECR	Q9NZ01	Q00059	TEF	Q10587	Q16236

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TEN1	Q86WV5	Q9H668, Q2NKG3	TENT4B	Q8NDF8	Q07955
TERF1	P54274, P54274-2	Q9NUX5	TES	Q9UGI8	P13569
TESMIN	Q9Y4I5	P08235	TET2	Q6N021	P19544
TEX264	Q9Y6I9	Q13370	TEX9	Q8N6V9	Q9UBU9
TFAP4	Q01664	P18846	TFB1M	Q8WVM0	Q00059
TFDP1	Q14186	O75496	TGFA	P01135	P00533
TGFB1I1	O43294	Q14289	TGFB2	P61812	P05067
TGFBRI	P36897	P62942	TGFBR3	Q03167	P22455
TGFBRAP1	Q8WUH2	Q9UBF6	TGIF1	Q15583	P53350
TGM1	P22735	Q9Y5Y2	THAP11	Q96EK4	P51610
THAP12	O43422	Q13158	THAP4	Q8WY91	Q6ZPD8
THAP6	Q8TBB0	Q9NPB3	THAP7	Q9BT49	P12830
THAP8	Q8NA92	Q9UBB9	THBS1	P07996- PRO_0000035842	P16671
THEM6	Q8WUY1	Q8TC07	THOC3	Q96J01	Q13838
THOC6	Q86W42	Q13838	THOP1	P52888	Q96CV9
THRAP3	Q9Y2W1	Q16629, Q07955	THSD4	Q6ZMP0	P11142
THUMPD1	Q9NXG2	Q96PZ0	TIA1	Q80ZW7	Q05397
TICAM2	Q86XR7	O00206	TIGD5	Q53EQ6	Q6VN20
TIGIT	Q495A1	P15151, Q9NQS3	TIMM10	P62072	Q9Y5J6
TIMM13	Q9Y5L4	Q99558	TIMM29	Q9BSF4	P16050
TIMM44	O43615	Q00403	TJP1	Q07157	P59637
TKT	Q16832	P29353	TLCD1	Q96CP7	Q9H2J7
TLCD4	Q96MV1	O75396	TLE1	Q04724	Q13761
TLK1	Q9UKI8	P50458	TLN2	Q9Y4G6	P00519
TLNRD1	Q9H1K6	Q9NRD5	TLR1	Q15399	O60603
TLR3	O15455	P12931	TLR6	Q9Y2C9	O60603
TM9SF3	Q9HD45	Q9H244	TMA16	Q96EY4	Q92973
TMBIM4	Q7Z782	O95231	TMBIM6	P55061	P46695
TMED1	Q13445	Q9H2A9	TMED10	P49755	Q9Y3A6
TMED3	Q9Y3Q3	Q15363	TMED5	Q9Y3A6	Q9H2A9
TMED7	Q9Y3B3	P31371	TMED8	Q6PL24	Q12893
TMEM11	P17152	Q16585	TMEM115	Q12893	Q86U10
TMEM120A	Q9BXJ8	P61073	TMEM120B	A0PK00	Q05329
TMEM126B	Q8IUX1	Q9P0J0	TMEM128	Q5BJH2-2	Q99551
TMEM131	Q92545	P0DTC3	TMEM132A	Q24JP5	Q01523
TMEM147	Q9BVK8	Q9UKG4	TMEM150A	Q86TG1	O60563
TMEM161A	Q9NX61	O15354, Q9HC97	TMEM165	Q9HC07	P13569
TMEM167B	Q9NRX6	Q9NRZ7	TMEM168	Q9H0V1	P46109
TMEM182	Q6ZP80	O00501	TMEM183A	Q8IXX5	O14879
TMEM184B	Q9Y519	Q96RK4	TMEM186	Q96B77	Q71RS6
TMEM187	Q14656	P21964	TMEM192	Q8IY95	Q9UBV7
TMEM199	Q8N511	O15342, P27449	TMEM203	Q969S6	Q12846
TMEM223	A0PJW6	Q6ZMH5	TMEM225B	P0DP42	O95070
TMEM229B	Q8NBD8	O75396	TMEM242	Q9NWH2	Q9NP91
TMEM243	Q9BU79	Q9H2K0	TMEM255B	Q8WV15	Q6ZPD8
TMEM259	Q4ZIN3	Q96A29	TMEM30A	Q9NV96	P27824
TMEM38A	Q9H6F2	O60496	TMEM38B	Q9NVV0	Q8WTS1
TMEM47	Q9BQJ4	P42858	TMEM51	Q9NW97	Q96J02
TMEM60	Q9H2L4	Q9UBD6	TMEM63B	Q5T3F8	P43119

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TMEM87A	Q8NBN3	Q9H8X2	TMF1	P82094	P53350
TMOD2	Q9NZR1	Q9UM54	TMOD3	Q9NYL9	P60709
TMSB4X	P20065	Q8WWQ8	TMTC3	Q6ZXV5	Q6PIU2
TMX1	Q9H3N1	Q9NPF0	TMX3	Q96JJ7	P27824
TNC	P24821	P0DTC8	TNFAIP8	O95379	Q96JC9
TNFRSF14	Q92956	Q7Z6A9	TNIP1	Q15025	Q00403
TNIP2	Q8NFZ5	Q15831	TNK2	Q07912	Q8WXH5
TNKS	O95271	Q9NTX7	TNKS2	Q9H2K2	Q9NTX7
TNPO1	Q92973	P52272	TNRC6A	Q8NDV7	P11940
TNRC6B	Q9UPQ9	P11940	TNS1	Q04205	P56945
TOLLIP	Q9H0E2	Q96PU5	TOM1L1	O75674	P56539
TOMM70	O94826	P59636	TOP2A	P11388	P78527
TOP2B	Q02880	P46531	TOP3A	Q13472	P23025
TOP3B	O95985	Q9Y2T1	TOR1AIP1	Q5JTV8	Q13257
TOR1B	O14657	P32971	TP53	P04637	P35232
TP53TG5	Q9Y2B4	P61289	TPD52	P55327	Q86T03
TPD52L1	Q16890	P31946	TPM1	P09493-10, P09493	Q9Y6D9
TPP1	Q96AP0	Q92900	TPP2	P29144	P06746
TPR	P12270	Q9Y6D9	TPRN	Q4KMQ1-2	O15247
TPST2	O60704	Q9BUV8	TPX2	Q9ULW0	O75330
TRAF3IP2	O43734	Q9Y4K3	TRAF5	O00463	Q92956
TRAF7	Q6Q0C0	Q9NWB7	TRAK1	Q9UPV9	P53350
TRAPPC11	Q7Z392	Q9P2M4	TRAPPC2L	Q9UL33	Q96QF0
TRAPPC3	O43617	Q96QF0	TRAPPC5	Q8IUR0	Q9P2M4
TRAPPC6B	Q86SZ2	Q9P2M4	TRAPPC9	Q96Q05	Q96QF0
TRIM25	Q14258	P03496	TRIM27	P14373	Q96E11
TRIM28	Q13263	O15397	TRIM3	O75382	P51668
TRIM32	Q13049	Q06124	TRIM36	Q9NQ86	Q14451
TRIM41	Q8WV44	Q05513	TRIM47	Q96LD4	Q9UPW6, Q01826
TRIM59	Q8IWR1	P0DTC3	TRIM65	Q6PJ69	P09917
TRIO	O75962	Q9NV70	TRIP11	Q15643	P31946
TRIP12	Q14669	Q8N726	TRIP4	Q15650	P63244
TRIR	Q9BQ61	O75533	TRMO	Q9BU70	P12931
TRMT12	Q53H54	P11142	TRMT2A	Q8IZ69	P05455
TRNT1	Q96Q11-3	O76024	TRPC1	P48995	Q03135
TRPS1	Q9UHF7	Q13257	TRUB1	Q8WWH5	Q96DP5
TRUB2	O95900	P09651	TSC22D3	Q99576-3, Q99576	Q9Y6D9
TSC22D4	Q9Y3Q8	Q8NG68	TSEN54	Q7Z6J9	Q92989
TSNARE1	A0AVG3	Q8N661	TSNAX	Q99598	Q00994
TSPAN10	Q9H1Z9	Q7Z5P4	TSPAN12	O95859	Q96AA3
TSPAN14	Q8NG11	P48509	TSPAN15	O95858	O14672
TSPAN3	O60637	Q99741	TSPAN31	Q12999	Q9BRI3
TSR3	Q9UJK0	Q9Y216	TSSC4	Q9Y5U2	P62306, P62304
TSTD2	Q5T7W7	Q9UGI0	TTC21B	Q7Z4L5	Q8NEZ3, Q96RY7, Q96FT9, Q9HBG6, Q9P2L0
TTC3	P53804	P31751	TTI2	Q6NXR4	P35613
TTK	P33981	P08173	TTL12	Q14166	P68104
TTL7	I6L9D0	Q96CV9	TTN	Q8WZ42	Q9GZQ8
TTYH3	Q9C0H2	P21453	TUBA1A	Q71U36	Q9UBX3

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TUBA4A	P68366	P30622	TUBB2A	Q13885	P36404
TUBB2B	Q9BVA1	P36404	TUBB4A	P04350	P22736
TULP4	Q9NRJ4	Q93034, Q9UBF6	TUSC2	O75896	P11142
TUT4	Q5TAX3	Q9H9Z2	TWF1	Q12792	P67870
TXLNA	P40222	Q96LB3	TXLNG	Q8BHN1	Q9UHD2
TXNDC11	Q6PKC3	Q99633	TXNDC9	O14530	O00628
TXNIP	Q9H3M7	P10599	TXNL4A	P83876	P62306, P14678
TYSND1	Q2T9J0	P11142	TYW1	Q9NV66	Q8WTR4
U2SURP	O15042	Q16629	UACA	Q9BZF9	P04004
UBA2	Q9UBT2	P63279	UBA3	Q8TBC4	Q15843
UBA5	Q9GZZ9	Q15067	UBA52	P62990	P51668
UBA6	A0AVT1	P43353	UBA7	P41226	P05161
UBAC1	Q9BSL1	Q09327	UBAC2	Q8NBM4	Q8TC07, Q9HA65
UBAP2L	Q14157	P10636-8	UBASH3B	Q8TF42	P29353
UBE2D3	P61077	Q9Y4K3	UBE2D4	Q9Y2X8	Q9P0P0
UBE2I	Q7KZS0	P60709	UBE2J2	Q8N2K1	A1L3X0
UBE2K	P61086	Q15843	UBE2Q1	Q7Z7E8	Q9UNE7
UBE2Q2	Q8WVN8	P04792	UBE2W	Q96B02	Q6ZNA4
UBE3A	Q05086	Q06413	UBE4A	Q14139	P55072
UBL4A	P11441	O15055	UBLCP1	Q8WVY7	P62191, Q13200
UBN1	Q9NPG3	P54198	UBOX5	O94941	P68036
UBQLN1	Q9UMX0	Q8WUX9	UBQLN2	Q9UHD9	Q9UNW1
UBQLN4	Q9NRR5	P10451	UBR1	Q8IWV7	P06850
UBR2	Q8IWV8	O14786	UBR3	Q6ZT12	Q13304
UBR4	Q5T4S7	P0DP23	UBR5	O95071	P49841
UBTD1	Q9HAC8	P51668	UBXN4	Q92575	P21802
UBXN6	Q9BZV1, Q9BZV1-1, Q9BZV1-2	P55072	UBXN7	O94888	Q15843
UBXN8	O00124	Q8TC07	UCHL5	Q9Y5K5	P60896
UCK1	Q9HA47-2	P06396	UCN	P55089	P06850
UCP2	P55851	P20930	UFL1	O94874	P21462
UFSP1	Q6NVU6	P35680	UFSP2	Q9NUQ7	Q96FT7
UGCG	Q16739	P43119	UGGT1	Q9NYU2	P04578
UGGT2	Q9NYU1	P11021	UGP2	Q16851	O76003
UHK1	Q8TAS1-2	Q9BU61	UIMC1	Q96RL1	P38398
ULK3	Q6PHR2	Q13451	UNC119	Q13432	P36404
UNC45A	Q9H3U1	P48729	UPF1	Q92900	Q14493
UPF2	Q9HAU5	Q92900	UQCRC1	P31930	P26641
UQCRC2	P22695	P31930, O14949, P14927	UQCRFS1	P47985	Q8IWL3
USF1	P22415	P78527	USF2	Q15853	Q15562
USO1	O60763	Q9Y6L6	USP11	P51784	Q86YC2
USP12	O75317	Q96B01	USP13	Q92995	O60216
USP14	P54578	P60896	USP15	Q9Y4E8, Q9Y4E8-2	Q9Y6D9
USP16	Q9Y5T5	O95714	USP19	O94966	P07900
USP2	O75604-3	Q00403	USP21	Q9UHP3	Q9UKG9
USP24	Q9UPU5	O75695	USP25	Q9UHP3	Q9UKG9
USP28	Q96RU2	Q12888	USP30	Q70CQ3	Q9H5K3
USP33	Q8TEY7	P00533	USP34	Q70CQ2	P48729
USP36	Q9P275	P06748	USP44	Q9H0E7	P41208
USP47	Q96K76	O60216	USP53	Q70EK8	P46109

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
USP8	P40818	Q16665, Q99814	USP9X	Q93008	P29353
USPL1	Q6DRC5, Q5W0Q7	P55854, P63165, P61956	UTP14C	Q5TAP6	Q9NRX1
UTP23	Q9BRU9	Q96RU7	UTRN	P46939	Q14118
UTY	O14607	Q14686	UVRAG	Q9P2Y5	P0DTC3
UXT	Q9UBK9	P36873	VAC14	Q08AM6	Q53HV7
VAMP2	P63027	P55072	VAMP4	O75379	Q13190
VAMP7	P51809	P52594	VANGL1	Q8TAA9	P18827
VASN	Q6EMK4	P78504	VAT1	P54219	Q7L8W6
VBP1	P61758	O00628	VCL	P18206	Q9Y490
VDAC3	Q9Y277	P21796	VEGFB	P49765	P15692
VGLL3	A8MV65-2	O43559	VGLL4	Q14135	P46937
VIPAS39	Q9H9C1	P12273	VIRMA	Q69YN4	Q86VM9
VLDLR	P98155	P21579	VMP1	Q96GC9	Q07157
VOPP1	Q96AW1	Q9NZC7	VPS11	Q9H270	P0DTC3
VPS13B	Q7Z7G8	P49674	VPS13D	Q5THJ4	P31946
VPS18	Q9P253	P0DTC3	VPS29	Q9UBQ0-2	P60709
VPS33A	Q96AX1	Q14790	VPS36	Q86VN1	P0CG47
VPS37B	Q9H9H4	P07992	VPS45	P38932	Q13190
VPS4B	O75351	Q9UN37	VPS54	Q9P1Q0-4	P06396
VSIR	Q9H7M9	P27449	VTa1	Q9NP79	Q9UN37, O43633
VWA8	A3KMH1	Q12824	WAPL	Q7Z5K2	O60216, Q7Z5K2
WASF1	Q92558	Q8IY31	WASHC2A	Q641Q2	P51530
WASHC2C	Q9Y4E1	P47756	WASHC3	Q9Y3C0	O75385
WBP1L	Q9NX94	Q8N2W9	WBP2	Q969T9	P46937
WDCP	Q9H6R7	P53350	WDFY3	Q8IZQ1	Q9GZQ8
WDR13	Q9H1Z4	P21709	WDR18	Q9BV38	Q8IZL8
WDR24	Q96S15	P55735	WDR31	Q8NA23-2	P0CG47
WDR35	Q9P2L0-2, Q9P2L0	Q7Z4L5, Q8NEZ3, Q96RY7, Q96FT9, Q9HBG6	WDR4	P57081	P39748
WDR44	Q5JSH3	P16284	WDR48	Q8TAF3	Q96B01
WDR54	Q9H977	Q9Y316	WDR6	Q9NNW5	P03496
WDR7	Q9Y4E6	P21281	WDR70	Q9NW82	Q06330
WDR75	Q8IWA0	Q13901, Q9NQT4	WDR81	Q562E7	Q14508
WDR82	Q6UXN9	P36873	WDR83	Q9BRX9	O75695
WDR90	Q96KV7	Q13451	WDR91	A4D1P6	P42858
WHRN	Q9P202	Q9UMD9	WIZ	O95785	P08311
WLS	Q5T9L3	P41221	WNT2B	Q93097	Q70UQ0
WNT5A	P41221	Q5T9L3	WRAP53	Q9BUR4	O60832, Q9NY12, Q9NPE3
WRAP73	Q9P2S5	Q9Y266	WRN	Q14191	P39748
WRNIP1	Q96S55	O95786	WSB1	Q9Y6I7	Q93034, Q9UBF6
WWC1	Q8IX03	Q05513	WWC2	Q6AWC2	P31946
WWP1	Q9H0M0	Q96J02	WWP2	O00308	O14641
WWTR1	Q9EPK5	Q07157	XBP1	P17861	Q16665
XIRP1	Q702N8	P50552	XKR8	Q9H6D3	P43119
XPA	P23025	Q9P0N9	XPO1	O14980	Q9UKV8
XPR1	Q9UBH6	P21802	XRN1	Q8IZH2	P09651
XXYLT1	Q8NBI6	Q04721	YBX1	P67809	Q86U42
YEATS2	Q9ULM3	Q9Y2G2	YES1	P07947	P46109
YIPF3	Q9GZM5	Q9NRZ7	YIPF5	Q969M3	O95070, O75460

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
YME1L1	Q96TA2	Q12824	YRDC	Q86U90	Q5VWX1
YTHDC1	Q96MU7	Q16629	YTHDC2	Q9H6S0	P40429
YTHDF1	Q9BYJ9	P08311	ZBED1	O96006	Q05048
ZBTB12	Q9Y330	O43639	ZBTB17	Q13105	P04198
ZBTB2	Q8N680	P08047	ZBTB26	Q9HCK0	P63279
ZBTB4	Q9P1Z0	Q08379	ZBTB40	Q9NUA8	Q99435
ZBTB43	O43298	P49407	ZBTB45	Q96K62	O95429
ZBTB46	Q86UZ6	Q9UF56	ZBTB47	Q9UFB7	Q9UGI0
ZBTB7A	O95365	Q13547	ZBTB9	Q96C00	Q04721
ZC2HC1A	Q96GY0	O43639	ZC3H10	Q96K80	P40429
ZC3H13	Q5T200	Q86VM9	ZC3H14	Q6PJT7	O75152
ZC3H18	Q86VM9	Q9C0J8, Q7Z6E9	ZC3H3	Q8IXZ2	P36941
ZC3H7A	Q8IWR0	P12273	ZC3HAV1	Q7Z2W4, Q7Z2W4-2	O95786
ZCCHC3	Q9NUD5	Q86VM9	ZCCHC8	Q6NZY4	P38398
ZDBF2	Q9HCK1	P48431	ZDHHC15	Q96MV8	Q9H2K0
ZDHHC16	Q969W1	O95070	ZDHHC20	Q5W0Z9-4	O76024
ZDHHC21	Q8IVQ6	Q16581	ZDHHC7	Q9NXF8-2	O43561-2
ZEB1	P37275	Q92769	ZFAT	Q9P243	Q9UER7
ZFP41	Q8N8Y5	P98164	ZFP90	Q8TF47	Q9Y2W6
ZFR	Q96KR1	P09651	ZFTRAF1	P0DTL6	P18827
ZFYVE19	Q96K21-3	O15169	ZFYVE26	Q68DK2-5	Q9Y5V3
ZGRF1	Q86YA3	Q06609	ZHX1	Q9UKY1	P25490
ZKSCAN1	P17029	Q13164	ZMAT3	Q9HA38	Q9UKV8
ZMYM2	Q9UBW7	Q16822	ZMYM4	Q5VZL5-4	Q96CV9
ZMYND11	Q15326	P36941	ZMYND19	Q96E35	Q6UN15
ZMYND8	Q9ULU4	Q92769, Q13547	ZNF10	P21506	Q13263
ZNF101	Q8IZC7	Q99435	ZNF12	P17014	Q13263
ZNF121	P58317	P01106	ZNF148	Q9UQR1	P52294
ZNF175	Q9Y473	P13196	ZNF195	O14628	Q13263
ZNF202	O95125	Q9H2K0	ZNF205	O95201	Q9NRD5
ZNF217	O75362	Q92769, Q13547	ZNF219	Q9P2Y4	Q04721
ZNF23	P17027	Q96EZ8	ZNF232	Q9UNY5	P12931
ZNF233	Q9NZL3	P98164	ZNF263	O14978	Q9UMX1
ZNF266	Q14584	Q9HD47	ZNF268	Q14587	O15431
ZNF280C	Q8ND82	Q13185	ZNF280D	Q6N043, Q6N043-2	Q13185
ZNF292	O60281	P83916	ZNF316	A6NFI3	Q04721
ZNF324B	Q6AW86	Q9H6S0	ZNF326	Q5BKZ1	P09651
ZNF330	Q9Y3S2	Q9UMX1	ZNF341	Q9BYN7	Q13882
ZNF354C	Q86Y25	O15397	ZNF358	Q9NW07	Q04721
ZNF384	EBI-11743072	P14373	ZNF398	Q8TD17	Q96CV9
ZNF408	Q9H9D4	Q99611	ZNF41	P51814-6	P55081
ZNF426	Q9BUY5	P57764	ZNF44	M0R160	O14964
ZNF441	Q8N8Z8	Q16623	ZNF451	Q9Y4E5	P63279
ZNF48	Q96MX3	Q13895	ZNF484	Q5JVG2	Q13263
ZNF503	Q96F45	O75593	ZNF507	Q8TCN5	O94901
ZNF511	Q8NB15	P50458	ZNF518A	Q6AHZ1	Q13185
ZNF518B	Q9C0D4	Q13185	ZNF524	Q96C55	Q9NWX6
ZNF526	Q8TF50	Q04721	ZNF536	O15090	P48431
ZNF558	Q96NG5	P57764	ZNF561	Q8N587	Q13263

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ZNF580	Q9UK33	P14373	ZNF584	Q8IVC4	P05067
ZNF592	Q92610	Q13547	ZNF593	O00488	Q9Y4Z2
ZNF596	Q8TC21	P60900	ZNF598	Q86UK7	P63244
ZNF621	Q6ZSS3	P45973	ZNF622	Q969S3	Q6NZI2
ZNF628	Q5EBL2	Q8IYI6	ZNF629	Q9UEG4	P40429
ZNF646	O15015	A8MZ59	ZNF655	Q8N720	P23025
ZNF668	Q96K58-2, Q96K58	Q9UGY1, Q13895	ZNF672	Q499Z4	Q04721
ZNF687	Q8N1G0	Q13547	ZNF689	Q96CS4	P45973
ZNF70	Q9UC06	Q8IXZ2	ZNF74	Q16587	Q8N2S1
ZNF768	Q9H5H4	Q04721	ZNF771	Q7L3S4	P40429
ZNF772	Q68DY9	P50542	ZNF785	A8K8V0	Q9NPB3
ZNF786	Q8N393	Q9NS56	ZNF791	Q3KP31	O76024
ZNF793	Q6ZN11	Q96EB6	ZNF865	P0CJ78	P36941
ZNF92	Q03936	P05067	ZNHIT2	Q9UHR6	Q9Y265, Q9Y230
ZPR1	O75312	P68104	ZRANB2	O95218	P32121, P49407
ZRANB3	Q5FWF4	P12004	ZSCAN22	P10073	Q00403
ZSCAN23	Q3MJ62	P54646	ZSCAN26	Q16670	O76024
ZSCAN30	Q86W11	O43559	ZSWIM8	A7E2V4	P49674
ZSWIM9	Q86XI8	Q13185	ZW10	O43264	Q12981, Q9NZ43, Q9P2W9, O75396
ZYG11B	Q9C0D3	P15907			

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ACBD6	Q9BR61	15530	BMP1	P13497	29108
BOLA1	Q9Y3E2	33737	GCC2	Q8IWJ2	15996
LRRK2	Q5S007	17552	MMUT	P22033	18408
NMT2	O60551	15530	NPC2	P61916	16113
PLEKHM1	Q9Y4G2	15996	RAB32	Q13637	15996
RAB38	P57729	15996	RAB9A	P51151	15996
RCC1	P18754	17552	RHOG	P84095	16152
RILP	Q96NA2	15996	RILPL2	Q969X0	15996
RRAGD	Q9NQL2	15996	S100A13	P97352	29036
SYT1	P21707	29036	THBS1	P07996	28815

7. Identifiers not found

These 857 identifiers were not found neither mapped to any entity in Reactome.

AASDH	ABALON	ABCA11P	ABHD16B	ABHD8	ABRACL	ACAP2	ACTR3B
ACYP2	ADCK1	AFAP1	AFAP1L2	AFG3L1P	AGAP1	AGAP9	AJAP1
AJM1	AKTIP	ALKBH6	ALKBH7	ALPK2	ALPK3	ANGEL2	ANKRD13B
ANKRD16	ANKRD30B	ANKRD37	ANKS3	ANXA11	ANXA8L1	ARHGAP11A-SCG5	ARHGAP29.1
ARMCX4	ARMCX6	ARRDC1-AS1	ASB3.1	ATF7IP2	ATOH8	ATP1A1-AS1	ATP5MF-PTCD1
ATP6V0E2-AS1	ATP8B1-AS1	ATPSCKMT	BAALC-AS1	BCL2L2-PABPN1	BCLAF3	BICC1	BIVM-ERCC5
BLTP1	BMP8B	BNC2	BOLA2B.1	BRD10	BRD3OS	BTBD10	BZW1
BZW1-AS1	C10orf90	C10orf95-AS1	C14orf132	C15orf61	C17orf75	C1QTNF9B	C1orf115
C1orf167	C1orf174	C1orf43	C22orf23	C2CD2	C2orf69	C6orf132	C6orf136
C6orf62	C8orf44-SGK3	C8orf82	C9orf40	CA5BP1	CACHD1	CALB2	CALML4
CAMK2N1	CAMTA1	CAPN1-AS1	CAPS2	CASZ1	CCDC142	CCDC171	CCDC183-AS1
CCDC191	CCDC80	CCDC91	CCDST	CCER2	CCNQ	CD276	CD63-AS1
CELF6	CELSR1	CELSR3	CEMP1	CERS6-AS1	CFAP300	CFAP97	CHIC1
CHROMR	CKAP2L	CKMT2-AS1	CNN3-DT	CORO2B	CPEB4	CPED1	CPLANE1
CRIM1	CRYZ	CRYZL2P	CTBP1-AS	CTIF	CTXN1	CYLD-AS1	DAAM2
DANCR	DAW1	DBNDD2	DCDC2C	DDX11L16.3	DDX11L2	DDX11L2.1	DDX59
DDX60	DDX60L	DEPDC1	DEPDC7	DIRC3	DLK2	DMTF1	DNAAF9
DNAH14	DNAH5	DNAJC21	DNAJC22	DOP1A	DRGX	DSP-AS1	DTD1
DUS1L	DUS4L-BCAP29	DUSP22	DYRK3-AS1	ECHDC3	EEF1AKMT4	EFR3A	EHBP1-AS1
EHMT2-AS1	EID2	EID2B	EIF2S3B	ELAPOR2	ELK4	ELMOD2	ELOF1
ENDOV	EPG5	EPHX4	ERCC6L2	FAM107B	FAM131A	FAM131B-AS2	FAM149B1
FAM174B	FAM174C	FAM21EP	FAM229B	FAM43B	FAM53C.1	FAM86B3P	FANCC.1
FAT4	FBXL19-AS1	FBXL9P	FBXO36	FBXO8	FILIP1L	FJX1	FLNB-AS1
FLVCR2	FOCAD	FRMD4B	GALNT7-DT	GAS6-DT	GATD1-DT	GBA1LP	GCNT2
GDF11	GFOD2	GFOD3P	GK5	GLMP	GLOD4	GNPTG	GOLGA6L10
GOLGA6L4	GOLGA8A	GOLGA8B	GOLM2	GPALPP1	GPATCH3	GPR137	GPR137B
GREB1L	GRINA	GRTP1	GSAP	GTF2IRD1P1	GTPBP10.1	GTPBP6	GULP1
GUSBP11	H2BC12L.1	HAGHL	HARBI1	HCP5	HDGFL3	HDHD2	HMGXB3
HPS5	HPS6	HRK	HSD17B1-AS1	HSDL2	IFITM10	IGFL2-AS1	IKZF2
ILF3-DT	ILRUN-AS1	INE1	INKA2	INSYN2B	INVS	IQCG	IQCH
IQCH-AS1	IRAK1BP1	IRX3	JAKMIP3	JKAMP	JMJD8	JRKL	JTB
KBTBD11	KBTBD11-OT1	KCP	KCTD2	KCTD8	KIAA1586	KIF23-AS1	KRT7-AS
LANCL3	LDLRAD2	LDLRAD3	LINC00205	LINC00520	LINC00654	LINC00667	LINC00847
LINC00863	LINC01111	LINC01123	LINC01128	LINC01134	LINC01239	LINC01278	LINC01444
LINC01561	LINC01833	LINC02009	LINC02159	LINC02693	LINC02802	LMBR1	LMLN
LRP3	LRRCC1	LRRIQ1	LSP1P5	LSP1P5.1	LYPLAL1	LYRM9	LYSET
LYSMD4	MAMSTR	MAP3K10	MAP6	MAP9	MARCHF9	MAST4	MATCAP1
MCTS1	MDFIC	MDM1	MEGF9	METRNL	METTTL25B	MFRP	MFSD12
MINCR	MINDY2	MIR100HG	MIR1915HG	MIR210HG	MIR22HG	MIR3064	MIR3652
MIR4683	MIR4709	MIR5047	MKLN1-AS	MMAA.1	MPDU1-AS1	MPV17L2	MREG
MRFAP1L2	MRPL20-AS1	MSANTD2	MSANTD3-TMEFF1	MTX3	MYCBP	MYL12-AS1	MYRF
MZF1	MZF1-AS1	MZF1-AS1.1	N4BP2	NAA25	NAGPA	NALF1	NAT9

NAV3	NCBP2AS2	NEAT1	NFKBIZ	NORAD	NPIPB13	NPTXR	NRAV
NRBP2	NSUN5	NSUN5P1	NSUN5P2	NSUN7	NT5DC1	NT5DC3	None.10101
None.10162	None.10163	None.10216	None.10218	None.1027	None.10284	None.10425	None.1052
None.10701	None.10893	None.10980	None.11016	None.1104	None.11104	None.11138	None.11139
None.11157	None.11163	None.11164	None.11198	None.11224	None.11341	None.11349	None.11469
None.11625	None.11647	None.1167	None.11670	None.11674	None.11705	None.11712	None.11715
None.11849	None.1185	None.12042	None.12076	None.12077	None.12078	None.12079	None.12080
None.12085	None.12217	None.12321	None.12563	None.12584	None.12585	None.12606	None.12610
None.12711	None.12736	None.12750	None.12764	None.12873	None.12889	None.1305	None.13054
None.13100	None.13142	None.13189	None.13214	None.13245	None.13263	None.13282	None.13283
None.13284	None.13288	None.13289	None.13290	None.13291	None.13294	None.13296	None.13297
None.13409	None.13411	None.13457	None.13466	None.13714	None.13794	None.13856	None.13857
None.13858	None.13859	None.13873	None.13878	None.13919	None.14056	None.1414	None.14161
None.14244	None.14245	None.14246	None.14247	None.14279	None.14280	None.14281	None.14282
None.14283	None.14284	None.14285	None.14286	None.14295	None.14297	None.14298	None.14299
None.14301	None.165	None.1687	None.1745	None.1798	None.1799	None.1937	None.2101
None.2103	None.2160	None.22	None.2231	None.2246	None.2372	None.2399	None.254
None.256	None.2743	None.2796	None.2797	None.2886	None.29	None.2939	None.2941
None.2986	None.3018	None.3055	None.3176	None.3363	None.3377	None.347	None.3484
None.373	None.38	None.3851	None.3909	None.394	None.3956	None.3989	None.4074
None.4105	None.4167	None.4212	None.4346	None.4549	None.4601	None.4603	None.4764
None.481	None.483	None.4946	None.5086	None.5237	None.5372	None.5693	None.5743
None.5868	None.5980	None.5986	None.5996	None.6119	None.6238	None.6294	None.6385
None.6432	None.6476	None.6555	None.6905	None.6928	None.6971	None.7065	None.7131
None.7210	None.7220	None.725	None.7250	None.7428	None.7465	None.7482	None.7560
None.7595	None.766	None.7835	None.7836	None.7846	None.789	None.7908	None.7909
None.7920	None.8020	None.8102	None.8131	None.8219	None.8233	None.8340	None.835
None.8352	None.8369	None.8372	None.8431	None.8450	None.8458	None.8479	None.8506
None.8529	None.8531	None.8540	None.8542	None.8635	None.88	None.8814	None.883
None.92	None.9207	None.9348	None.9358	None.9567	None.9615	None.9621	None.9775
None.9801	None.9901	None.998	OCIAD2	OGA	OPA3	ORAI3	P4HTM
PARP3	PAX8-AS1	PCDH9	PCIF1	PCNX1	PCNX3	PCNX4	PDE7B-AS1
PDLIM2	PEDS1	PELATON	PEPD	PHEX	PHF6	PHLDA3	PHTF2
PIM3	PINK1-AS	PKD1L2	PKD1P6-NPIPP1	PLD5	PLEKHF1	PLEKHH2	PLEKHM2
PLPBP	PLS3-AS1	PNMA8A	POLR2J4	PPAN-P2RY11	PPDPF	PPP1R35	PPP1R35-AS1
PRC1-AS1	PRDM15	PREPL	PRICKLE2	PROB1	PRR12	PRRT3	PRTG
PSMG3-AS1	PXDC1	PXDNL	PXK	PYROXD1	PYROXD2	QRICH2	RABL2B.1
RAD54B	RANBP17	RARA-AS1	RBBP9	RBMS3	RBPMS-AS1	RC3H1	RC3H2
RELA-DT	REXO1	RFLNB	RGPD1	RGPD4	RGPD5.1	RGS7BP	RN7SK
RN7SL1	RN7SL2	RN7SL3	RNA5-8SN1	RNA5-8SN2	RNA5-8SN3	RNASEH1-DT	RNF180
RNF227	RNF32-DT	RNFT2	RNPEPL1	RNU1-2	RNU1-3	RNU1-4	RNU4-2
RNVU1-18	RNVU1-7	ROCK1P1	RPLP0P2	RPS10-NUDT3	RPS6KC1	RPUSD1	RSAD1
RTCA	RTL8A	RTTN	RUFY3	S100A3	SAC3D1	SAMD9	SAMD9L
SANBR	SAYSD1	SBDSP1	SBF2-AS1	SCAMP1-AS1	SCG5	SDCBP2-AS1	SDR39U1
SEC14L2	SECISBP2L	SELENOW	SENP6	SERF2	SFT2D3	SH2D5	SH3BGRL2
SH3BP5-AS1	SHBG	SHE	SHROOM4	SIDT2	SIX3-AS1	SLC16A12	SLC16A4
SLC25A25	SLC25A33	SLC25A35	SLC25A36	SLC25A38	SLC25A39	SLC35E2B	SLC35F3
SLC46A3	SLC49A4	SLC4A11	SLC66A3	SLC6A10P	SLC6A17	SLC9A3-OT1	SLCO5A1
SLFN5	SMCR8	SMIM12	SMIM14	SMIM29	SMIM30	SMPDL3A	SNCG
SNHG17	SNN	SNX19	SPAG16	SPAG7	SPATA25	SPEF2	SPEN-AS1

SPMIP1	SRPX2	SRRD	ST7L	STEAP2	STK17A	STK33	STPG1
STYX	SUN3	SWI5	TALAM1	TATDN1	TATDN3	TBC1D19	TBC1D23
TBC1D32	TBC1D7.1	TBCK	TCEAL3	TEN1-CDK3	TENT2	TENT5C	TEX56P
TFAP2A-AS2	TGFB2-OT1	TGIF2-RAB5IF	THAP5	TIGD6	TIPARP-AS1	TJAP1	TM4SF1
TMED7-TICAM2	TMEM104	TMEM123	TMEM127	TMEM131L	TMEM158	TMEM161B-DT	TMEM175
TMEM178B	TMEM209	TMEM233	TMEM245	TMEM249	TMEM250	TMEM263	TMEM268
TMEM273	TMEM70	TMEM71	TMEM8B	TP53I11	TPBG	TPRA1	TPRG1L
TRABD	TRAF3IP2-AS1	TRIQK	TSKU	TTC9B	TVP23A	TWSG1	TXNDC12
UAP1L1	UBALD2	UBE2Q1-AS1	UBE2Q2P1	UBL3	UEVLD	UFM1	URGCP-MRPS24
USF3	USP40	USP9Y.1	UXT-AS1	VAC14-AS1	VEPH1	VESTAR	VEZT
VPS13A	VPS13C	VWA5A	WDFY3-AS2	WDPCP	WDR17	WDR41	WDR47
WFDC3	WSCD1	XKR3	XKR4	YJEFN3	ZAR1	ZBED4	ZBTB20
ZBTB37	ZBTB41	ZC3H12C	ZCCHC14	ZCWPW2	ZFAND1	ZFX	ZGLP1
ZMAT1	ZMIZ1-AS1	ZNF107	ZNF174	ZNF185	ZNF433-AS1	ZNF503-AS2	ZNF532
ZNF654	ZNF667-AS1	ZNF780B	ZNF787	ZNF8-ERVK3-1	ZNF804A	ZNF90	ZSWIM1
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